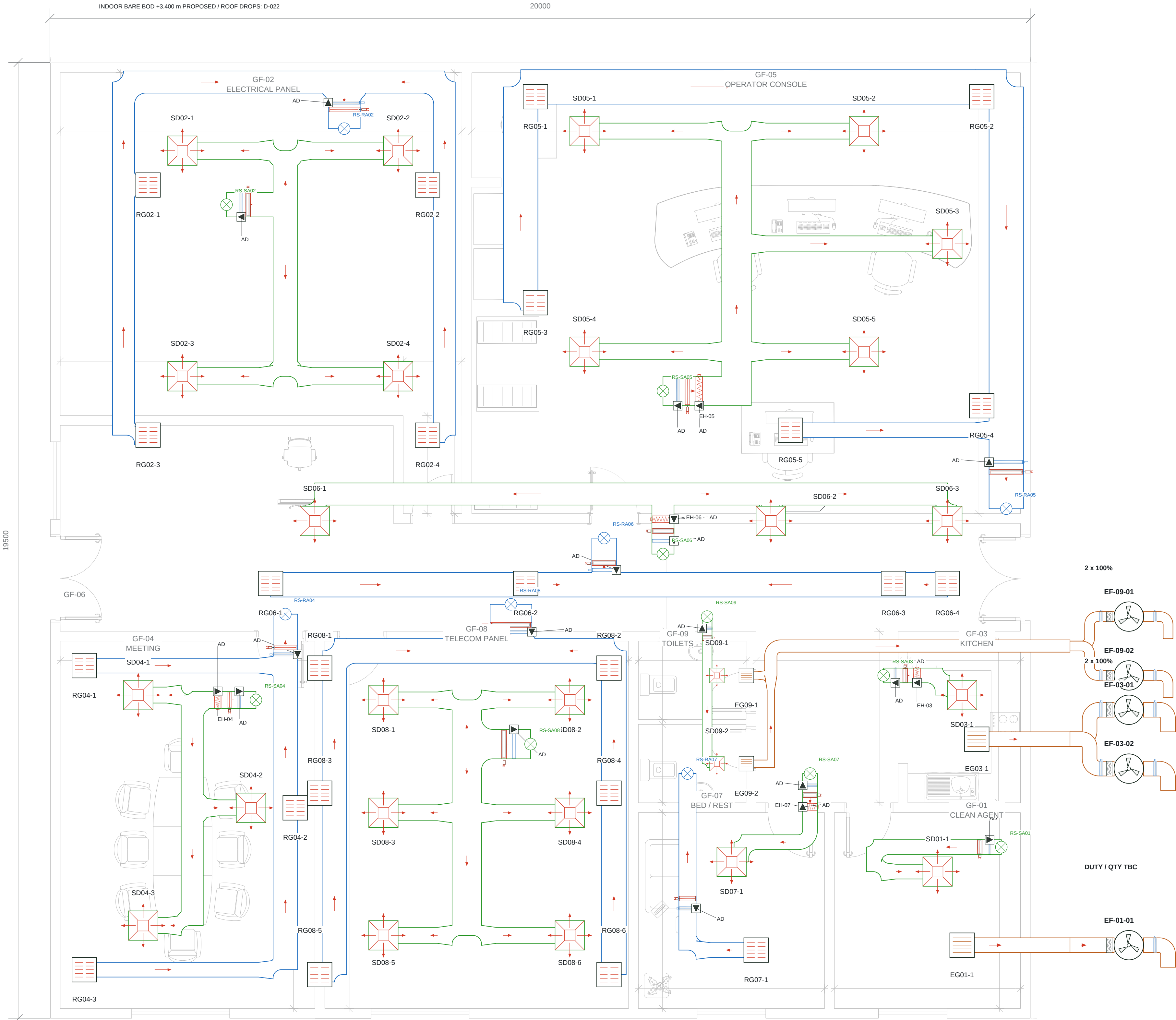




ROOM AIRFLOW SCHEDULE

GF	Room	SA L/s	RA L/s*	EA L/s*
02	ELECTRICAL PANEL	664.8	605.5	-
05	OPERATOR CONSOLE	777.7	718.4	-
04	MEETING	449.2	389.9	-
08	TELECOM PANEL	884.6	825.3	-
06	CORRIDOR	418.2	358.9	-
07	BED / REST	168.2	108.9	-
09	TOILETS	46.2	-	64.9
03	KITCHEN	147.6	-	147.6
01	CLEAN AGENT	141.3	-	TBC

DUCT / SYMBOL LEGEND

	SA - supply duct
	RA - return duct
	EA - dedicated extract
	SD - four-way supply diffuser
	RG - ducted return grille
	EG - extract grille
	VCD - balancing damper
	MFD* - conditional fire damper
	RCD - return control damper
	EH - inline electric heater
	AD - access door
	RS - proposed roof supply / return drop

Terminal aliases: SD02-1 = SD-02-01; RG / EG follow the same format. Full tags and quantities are in the registers.

Roof-main OPTION: 15 independent room drops RS-SA / RS-RA. No indoor service crossings. Finished ceiling +3.300 m; proposed bare BOD +3.400 m. Envelope: D-020; roof/drop holds: D-022.

* Return quantities are assumed balances. Extract references and +50 / +25 Pa targets require final airflow and pressure verification.

Units: mm / L/s / m/s. Terminal faces and clear duct widths are shown. Duct levels, selected fitting development and manufacturer clearances remain pending. No shared return from kitchen, toilet or clean-agent room.

TAQA TRANSMISSION

N-19082-RO-012 | PUMPING STATION STANDARDIZATION

ADMINISTRATION HVAC

DESIGN BASIS: C-H-SS-001 Rev0 / HAP REFERENCE

DRAWING TITLE

COORDINATED DUCT PLAN

FOR COORDINATION | CHECKED / APPROVED: PENDING

DRAWING NUMBER

ADM-HVAC-D-001

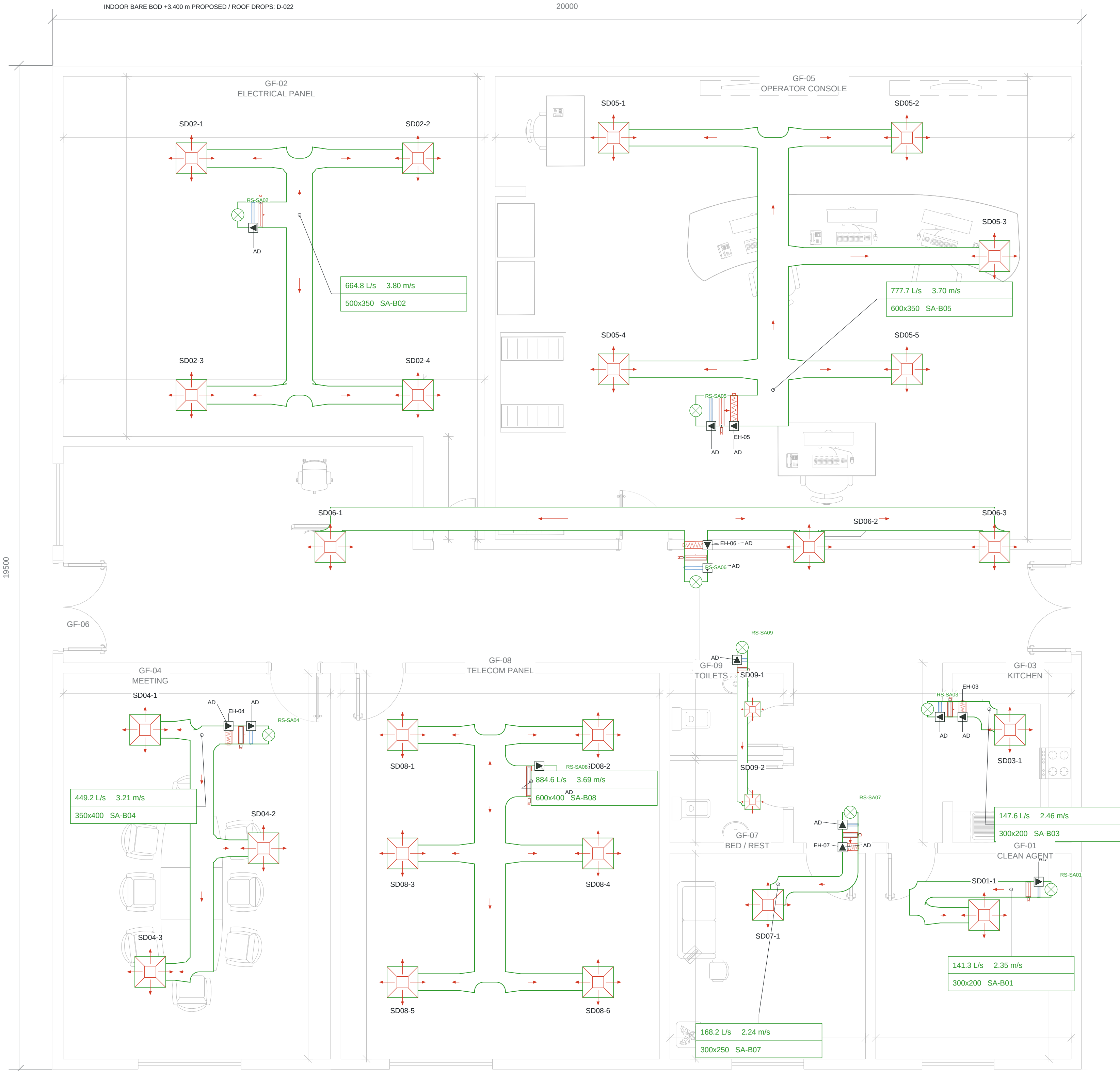
SCALE: 1:50 AT A1

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11

07 OCT 2026

SHEET 1 OF 22



PROPOSED ROOF SUPPLY MAIN BASIS

Section	W x H mm	L/s	m/s
SA-M01	1200x650	3697.8	4.74
SA-M02	1100x575	3033.0	4.80
SA-M03	650x750	2255.3	4.63
SA-M04	650x575	1806.1	4.83
SA-M05	650x300	921.5	4.73
SA-M06	500x250	503.3	4.03
SA-M07	450x250	457.1	4.06
SA-M08	400x200	288.9	3.61
SA-M09	300x200	141.3	2.35

ROOM SUPPLY BRANCHES

GF	W x H mm	L/s	m/s
02	500x350	664.8	3.80
05	600x350	777.7	3.70
04	350x400	449.2	3.21
08	600x400	884.6	3.69
06	450x300	418.2	3.10
07	300x250	168.2	2.24
09	200x150	46.2	1.54
03	300x200	147.6	2.46
01	300x200	141.3	2.35

Room headers retain their branch size. Diffuser tags identify individual outlets. Round runout / neck diameters and each terminal airflow are listed on D-004 to D-006 and D-010.

MFD* is a conditional rated-penetration provision. VCD-Snn and FSD-Snn correspond to the GF-rn supply branch. Inline heaters are upstream of that room's terminal take-offs.

Continuous main tapers, flared branch boots and terminal adaptors: D-011. AD tags identify maintenance access doors. Proposed taper lengths and device coordinates are in the registers.

Total HAP-reference supply: 3697.80 L/s. Three 50% PAUs: two duty, rotating available alternate. Boxed callouts show branch flow, velocity and clear size. No room sensors are overlaid on this duct plan.

TAQA TRANSMISSION

N-19082-RO-012 | PUMPING STATION STANDARDIZATION

ADMINISTRATION HVAC

DESIGN BASIS: C-H-SS-001 Rev0 / HAP REFERENCE

DRAWING TITLE

SUPPLY AIR DUCT PLAN

FOR COORDINATION | CHECKED / APPROVED: PENDING

DRAWING NUMBER

ADM-HVAC-D-002

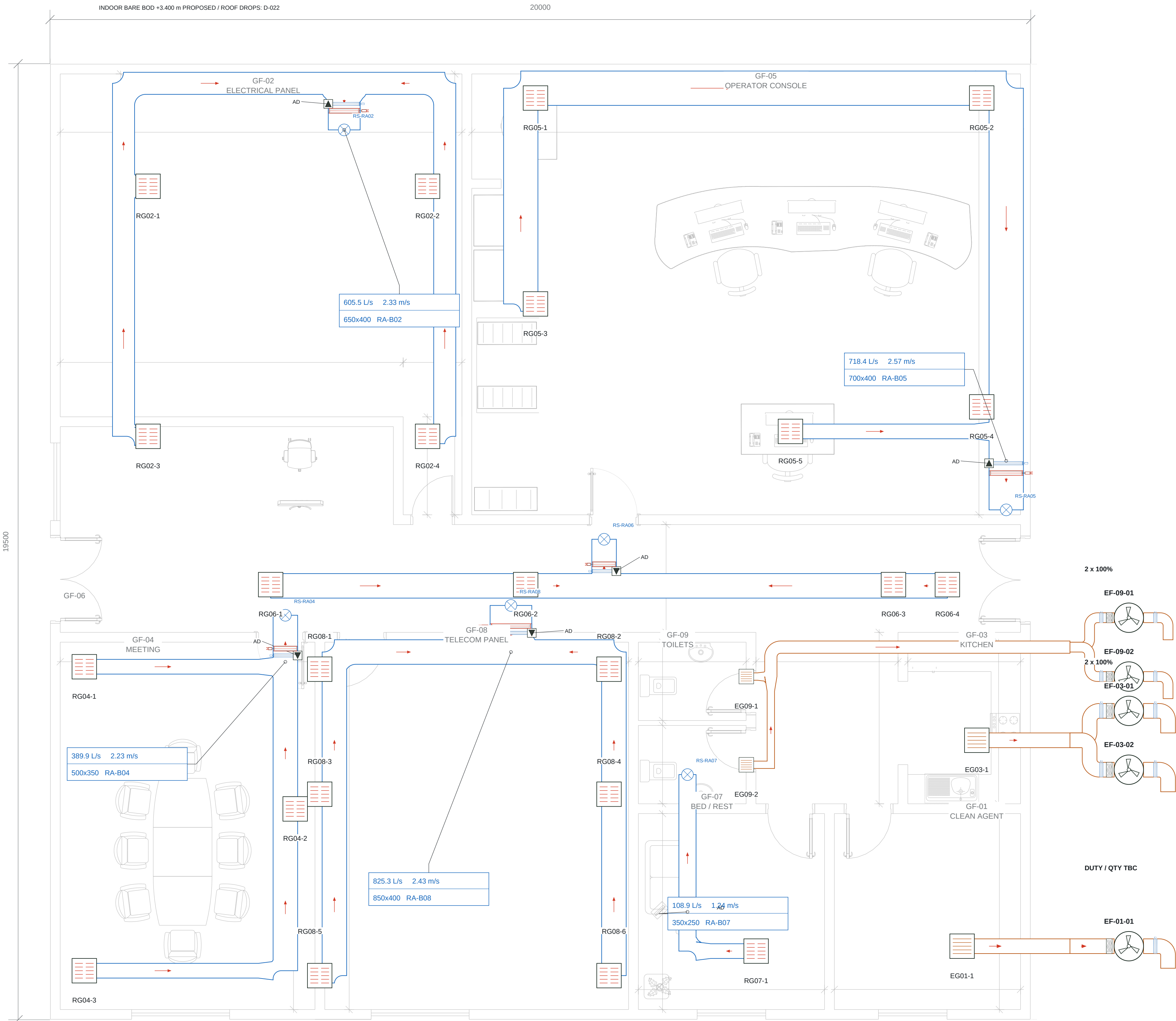
SCALE: 1:50 AT A1

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11

07 OCT 2026

SHEET 2 OF 22



PROPOSED ROOF RETURN MAIN BASIS

Section	W x H mm	L/s	m/s
RA-M01	1400x650	3006.7	3.30
RA-M02	1200x600	2401.2	3.34
RA-M03	600x800	1682.9	3.51
RA-M04	600x600	1293.0	3.59
RA-M05	350x575	467.7	2.32
RA-M06	350x300	108.9	1.04

ROOM RETURN / EXTRACT

GF	Air	W x H mm	L/s*
02	RA	650x400	605.5
05	RA	700x400	718.4
04	RA	500x350	389.9
08	RA	850x400	825.3
06	RA	500x400	358.9
07	RA	350x250	108.9
09	EA	250x200	64.9
03	EA	300x200	147.6
01	EA	TBC	TBC

No central return: GF-09 toilets, GF-03 kitchen and GF-01 clean-agent. Kitchen / toilet fans: 2 x 100% with duty rotation. CAG duty and quantity remain unverified.

Extract fan symbols are schematic equipment markers; actual fan and louver dimensions depend on manufacturer selection. Wall outlets face downwards.

* Return total 3006.68 L/s is an assumed starting balance. Existing extract references do not verify positive pressure or outdoor-air compliance. Confirmed fire shuts down all HVAC; automatic fire extraction is not enabled.

TAQA TRANSMISSION

N-19082-RO-012 | PUMPING STATION STANDARDIZATION

ADMINISTRATION HVAC

DESIGN BASIS: C-H-SS-001 Rev0 / HAP REFERENCE

DRAWING TITLE

RETURN AIR AND EXTRACT DUCT PLAN

FOR COORDINATION | CHECKED / APPROVED: PENDING

DRAWING NUMBER

ADM-HVAC-D-003

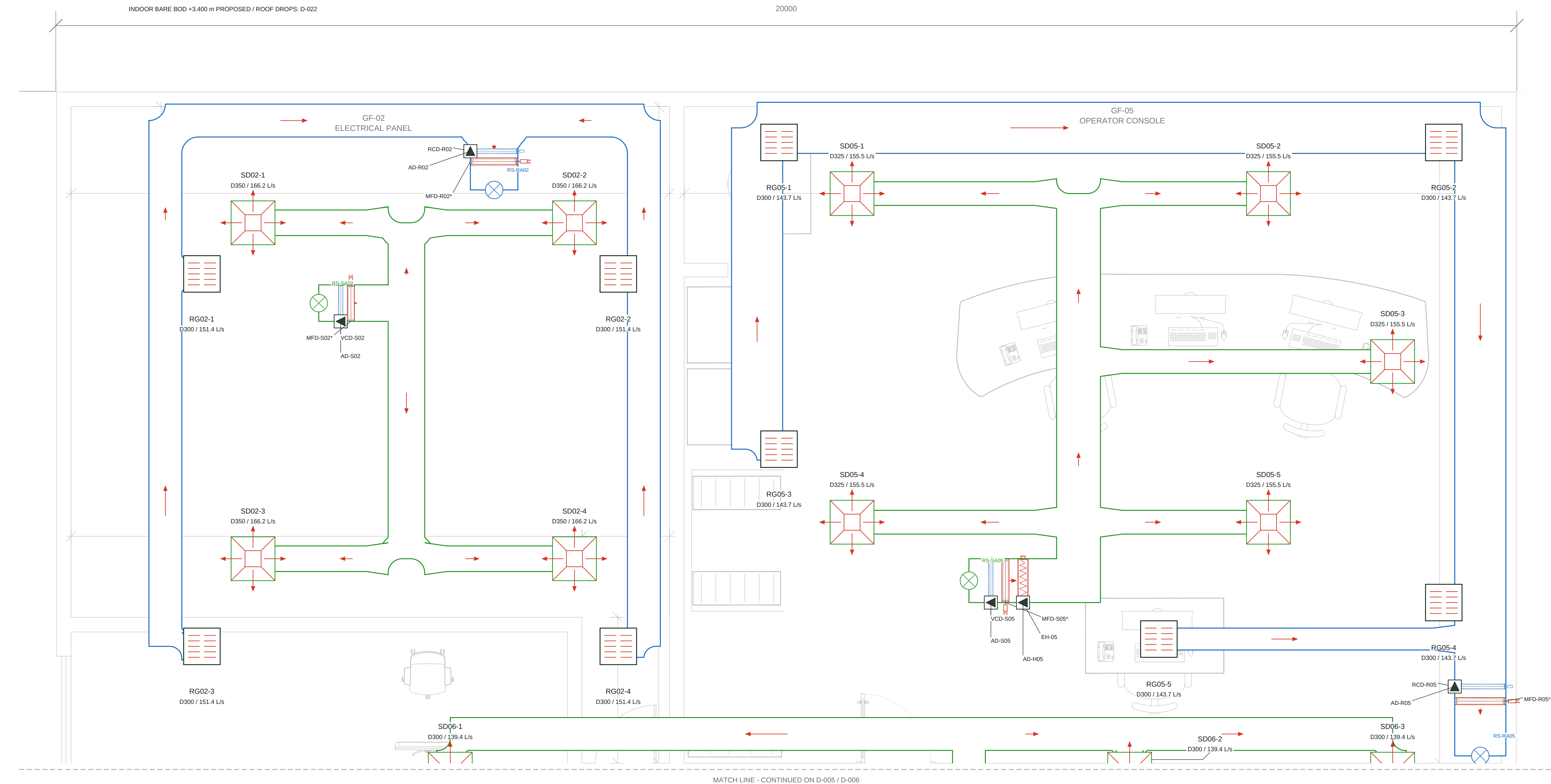
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REVISION

11

07 OCT 2026

SHEET 3 OF 22



NORTH-ROOM TERMINALS / BRANCHES

GF-02 ELECTRICAL PANEL

SA-B02 500x350 mm | 664.80 L/s | 3.80 m/s

RA-B02 650x400 mm | 605.46 L/s | 2.33 m/s

SD: 4 x 166.20 L/s; neck D350; face 600x600

RG: 4 x 151.37 L/s; neck D300; face 500x500

Room header: branch size. Terminal runout: listed neck.

GF-05 OPERATOR CONSOLE

SA-B05 600x350 mm | 777.70 L/s | 3.70 m/s

RA-B05 700x400 mm | 718.36 L/s | 2.57 m/s

SD: 5 x 155.54 L/s; neck D325; face 600x600

RG: 5 x 143.67 L/s; neck D300; face 500x500

Room header: branch size. Terminal runout: listed neck.

Plan symbols identify VCD-S02 / S05, conditional MFD-S02 / S05*, RCD-R02 / R05 and conditional return MFDs. The component register retains the FSD tag prefix for the conditional motorized fire/smoke damper symbols.

Deep common mains are proposed on the roof. Separate supply / return drops feed each room tree. Bare BOD +3.400 m is a proposed indoor level. Actual roof routes and structural penetrations require coordination; D-020 / D-022.

Access doors: AD-Snn / AD-Rnn beside branch dampers; AD-Hnn beside heaters. Taper / access arrangements: D-011. Instrumentation: D-007 / D-012. Catalogue selection remains pending.

TAQA TRANSMISSION

N-19082-RO-012 | PUMPING STATION STANDARDIZATION

ADMINISTRATION HVAC

DESIGN BASIS: C-H-SS-001 Rev0 / HAP REFERENCE

DRAWING TITLE

NORTH ROOMS - DUCT ENLARGEMENT

FOR COORDINATION | CHECKED / APPROVED: PENDING

DRAWING NUMBER

ADM-HVAC-D-004

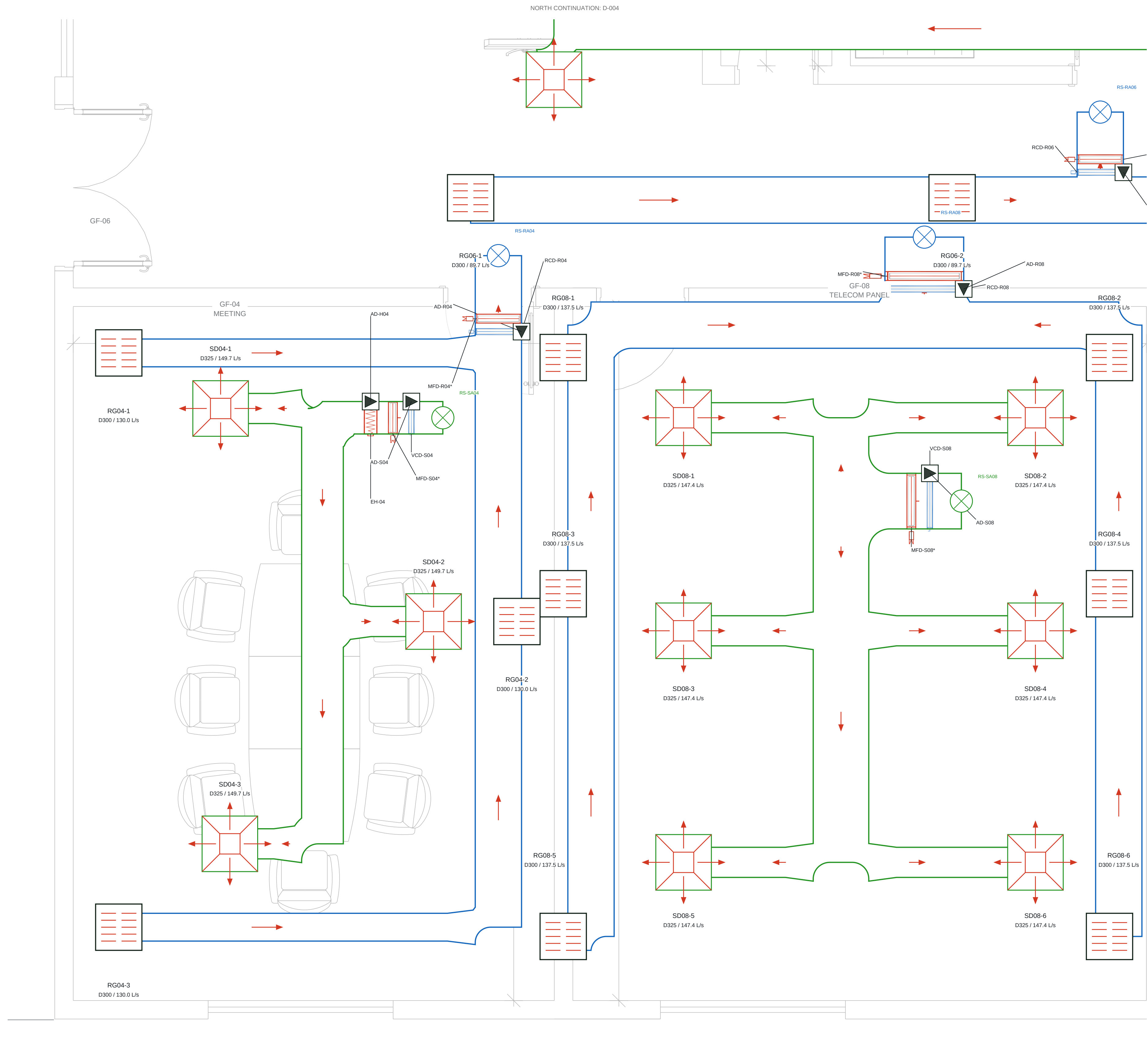
SCALE: 1:35 AT A1

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11

07 OCT 2026

SHEET 4 OF 22



ROOM BRANCH / TERMINAL DATA

GF-04 MEETING

SA-B04 350x400 mm | 449.20 L/s | 3.21 m/s

RA-B04 500x350 mm | 389.86 L/s | 2.23 m/s

SD: 3 x 149.73 L/s; neck D325; face 600x600

RG: 3 x 129.95 L/s; neck D300; face 500x500

Room header; branch size. Terminal runout; listed neck.

GF-08 TELECOM PANEL

SA-B08 600x400 mm | 884.60 L/s | 3.69 m/s

RA-808 850x400 mm | 825.26 L/s | 2.43 m/s

SD: 6 x 147.43 L/s; neck D325; face 600x600

RG: 6 x 137.54 L/s; neck D300; face 500x500

Room header: branch size. Terminal runout: listed neck.

GF-06 CORRIDOR

SA-B06 450x300 mm | 418.20 L/s | 3.10 m/s

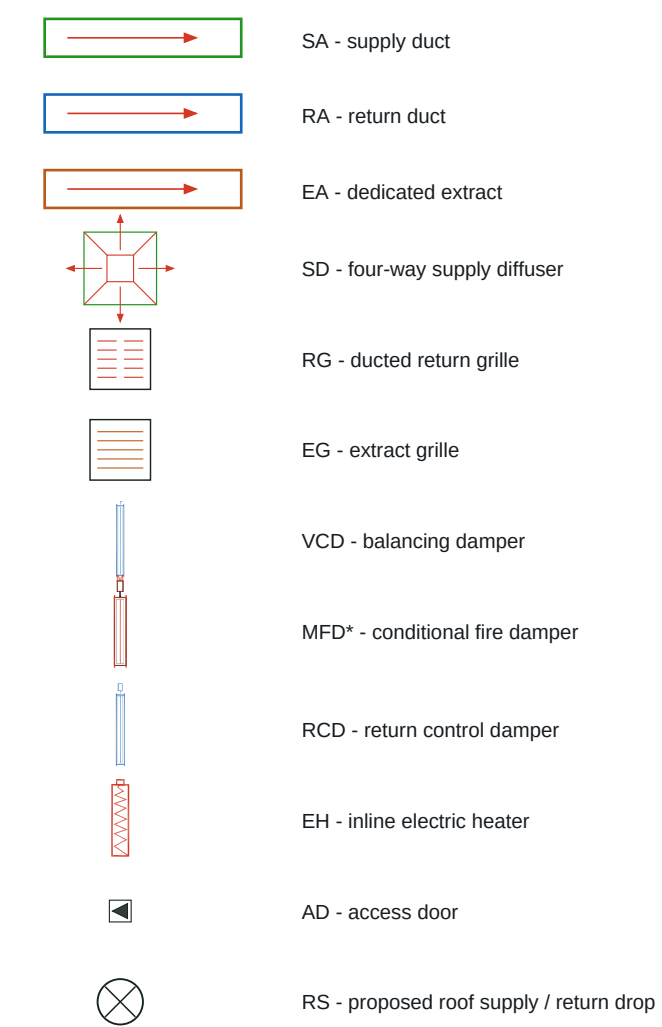
RA-B06 500x400 mm | 358.86 L/s | 1.79 m/s

SD: 3 x 139.40 L/s; neck D300; face 600x600

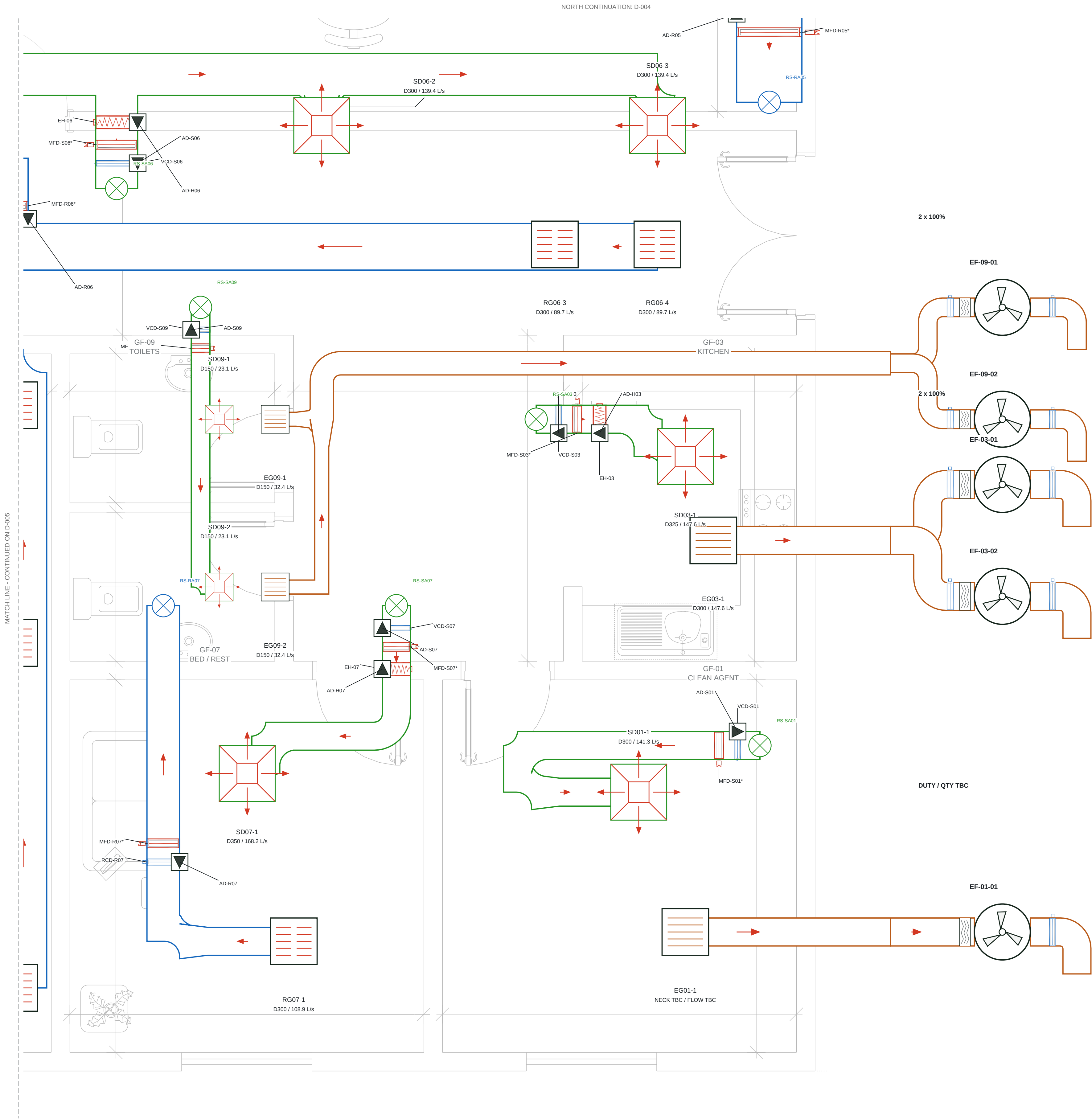
RG: 4 x 89.72 L/s; neck D300; face 500x500

Room header; branch size. Terminal runout; listed neck.

DUCT / SYMBOL LEGEND



Red arrows show air direction; green outlines are supply and blue outlines are return. Room drops refer to D-022. No indoor service crossovers. Same-service junctions are continuous open tees; headroom envelope: D-020.



ROOM BRANCH / TERMINAL DATA

GF-09 TOILETS

SA-B09 200x150 mm | 46.20 L/s | 1.54 m/s

EA-B09 250x200 mm | 64.89 L/s | 1.30 m/s

SD: 2 x 23.10 L/s; neck D150; face 300x300

Dedicated extract; no central return

Room header: branch size. Terminal runout: listed neck.

GF-07 BED / REST

SA-B07 300x250 mm | 168.20 L/s | 2.24 m/s

RA-B07 350x250 mm | 108.86 L/s | 1.24 m/s

SD: 1 x 168.20 L/s; neck D350; face 600x600

RG: 1 x 108.86 L/s; neck D300; face 500x500

Room header: branch size. Terminal runout: listed neck.

GF-03 KITCHEN

SA-B03 300x200 mm | 147.60 L/s | 2.46 m/s

EA-B03 300x200 mm | 147.60 L/s | 2.46 m/s

SD: 1 x 147.60 L/s; neck D325; face 600x600

Dedicated extract; no central return

Room header: branch size. Terminal runout: listed neck.

GF-01 CLEAN AGENT

SA-B01 300x200 mm | 141.30 L/s | 2.35 m/s

SD: 1 x 141.30 L/s; neck D300; face 600x600

EA / grille duty and pickup level: TBC

Room header: branch size. Terminal runout: listed neck.

EXTRACT ARRANGEMENTS

EF-09-01/02: toilets, 2 x 100%. EF-03-01/02: kitchen, 2 x 100%. EF-01-01: one CAG fan marker shown; final quantity, airflow and pickup elevation require confirmation.

Kitchen, toilet and CAG targets: +25 Pa to outdoors. Resting room: +50 Pa, with return air. Minimum ventilation: 10 ACH kitchen / toilet; 5 ACH CAG.

Toilet HAP supply is 46.2 L/s; minimum outdoor-air duty is not established by that mixed-air supply. Resolve the fresh-air duty before final HAP and pressure balance.

TAQA TRANSMISSION

N-19082-RO-012 | PUMPING STATION STANDARDIZATION

ADMINISTRATION HVAC

DESIGN BASIS: C-H-SS-001 Rev0 / HAP REFERENCE

DRAWING TITLE

SOUTH-EAST ROOMS - DUCT ENLARGEMENT

FOR COORDINATION | CHECKED / APPROVED: PENDING

DRAWING NUMBER

ADM-HVAC-D-006

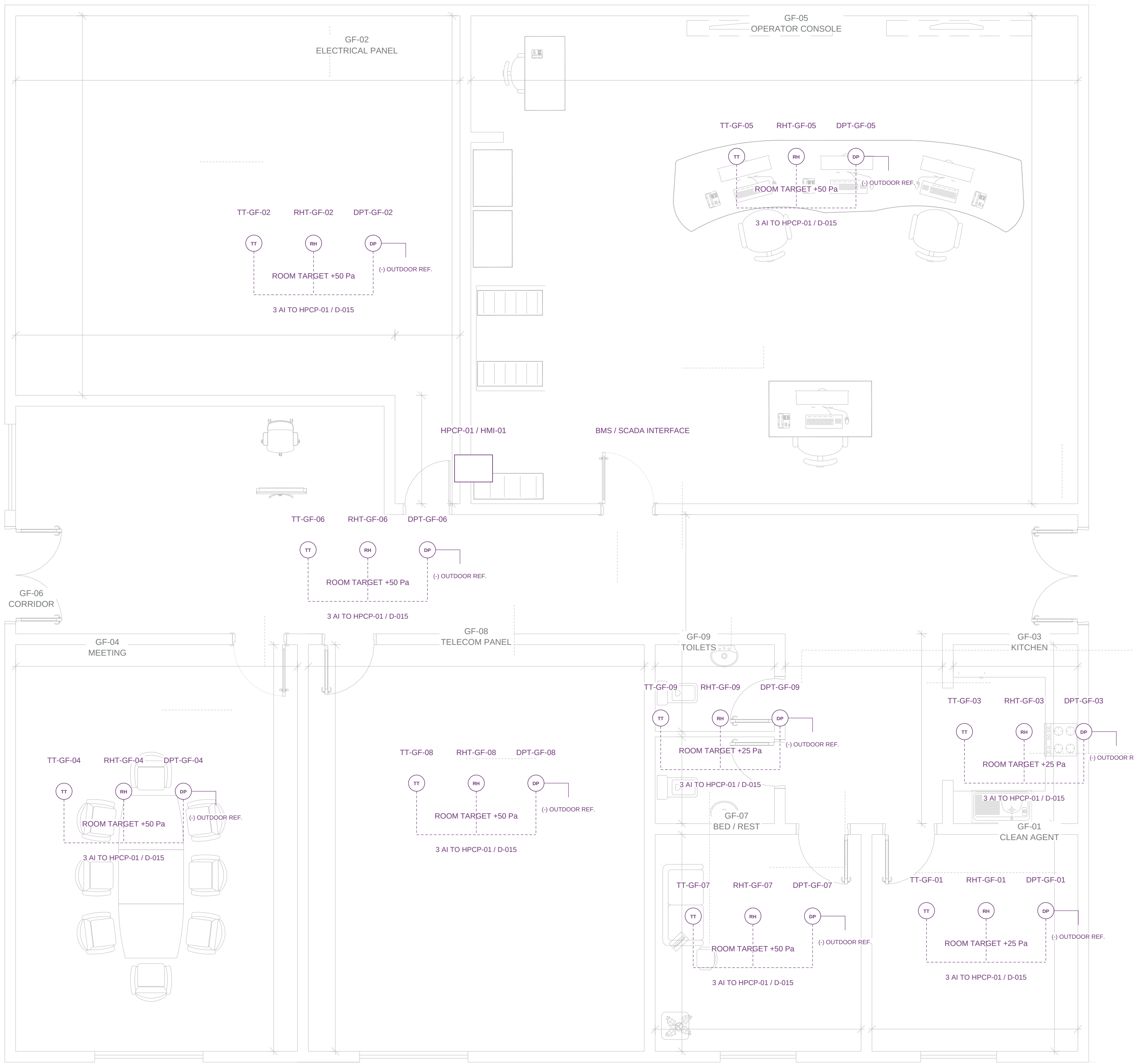
SCALE: 1:25 AT A1

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11

07 OCT 2026

SHEET 6 OF 22



ROOM FIELD INSTRUMENTS

GF	Temperature	Humidity	Pressure	Pa
02	TT-GF-02	RHT-GF-02	DPT-GF-02	50
05	TT-GF-05	RHT-GF-05	DPT-GF-05	50
04	TT-GF-04	RHT-GF-04	DPT-GF-04	50
08	TT-GF-08	RHT-GF-08	DPT-GF-08	50
06	TT-GF-06	RHT-GF-06	DPT-GF-06	50
07	TT-GF-07	RHT-GF-07	DPT-GF-07	50
09	TT-GF-09	RHT-GF-09	DPT-GF-09	25
03	TT-GF-03	RHT-GF-03	DPT-GF-03	25
01	TT-GF-01	RHT-GF-01	DPT-GF-01	25

MAIN DUCT / PLANT MONITORING

TT-SA-01 / TT-RA-01: main duct temperature

RHT-RA-01: main return humidity

FIT-SA-01 / FIT-RA-01: common airflow

Each PAU filter: PDT remote + PDI local

DPT-SF-A/B/C: PAU fan pressure proof

VS: PAU and ventilation-fan vibration

EM: feeders >7.5 kW, quantity to confirm

Signals report to HPCP-01 dedicated PLC and HMI-01. BMS / SCADA provides the specified monitoring, initial start and emergency-stop interfaces. Protocol, alarm limits, I/O module allocation and actuator fail positions remain pending.

DPT room references are outdoors, not the adjacent room. DPT-GF-06 is the corridor / building reference. Additional room DPTs and CAG humidity sensing are marked as added provisions in the register.

Room instrument locations are proposed. Plant/filter/fan instruments and heater AFS / TSHH protection: D-012. Final ranges, reference probes, mounting and routes require I&C coordination; operating sequence: S-004.

TAQA TRANSMISSION

N-19082-RO-012 | PUMPING STATION STANDARDIZATION

ADMINISTRATION HVAC

DESIGN BASIS: C-H-SS-001 Rev0 / HAP REFERENCE

DRAWING TITLE

ROOM INSTRUMENTATION AND CONTROL LOCATIONS

FOR COORDINATION | CHECKED / APPROVED: PENDING

DRAWING NUMBER

ADM-HVAC-D-007

SCALE: 1:50 AT A1

REVISION

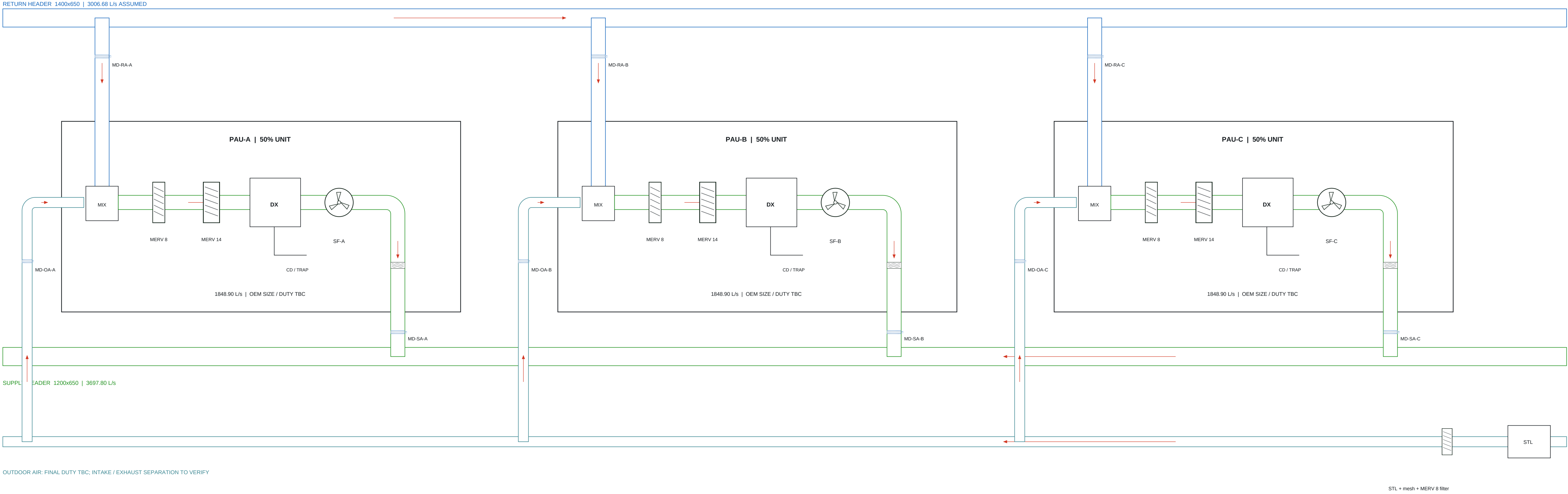
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07 OCT 2026

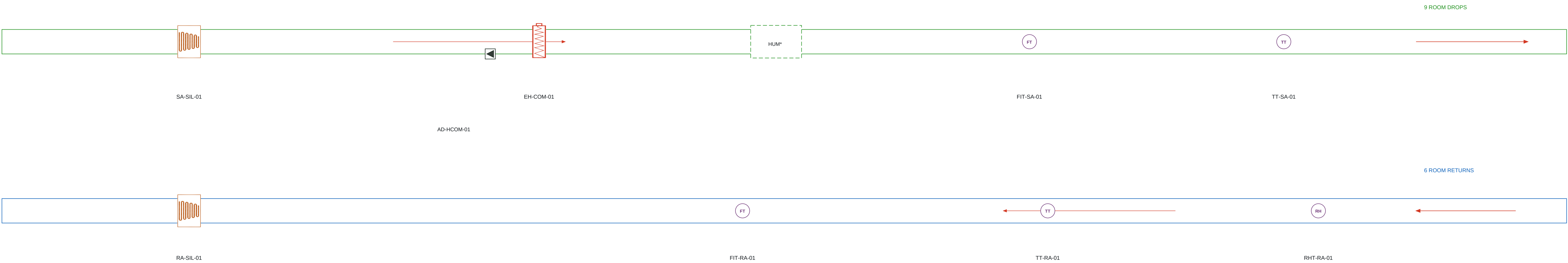
SHEET 7 OF 22

ADMINISTRATION BUILDING - PAU BANK AND PROPOSED ROOF COMMON DISTRIBUTION

A ROOF CONNECTION ARRANGEMENT - SCHEMATIC FOOTPRINTS



B ROOF COMMON PLANT / DISTRIBUTION OPTION - FUNCTIONAL NTS



Common heater / conditional humidifier move to a weather-protected roof plant enclosure in this option. Equipment suitability, safety, intake separation, maintenance space and outdoor insulation require OEM / project specification. Actual roof geometry and vertical development: TBC.

ROOF OPTION, NOT CONSTRUCTION ISSUE. Functional connections shown NTS; do not scale as roof routing. Each room drop and structural penetration is proposed, D-022. Common instrument signals remain on D-012 / S-001.

TAQA TRANSMISSION

N-19082-RO-012 | PUMPING STATION STANDARDIZATION

ADMINISTRATION HVAC

DESIGN BASIS: C-H-SS-001 Rev0 / HAP REFERENCE

DRAWING TITLE

PAU BANK AND PROPOSED ROOF COMMON DISTRIBUTION

FOR COORDINATION | CHECKED / APPROVED: PENDING

DRAWING NUMBER

ADM-HVAC-D-008

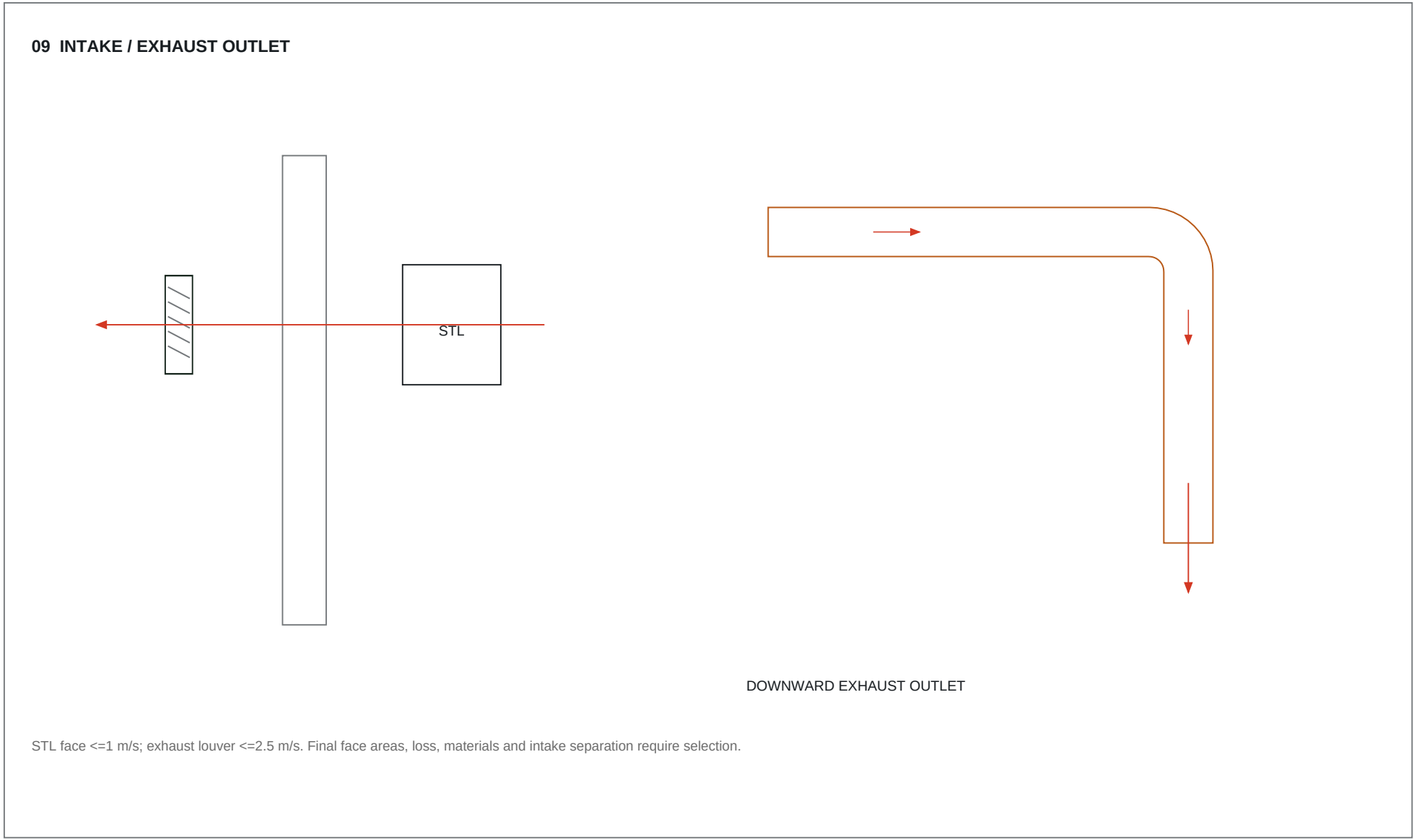
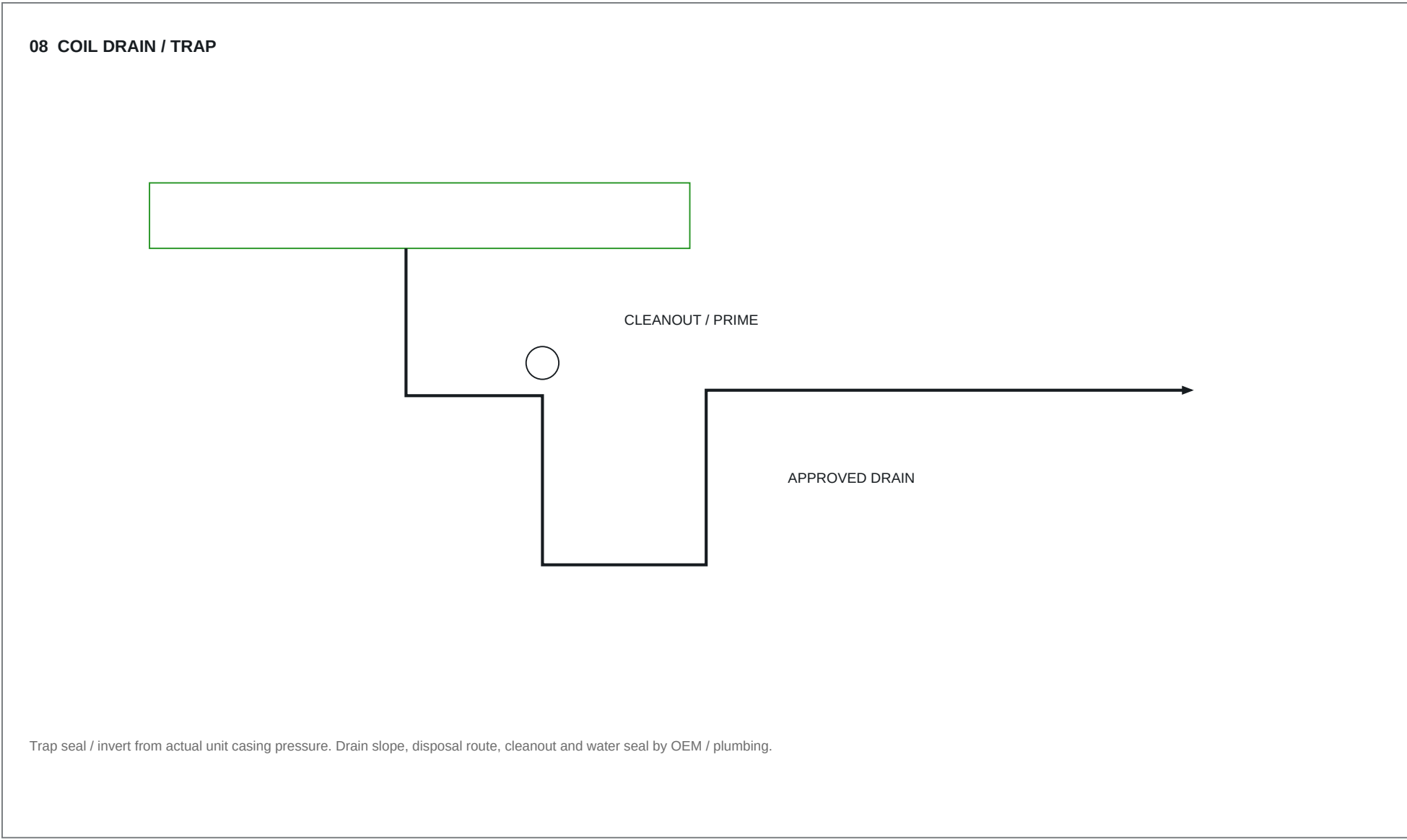
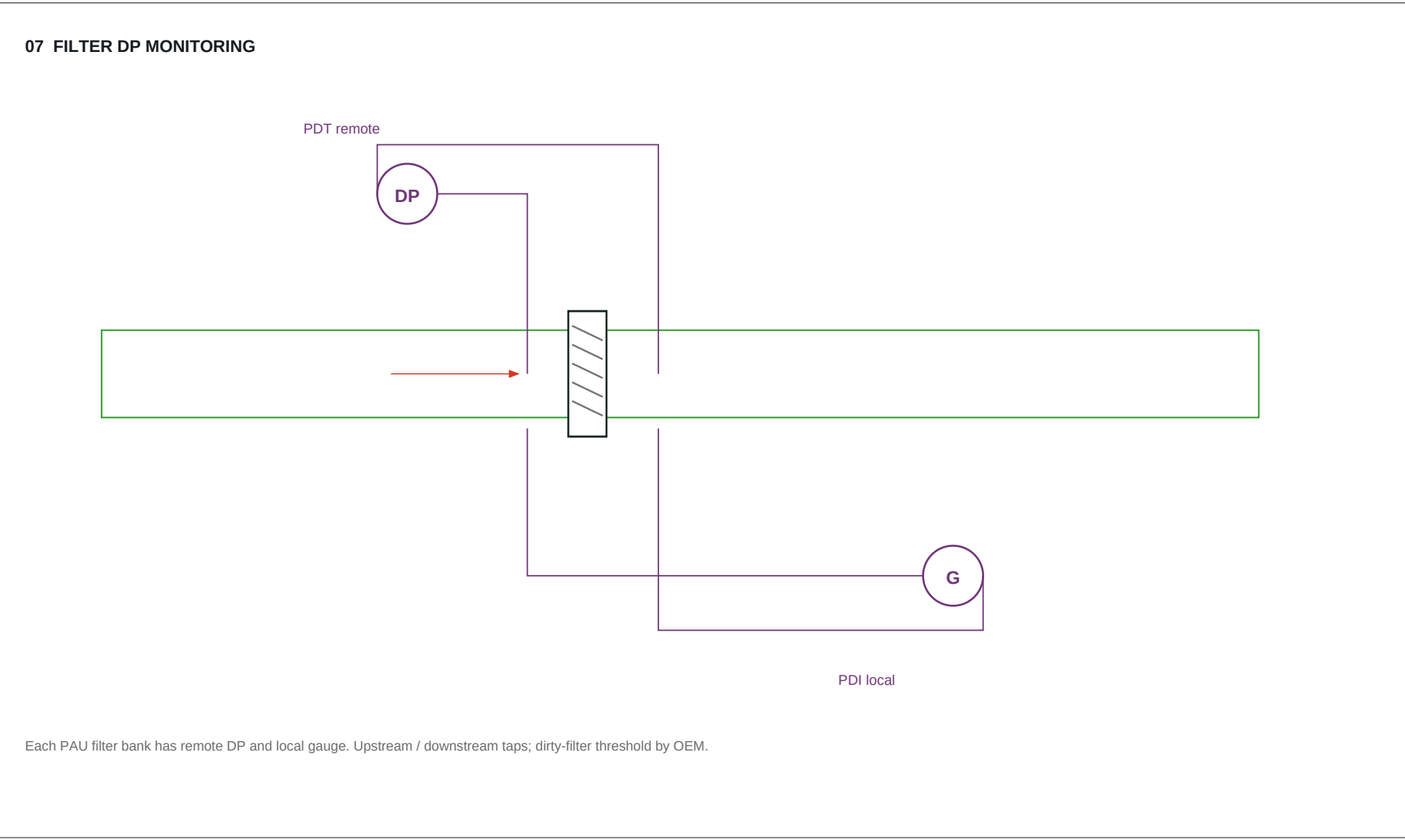
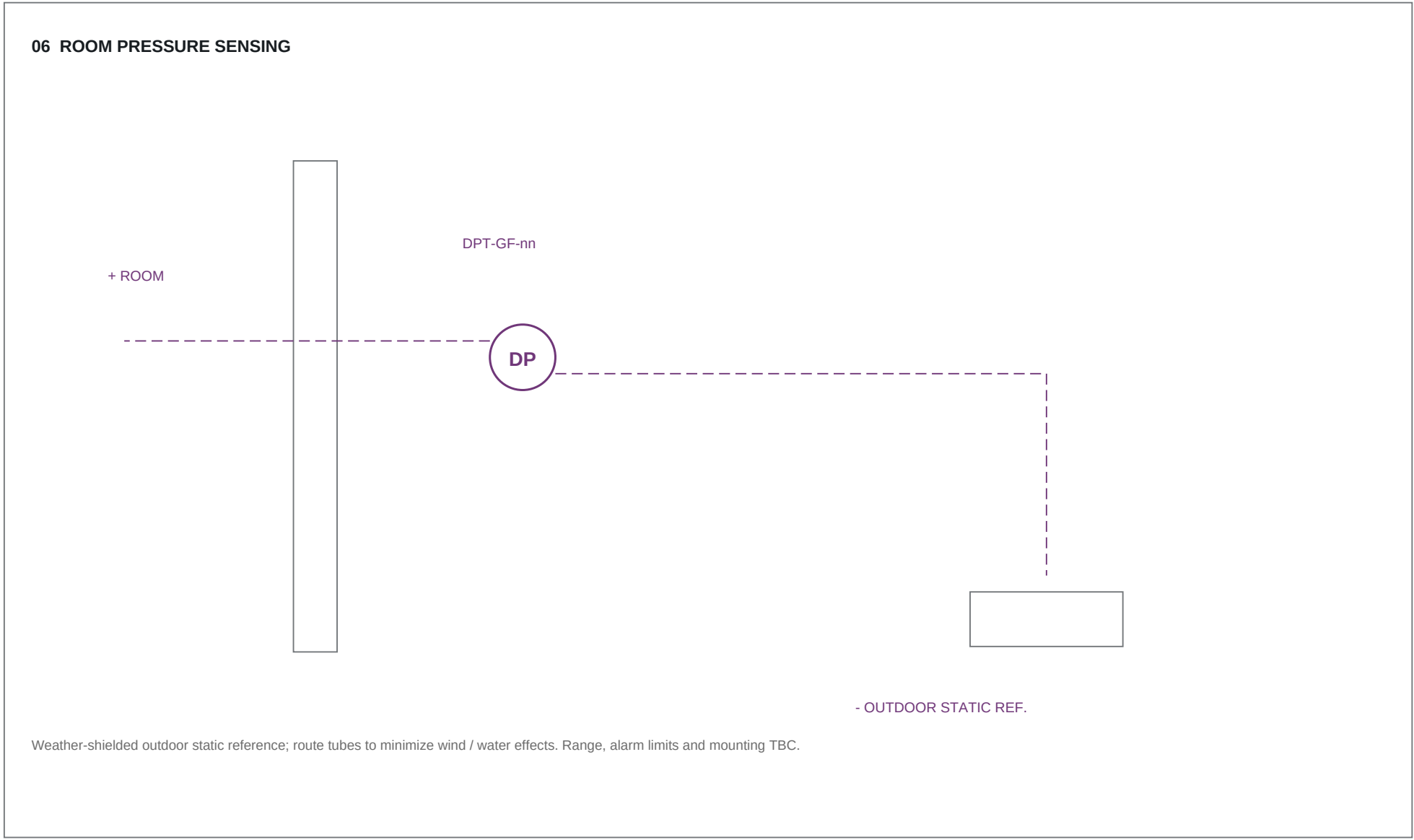
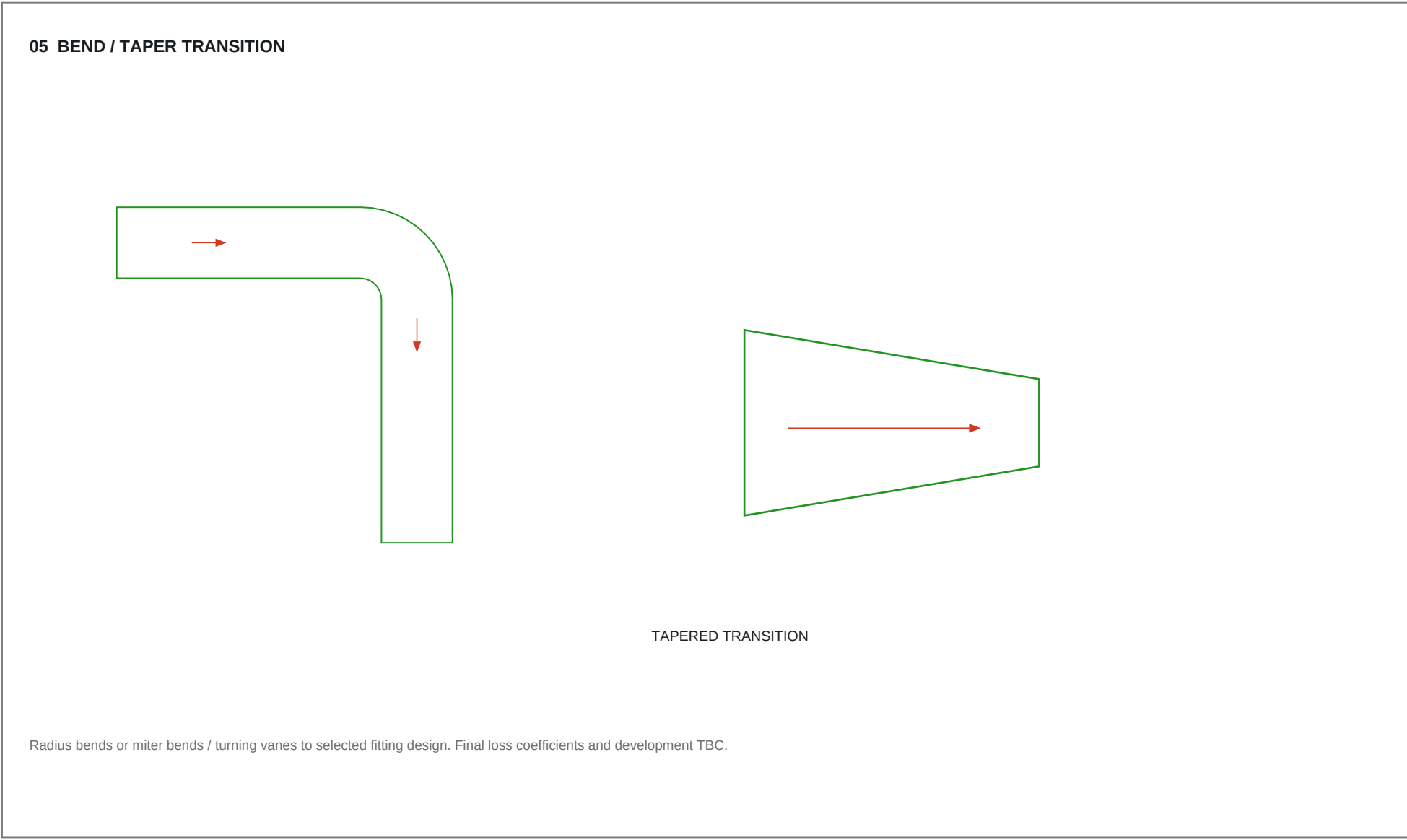
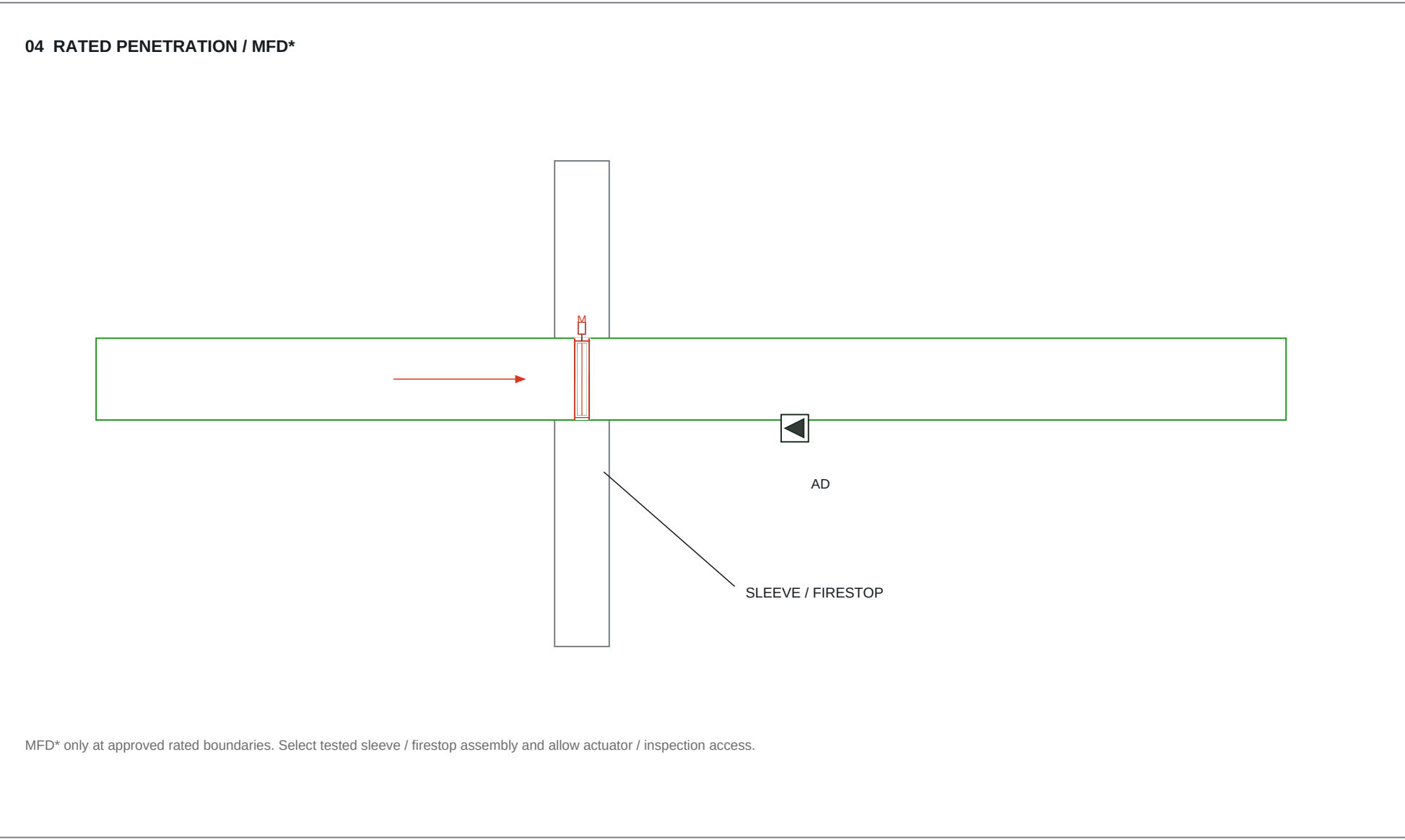
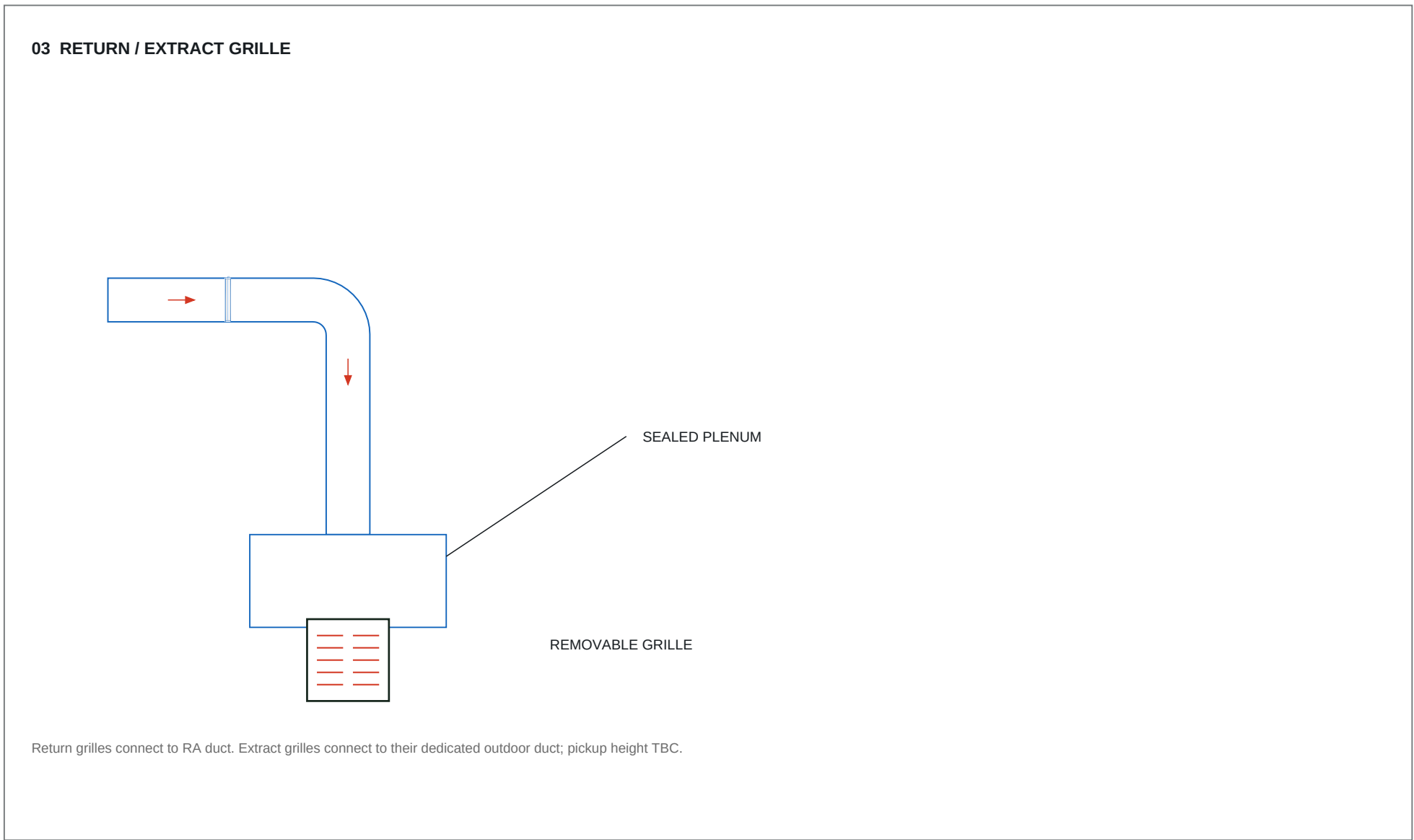
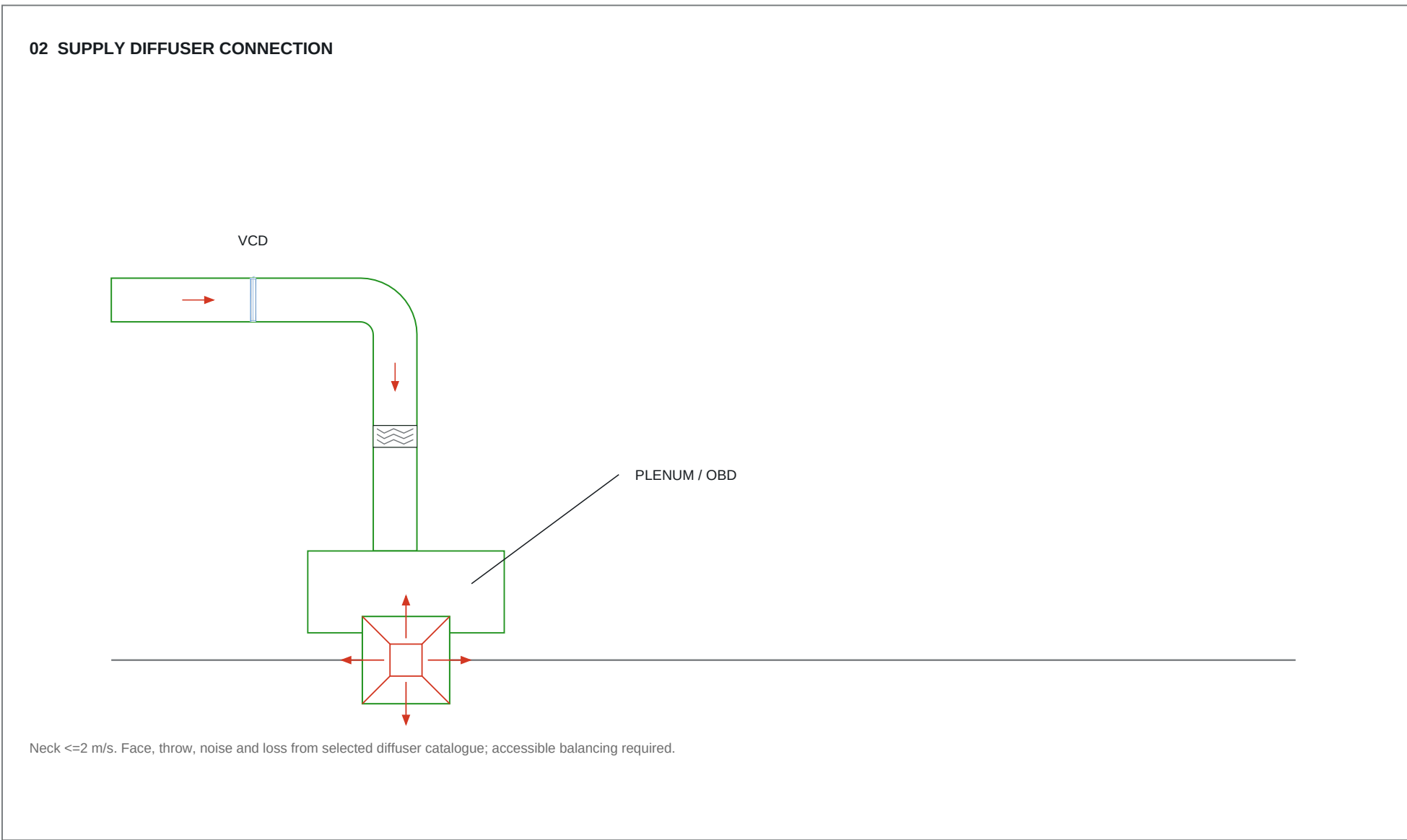
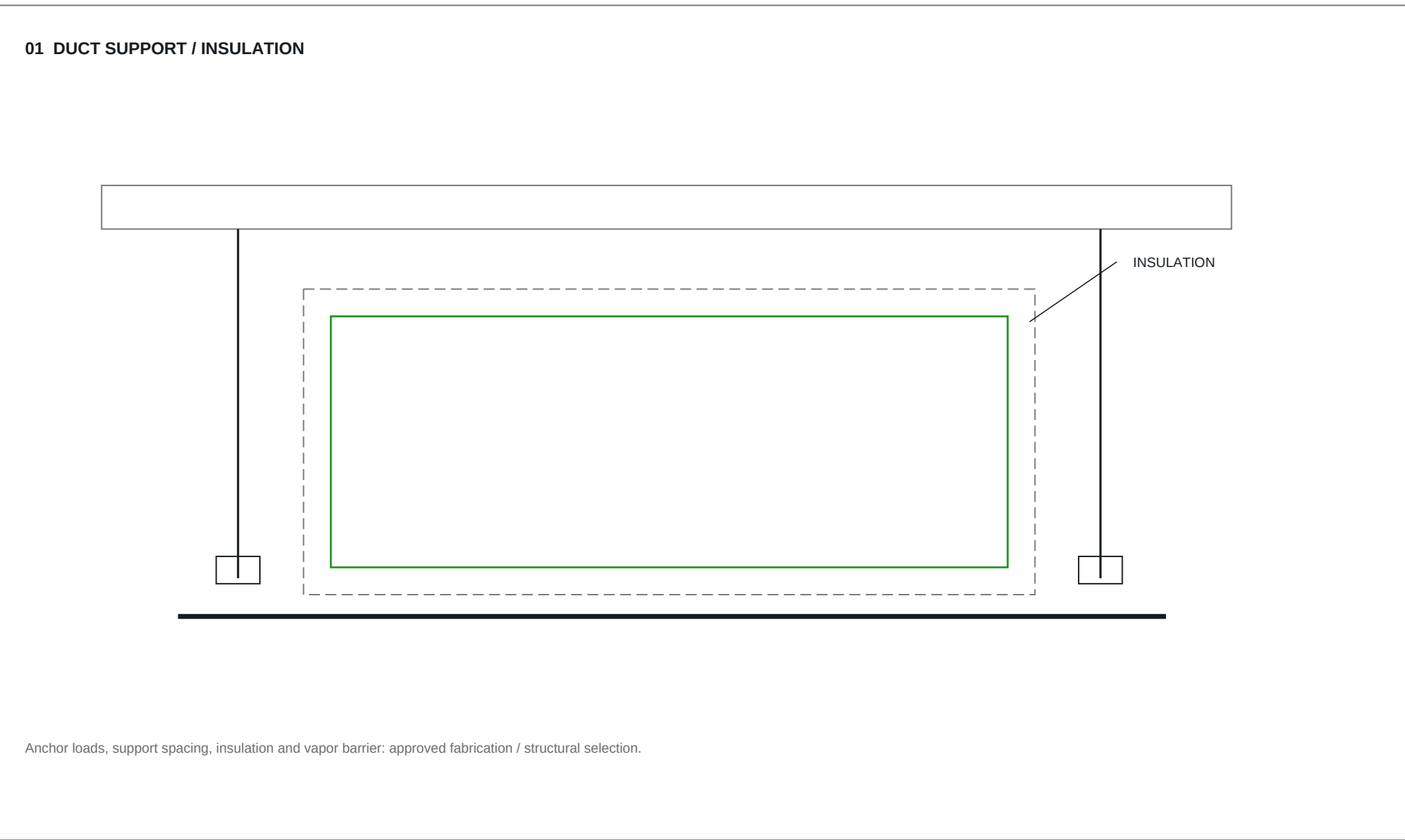
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11

07 OCT 2026

SHEET 8 OF 22



Details are NTS engineering concepts. Duct colours / symbols match the plans; body dimensions are not manufacturer selections. No room pressure, fire rating or equipment clearance is certified by this drawing.

TAQA TRANSMISSION

N-19082-RO-012 | PUMPING STATION STANDARDIZATION

ADMINISTRATION HVAC

DESIGN BASIS: C-H-SS-001 Rev0 / HAP REFERENCE

DRAWING TITLE

DUCT SYMBOLS AND INSTALLATION DETAILS

FOR COORDINATION | CHECKED / APPROVED: PENDING

DRAWING NUMBER

ADM-HVAC-D-009

SCALE: NTS AT A1

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11

07 OCT 2026

SHEET 9 OF 22

ADMINISTRATION BUILDING - DUCT AND TERMINAL SCHEDULES

MAIN AND ROOM BRANCH SECTIONS

Section	Size mm	L/s	m/s	Lh m*	Pa/m
SA-B02	500x350	664.80	3.80	1.20	0.399
RA-B02	650x400	605.46	2.33	0.95	0.130
SA-B04	350x400	449.20	3.21	1.30	0.330
RA-B04	500x350	389.86	2.23	0.95	0.151
SA-B08	600x400	884.60	3.69	1.30	0.313
RA-B08	850x400	825.26	2.43	0.95	0.125
SA-B05	600x350	777.70	3.70	1.90	0.349
RA-B05	700x400	718.36	2.57	1.00	0.150
SA-B06	450x300	418.20	3.10	1.23	0.324
RA-B06	500x400	358.86	1.79	0.95	0.093
SA-B09	200x150	46.20	1.54	0.60	0.231
SA-B07	300x250	168.20	2.24	2.80	0.254
RA-B07	350x250	108.86	1.24	2.80	0.081
SA-B03	300x200	147.60	2.46	1.20	0.351
SA-B01	300x200	141.30	2.35	2.60	0.325
EA-B09	250x200	64.89	1.30	6.70	0.123
EA-B03	300x200	147.60	2.46	1.90	0.351
EA-B01	TBC	TBC	TBC	TBC	TBC

* Lh is the source orthogonal horizontal route allowance. Final straight lengths, selected elbow development, risers / drops and fitting losses require coordination. Vertical lengths and BOD are unverified.

TERMINAL GROUPS

GF	Type	Qty	Each L/s	Neck mm	Face mm
02	SD	4	166.20	D350	600x600
02	RG	4	151.37	D300	500x500
05	SD	5	155.54	D325	600x600
05	RG	5	143.67	D300	500x500
04	SD	3	149.73	D325	600x600
04	RG	3	129.95	D300	500x500
08	SD	6	147.43	D325	600x600
08	RG	6	137.54	D300	500x500
06	SD	3	139.40	D300	600x600
06	RG	4	89.72	D300	500x500
07	SD	1	168.20	D350	600x600
07	RG	1	108.86	D300	500x500
09	SD	2	23.10	D150	300x300
09	EG	2	32.44	D150	300x300
03	SD	1	147.60	D325	600x600
03	EG	1	147.60	D300	500x500
01	SD	1	141.30	D300	600x600
01	EG	1	TBC	TBC	500x500

SYSTEM / DESIGN BASIS

Total supply: 3697.80 L/s. Assumed return: 3006.68 L/s.

Six return rooms: 59.34 L/s net outward allowance each, based on the accepted 0.01 m2 illustrative equivalent area.

Final room pressure, IAQ outdoor-air duty and revised HAP loads remain unverified. Current 420.70 L/s OA is below the 691.12 L/s assumed-balance makeup requirement.

All 116 duct sections, 37 fitting entries, 53 terminals, 176 components and 85 instruments are in the editable registers.

Rev11: routing / fitting topology revised, changed sizes and proposed terminal relocations are registered on D-021. Same-service overlaps and duplicated centerlines are rejected.

Sizes are clear internal dimensions. Catalogue free area, diffuser throw / noise, coil / filter / louver / silencer losses, fitting coefficients and OEM fan curves are pending. Confirmed fire stops the entire HVAC.

TAQA TRANSMISSION

N-19082-RO-012 | PUMPING STATION STANDARDIZATION

ADMINISTRATION HVAC

DESIGN BASIS: C-H-SS-001 Rev0 / HAP REFERENCE

DRAWING TITLE

DUCT AND TERMINAL SCHEDULES

FOR COORDINATION | CHECKED / APPROVED: PENDING

DRAWING NUMBER

ADM-HVAC-D-010

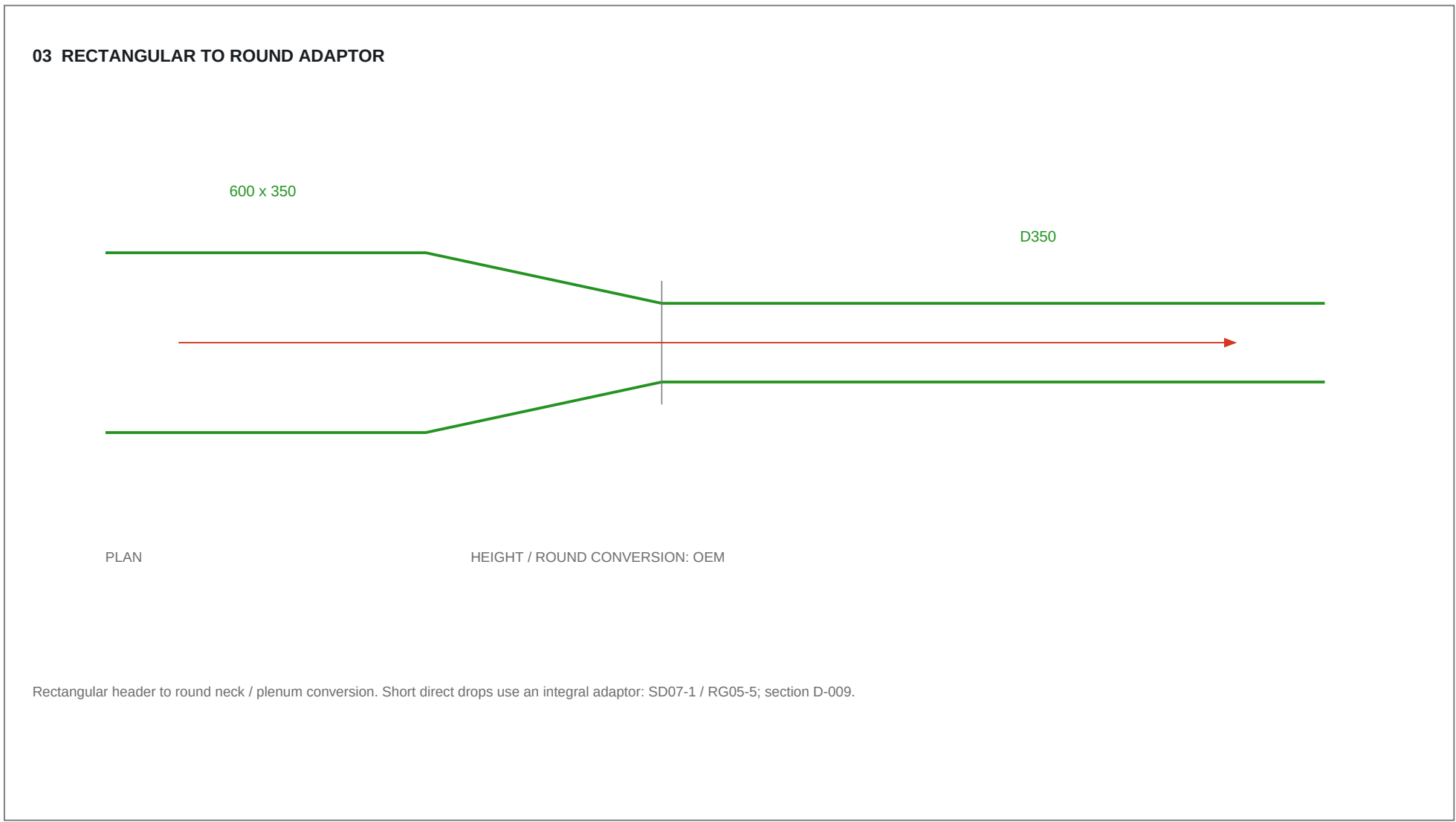
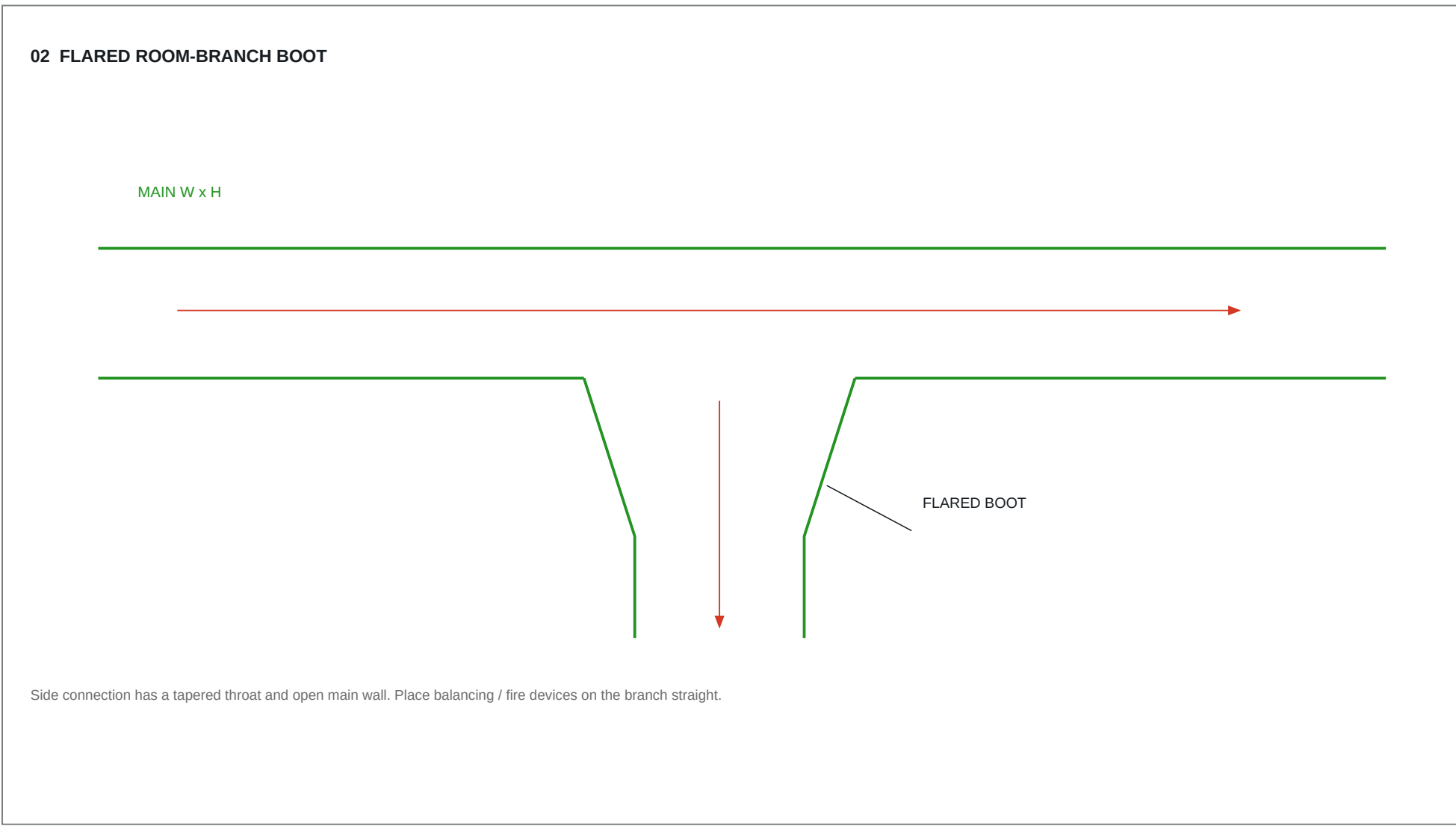
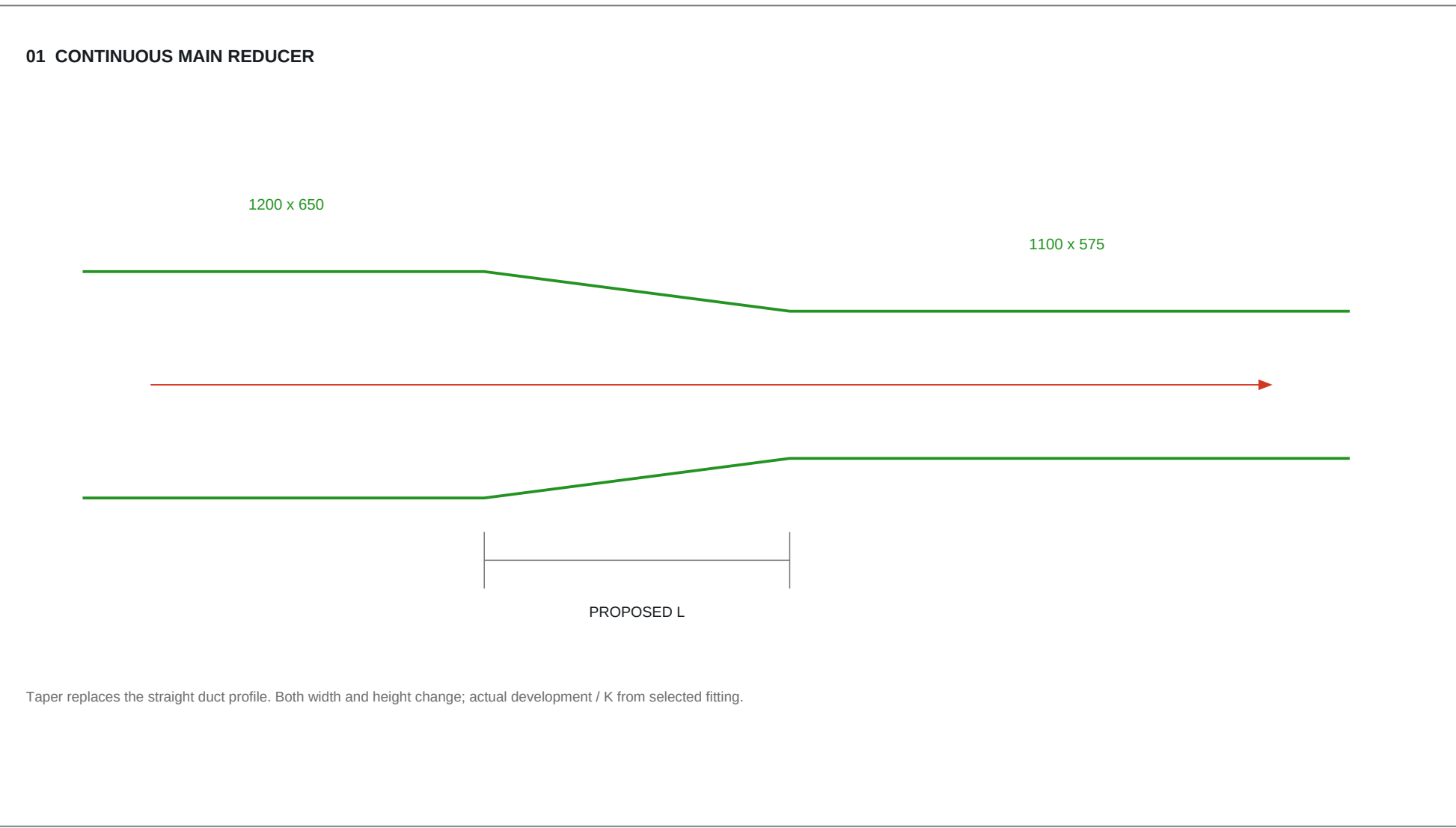
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REVISION

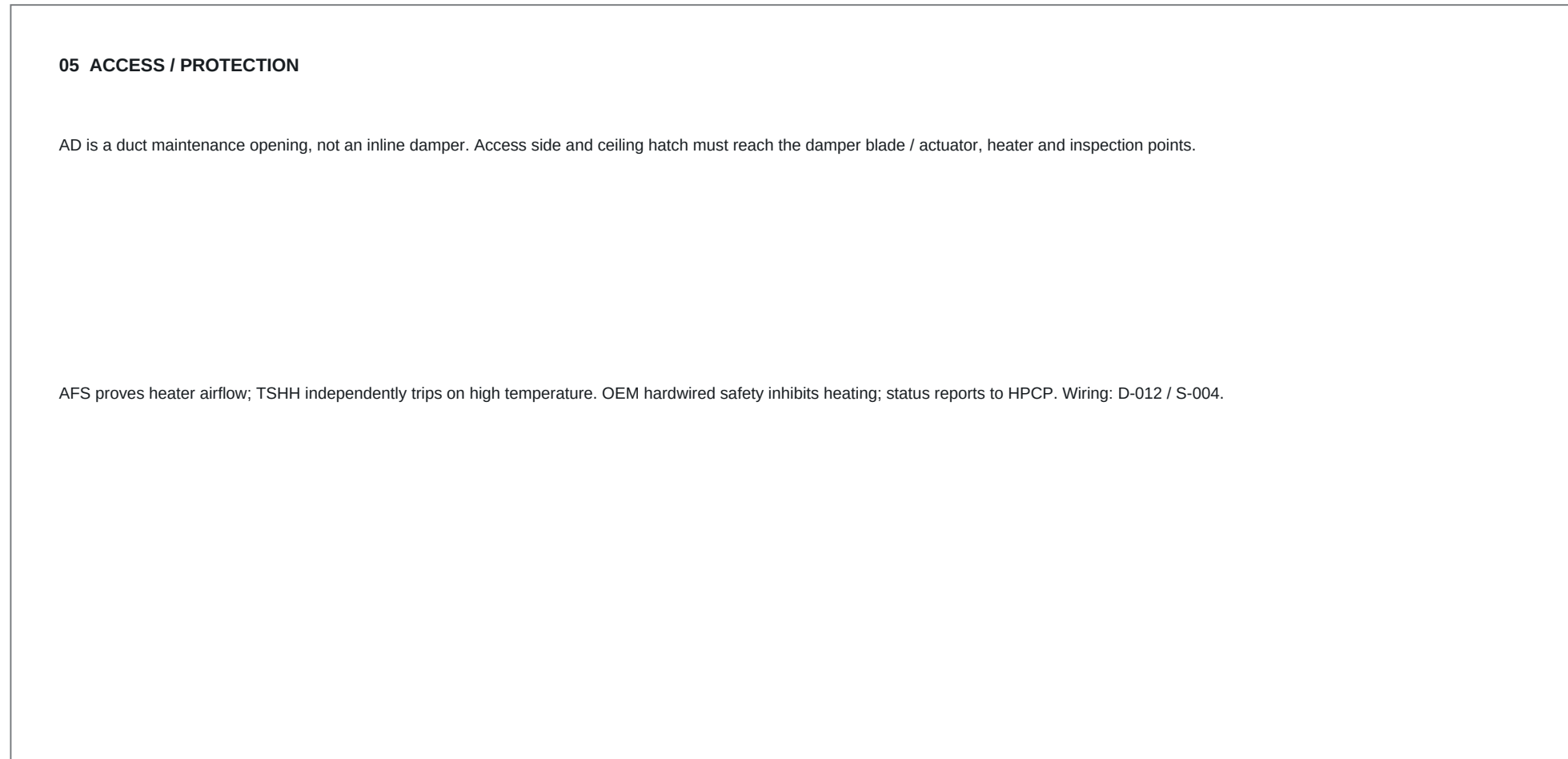
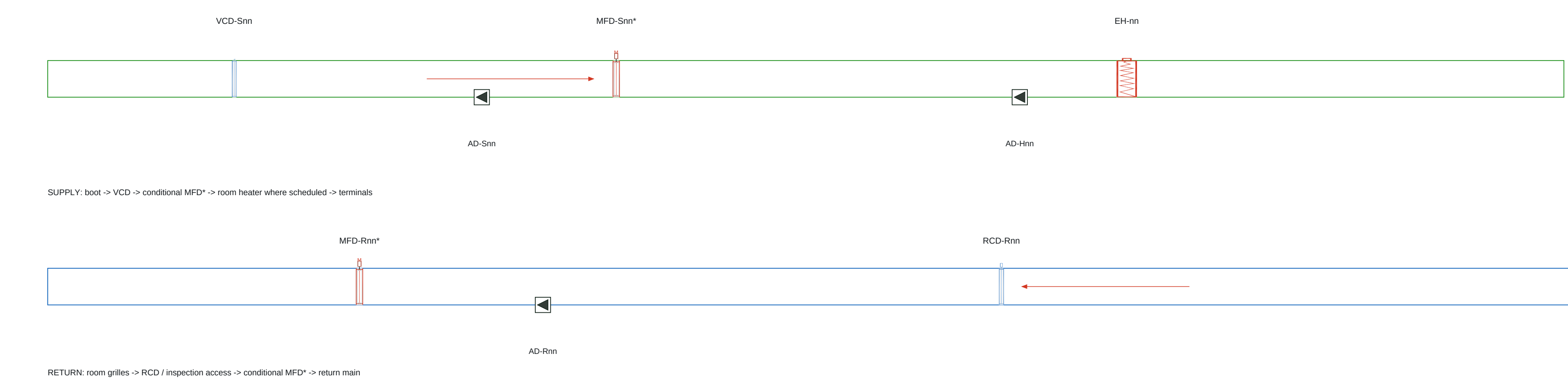
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07 OCT 2026

SHEET 10 OF 22



04 TYPICAL BRANCH STATIONS - FLOW DIRECTION / ACCESS



MAIN TAPER SCHEDULE - PROPOSED GEOMETRY

Fitting	Inlet mm	Outlet mm	L mm*
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ACCESS-DOOR TAGS / BRANCH MAP

GF	Supply access	Return access	Heater access
02	AD-S02	AD-R02	-
05	AD-S05	AD-R05	AD-H05
04	AD-S04	AD-R04	AD-H04
08	AD-S08	AD-R08	-
06	AD-S06	AD-R06	AD-H06
07	AD-S07	AD-R07	AD-H07
09	AD-S09	-	-
03	AD-S03	-	AD-H03
01	AD-S01	-	-
COM	AD-HCOM-01	-	Common heater

20 indoor AD locations plus AD-HCOM-01 roof plant access are registered. Door clear opening, sealing / insulation, access hatch and OEM working envelope remain to be selected.

* Taper lengths are explicit proposed coordination geometry, not approved fabrication lengths. The indoor register includes terminal adaptors. Main reducers and branch boots above are roof functional typicals; actual roof fittings and vertical drop details remain pending.

TAQA TRANSMISSION

N-19082-RO-012 | PUMPING STATION STANDARDIZATION

ADMINISTRATION HVAC

DESIGN BASIS: C-H-SS-001 Rev0 / HAP REFERENCE

DRAWING TITLE

TAPERED FITTINGS AND MAINTENANCE ACCESS

FOR COORDINATION | CHECKED / APPROVED: PENDING

DRAWING NUMBER

ADM-HVAC-D-011

SCALE: NTS AT A1

REVISION

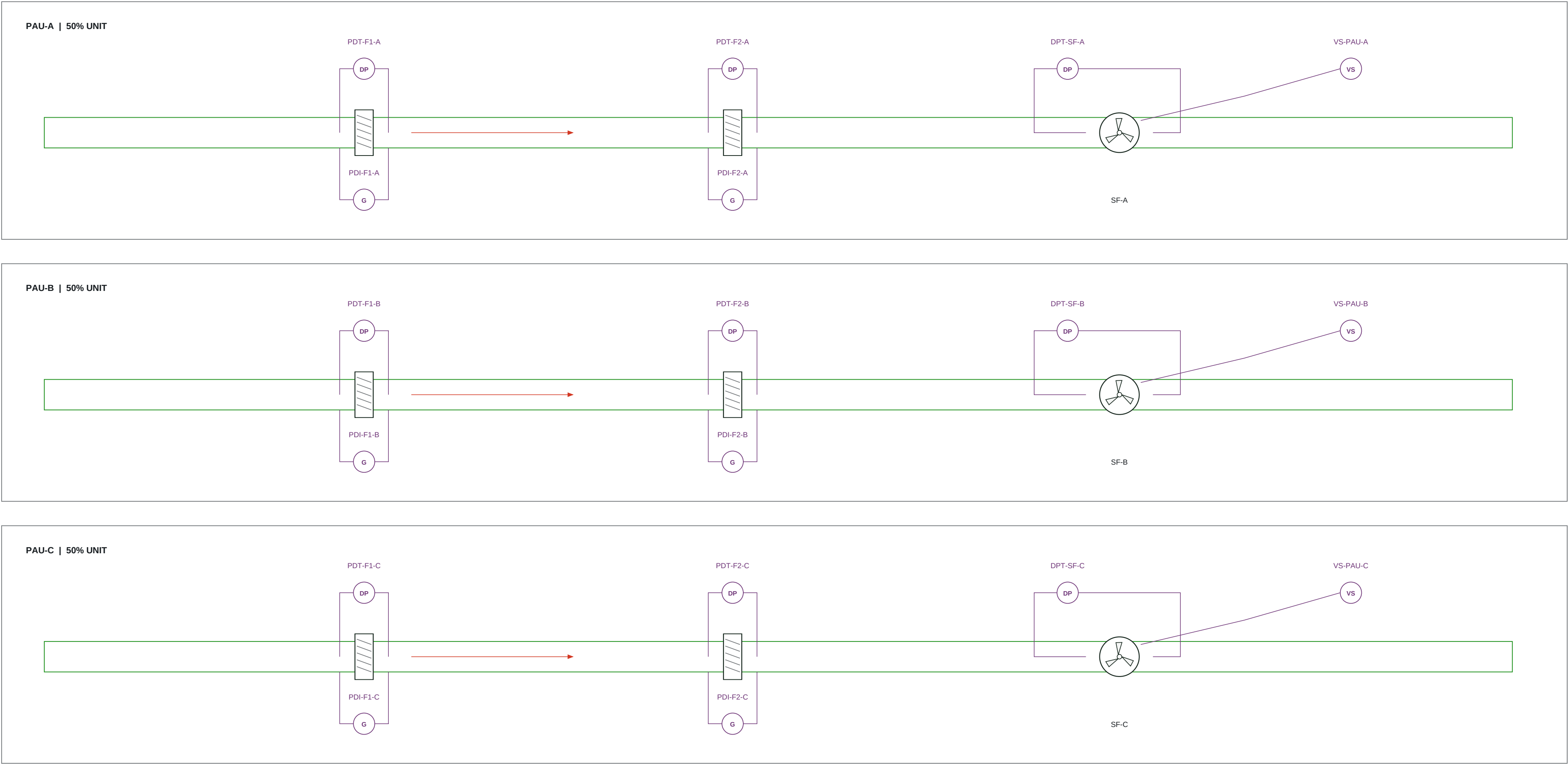
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07 OCT 2026

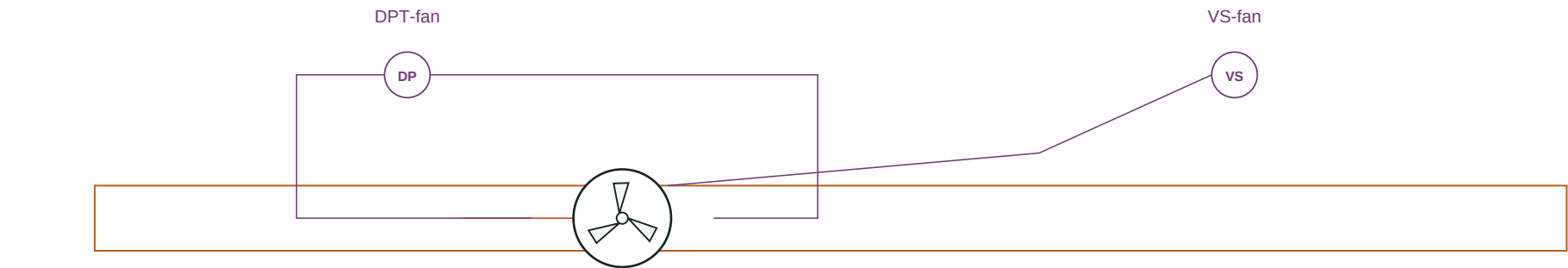
SHEET 11 OF 22

ADMINISTRATION BUILDING - PLANT INSTRUMENTATION AND HEATER PROTECTION

A EACH PAU: FILTER DIFFERENTIAL PRESSURE / FAN PROOF / VIBRATION



C EXTRACT / BASEMENT FAN PROOF

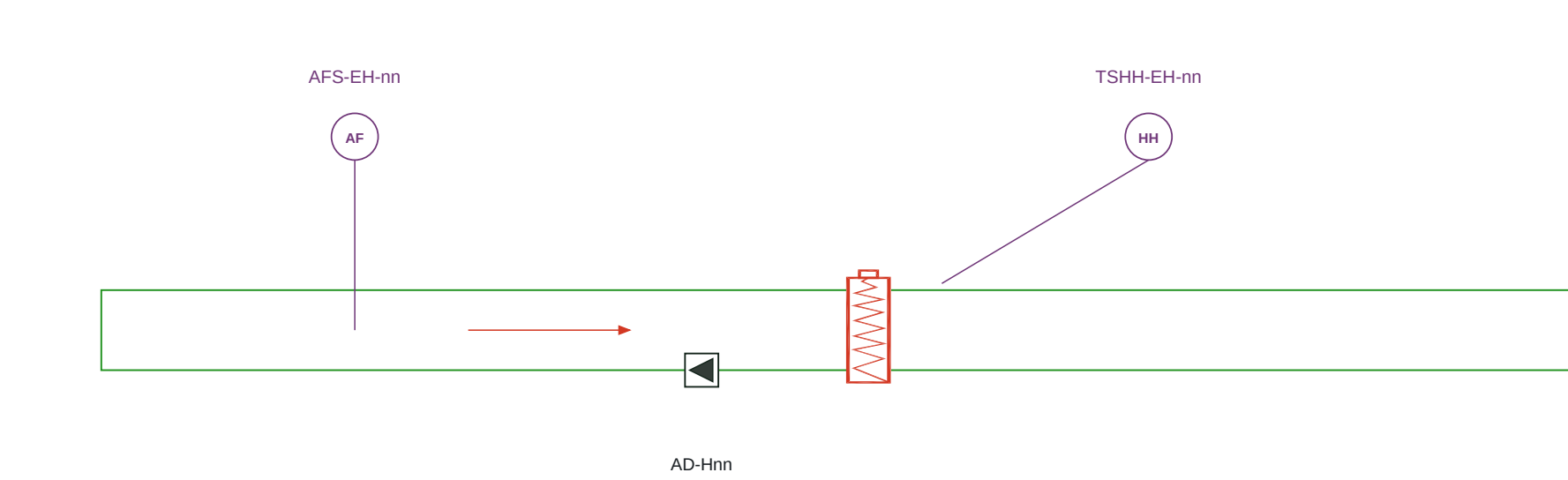


Tag suffixes: EF-03-01/02; EF-09-01/02; EF-01-01. Each has DPT- and VS- tags. Basement: SF-B1-01/02 and EF-B1-01/02 use the same convention.

Fan start / run / fault / ready / local-auto / E-stop and damper proofs are listed in I/O. CAG duty / quantity and basement duties remain pending.

Instrument glyphs are functional symbols, not equipment dimensions. AFS additions are explicit proposed OEM protection / monitoring provisions. Full tags: Excel / CSV registers. Room TT / RHT / DPT locations: D-007.

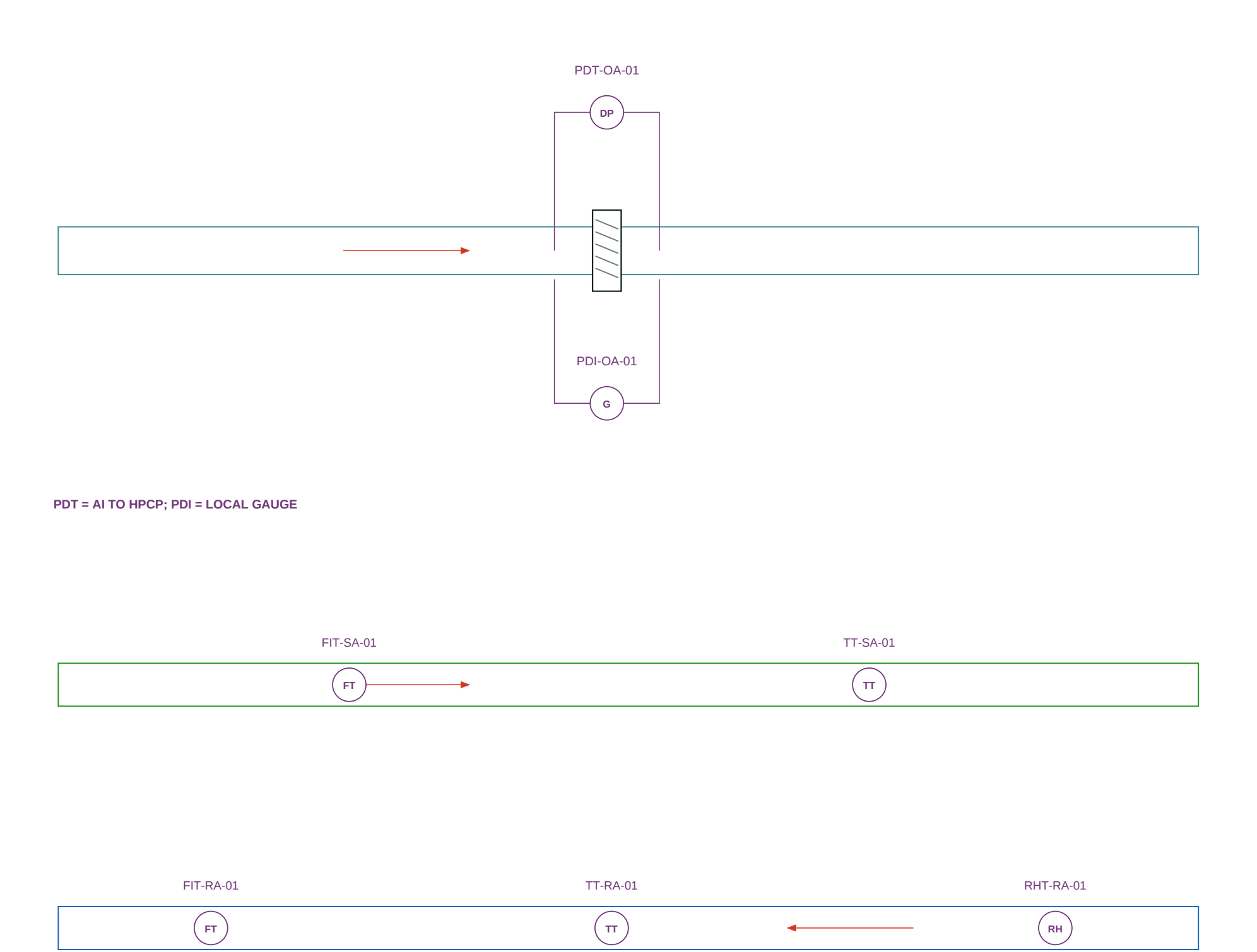
D HEATER AIRFLOW / HIGH-LIMIT SAFETY



nn = 04, 05, 06, 07, 03, COM-01. Six AFS DI points are retained from Rev07. Each heater retains independent TSHH; loss of proof / high limit switches heating off.

AFS proof technology, setpoint and OEM hardwired wiring to be selected. PLC demand cannot override local safety.

B OUTDOOR-AIR FILTER / COMMON DUCTS



Filter taps measure pressure across each filter, not between rooms. Fan DP proves the operating fan path; VS monitors fan / unit vibration. Alarm thresholds and OEM signal type TBC.

E PRELIMINARY HPCP I/O

Type	Points	Spare >=20%	Channels
AI	65	13	78
AO	22	5	27
DI	141	29	170
DO	44	9	53

COMM: 3 interfaces; separate from wired channel totals. Signals report to dedicated HPCP-01 PLC / HMI-01 and BMS / SCADA. Final protocol and allocation TBC.

CONFIRMED FIRE: stop every HVAC fan and heater. No automatic fire extraction / post-fire purge. Reset and damper fire positions follow the approved cause-and-effect.

TAQA TRANSMISSION

N-19082-RO-012 | PUMPING STATION STANDARDIZATION

ADMINISTRATION HVAC

DESIGN BASIS: C-H-SS-001 Rev0 / HAP REFERENCE

DRAWING TITLE

PLANT INSTRUMENTATION AND HEATER PROTECTION

FOR COORDINATION | CHECKED / APPROVED: PENDING

DRAWING NUMBER

ADM-HVAC-D-012

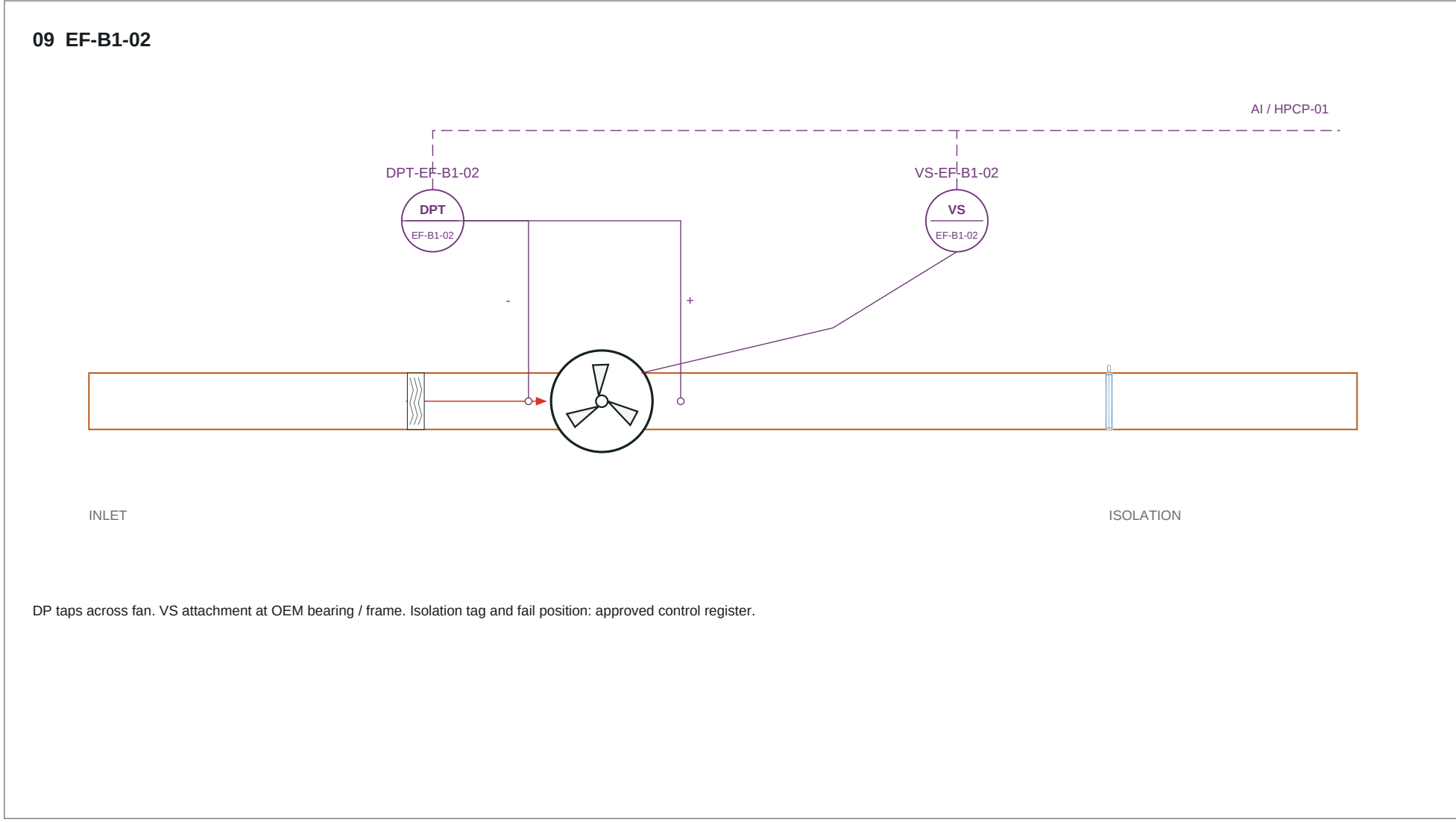
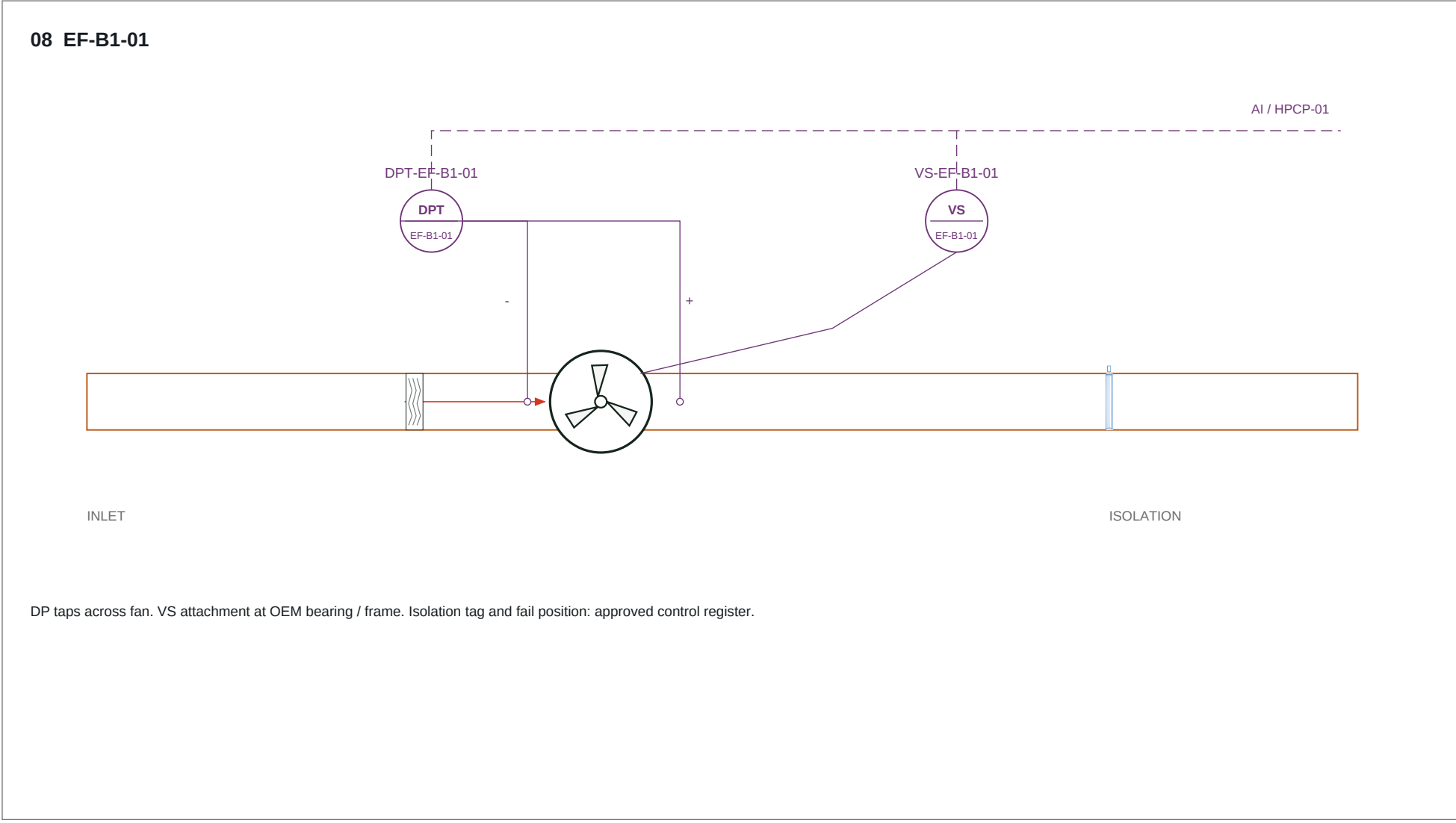
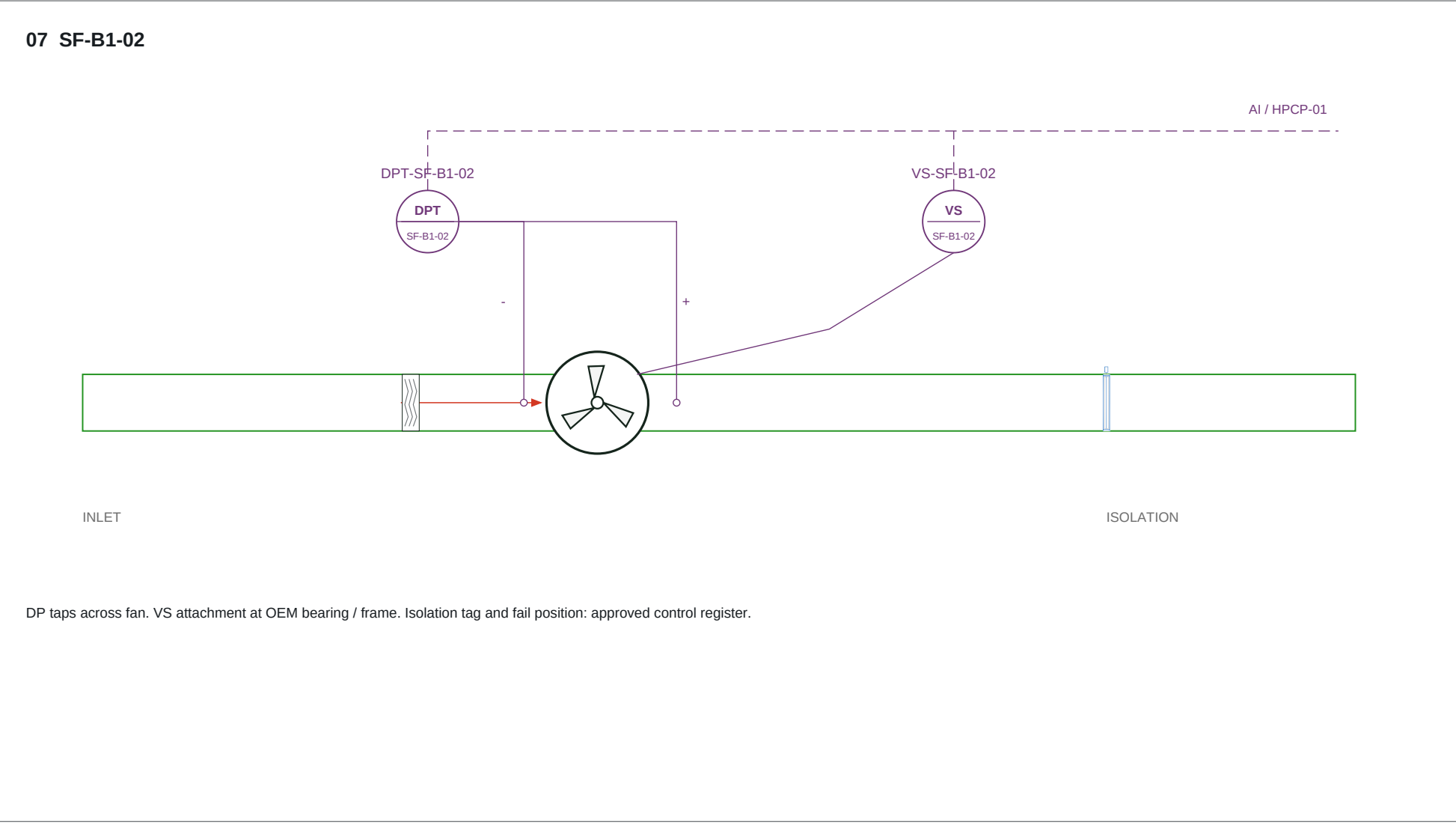
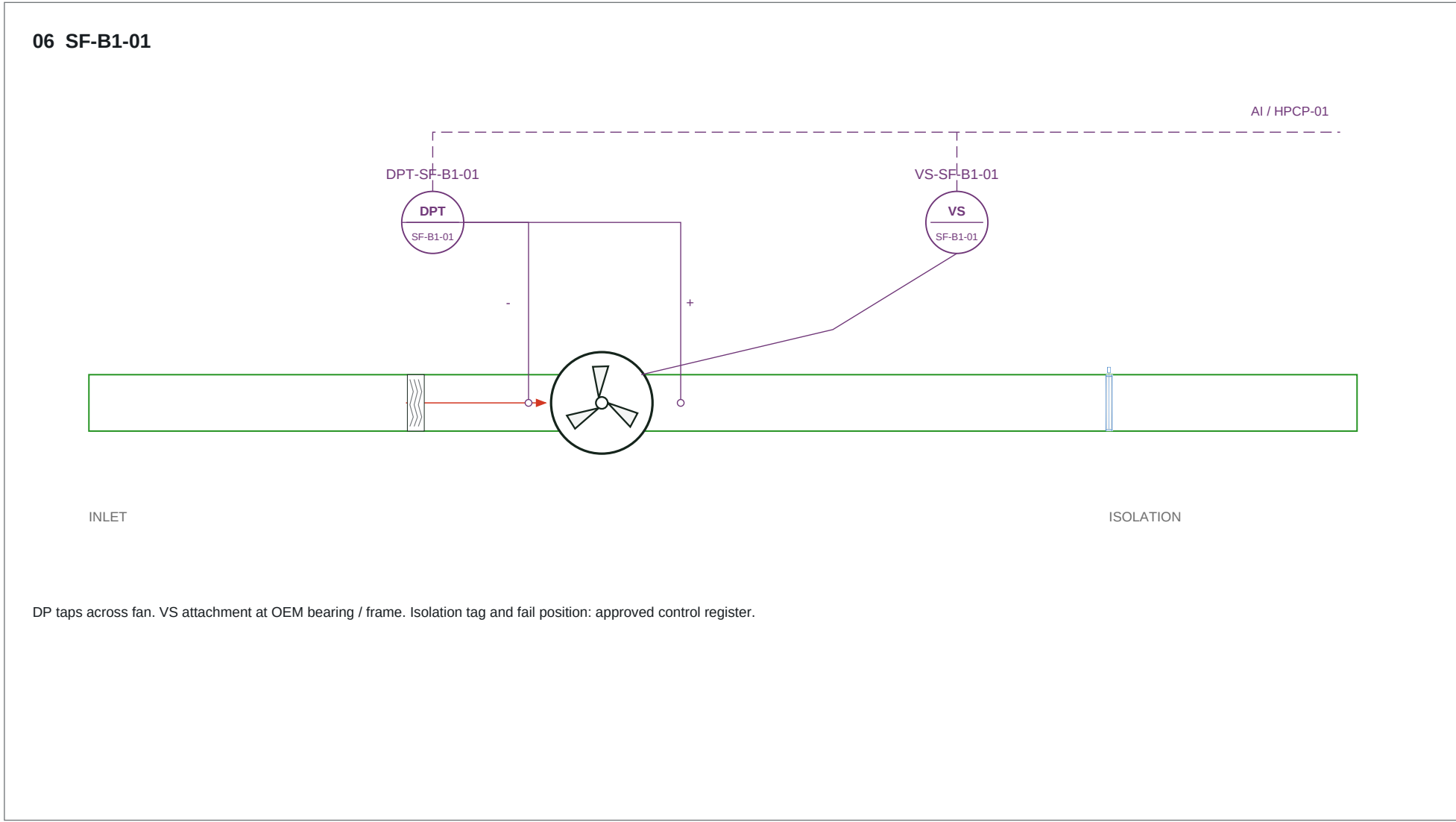
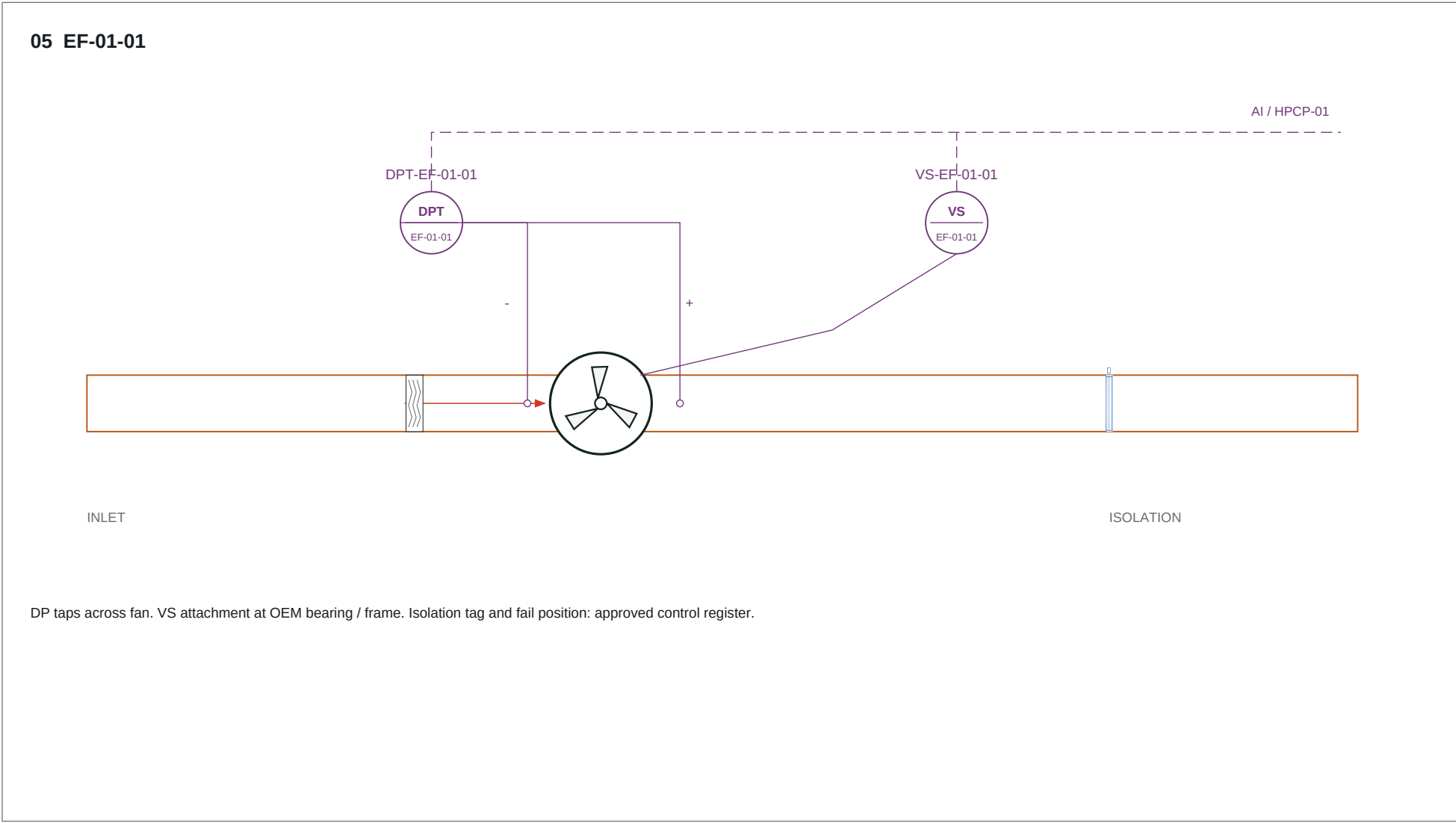
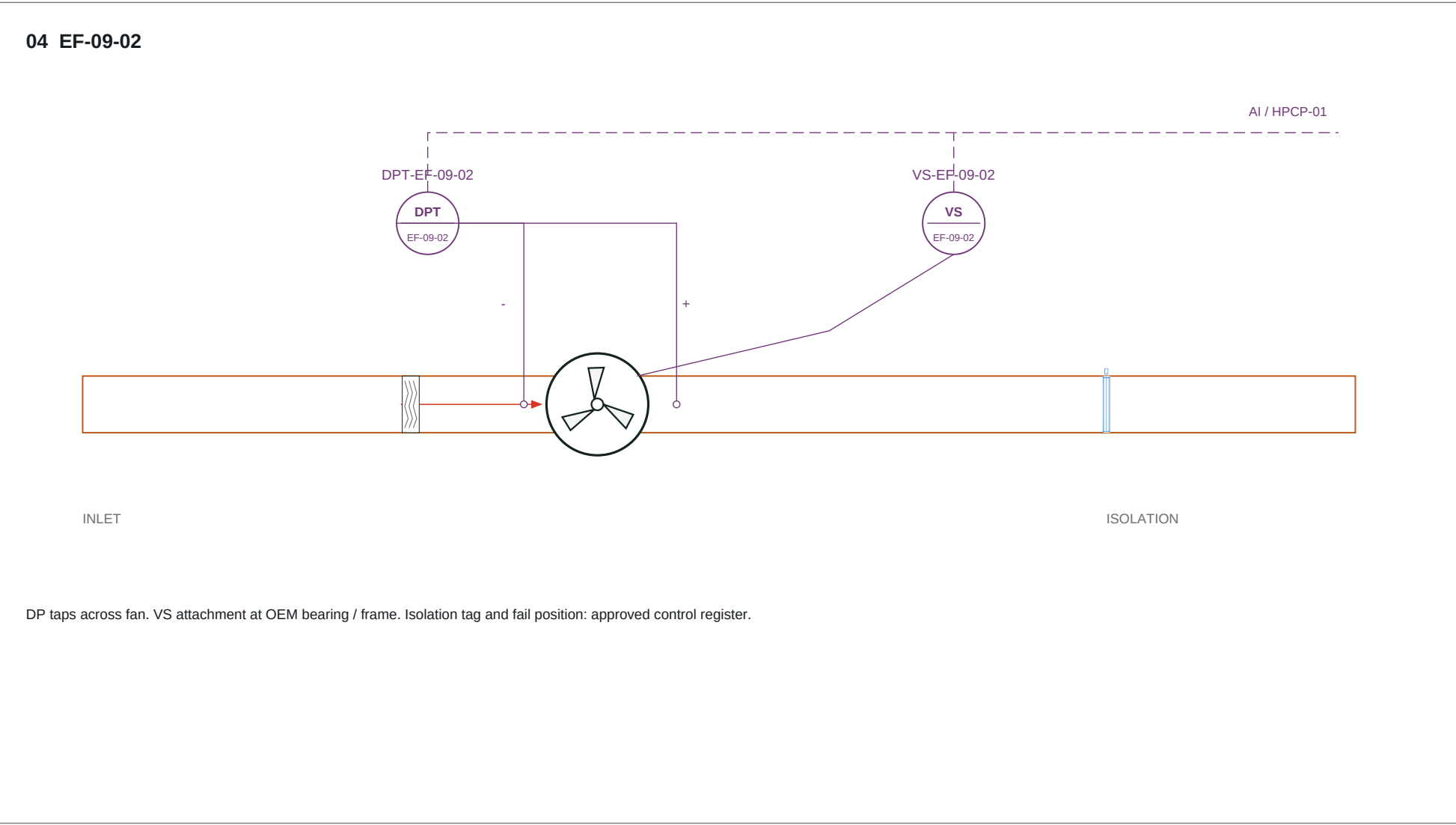
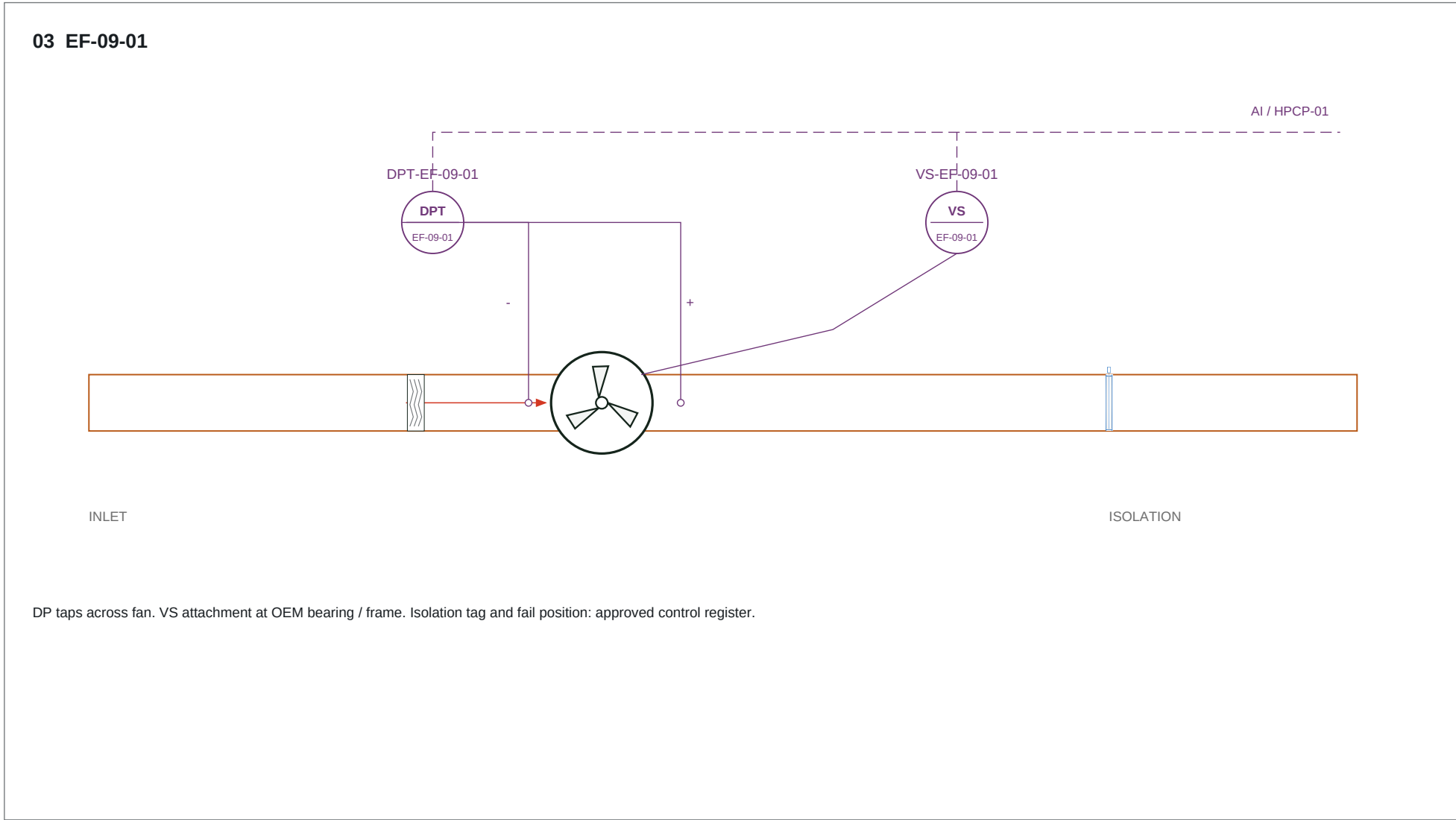
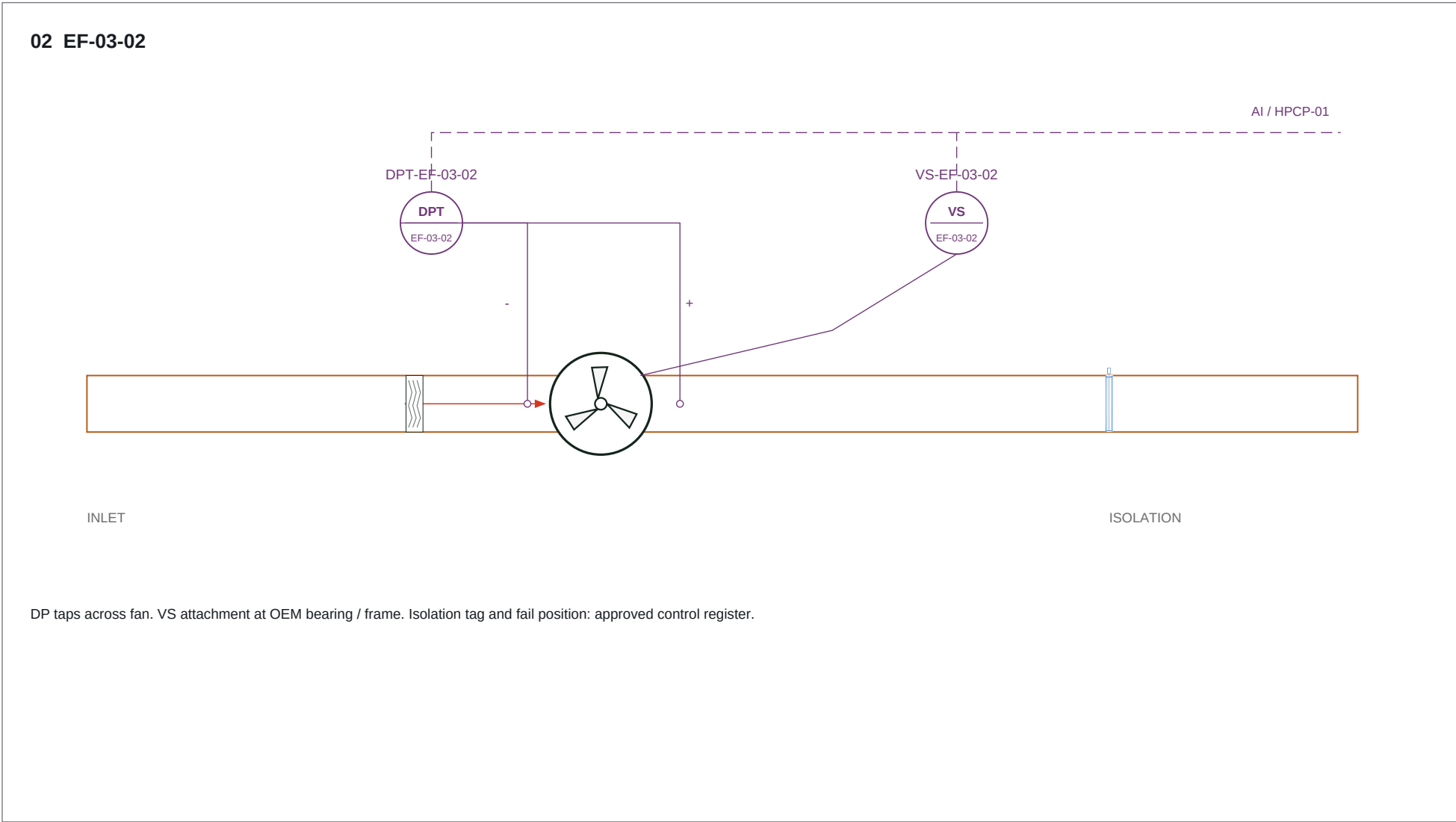
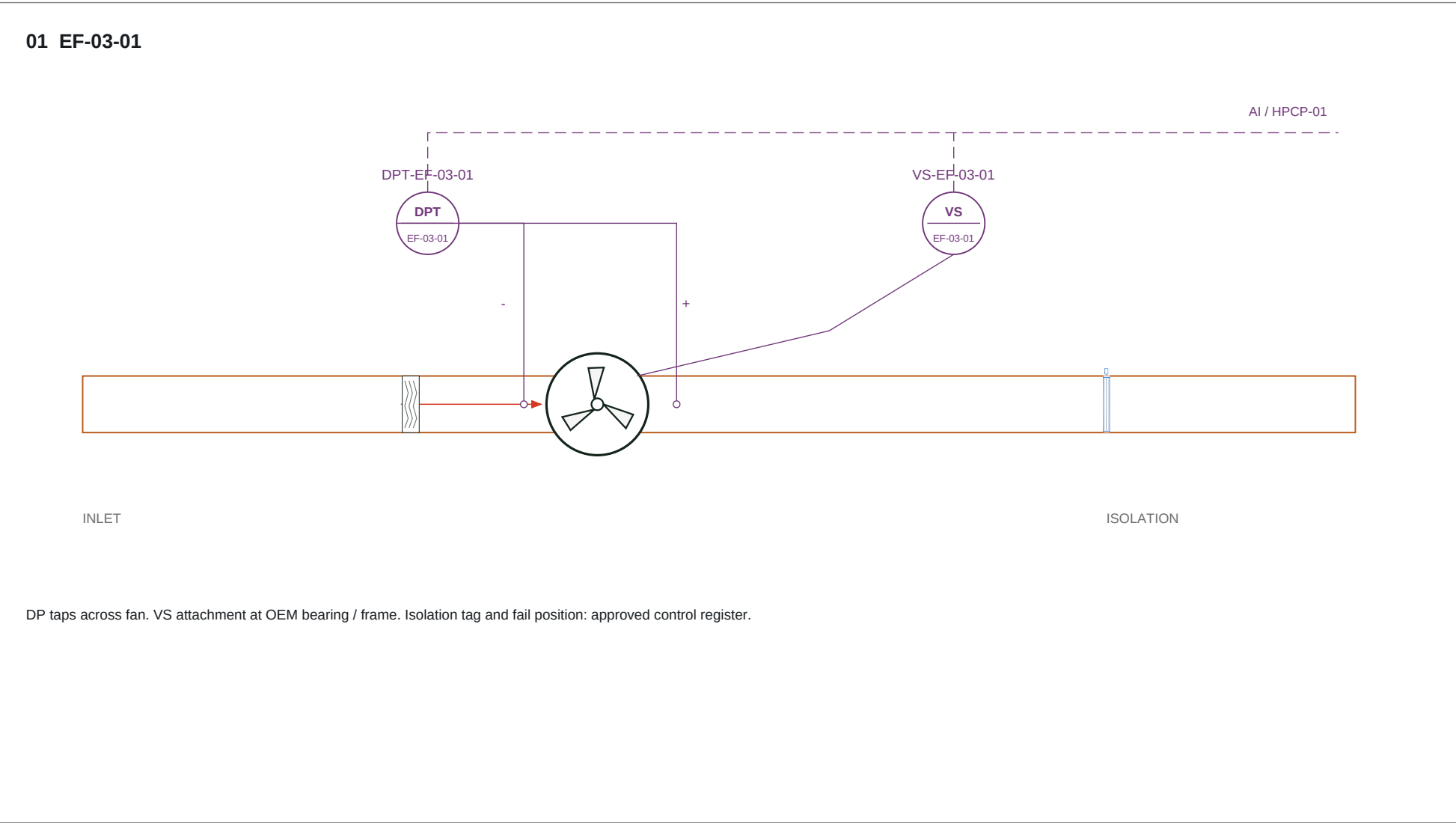
SCALE: NTS AT A1

REVISION

11

07 OCT 2026

SHEET 12 OF 22



NTS functional arrangements. Basement equipment positions are not shown on the Administration plan. CAG fan duty / quantity, tap locations, alarm limits and vibration mountings require OEM / I&C confirmation.

TAQA TRANSMISSION

N-19082-RO-012 | PUMPING STATION STANDARDIZATION

ADMINISTRATION HVAC

DESIGN BASIS: C-H-SS-001 Rev0 / HAP REFERENCE

DRAWING TITLE

EXTRACT AND BASEMENT FAN INSTRUMENTATION

FOR COORDINATION | CHECKED / APPROVED: PENDING

DRAWING NUMBER

ADM-HVAC-D-013

SCALE: NTS AT A1

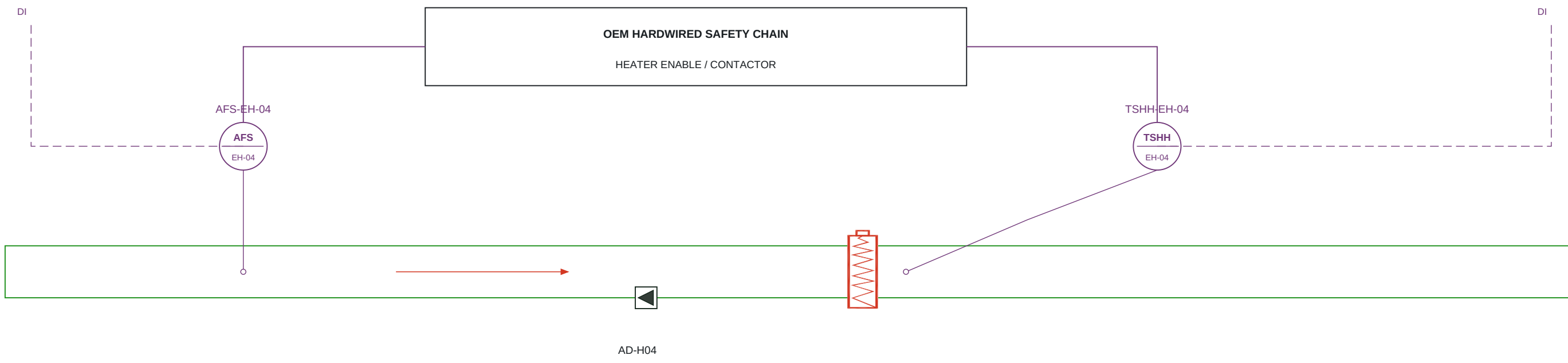
REVISION

11

07 OCT 2026

SHEET 13 OF 22

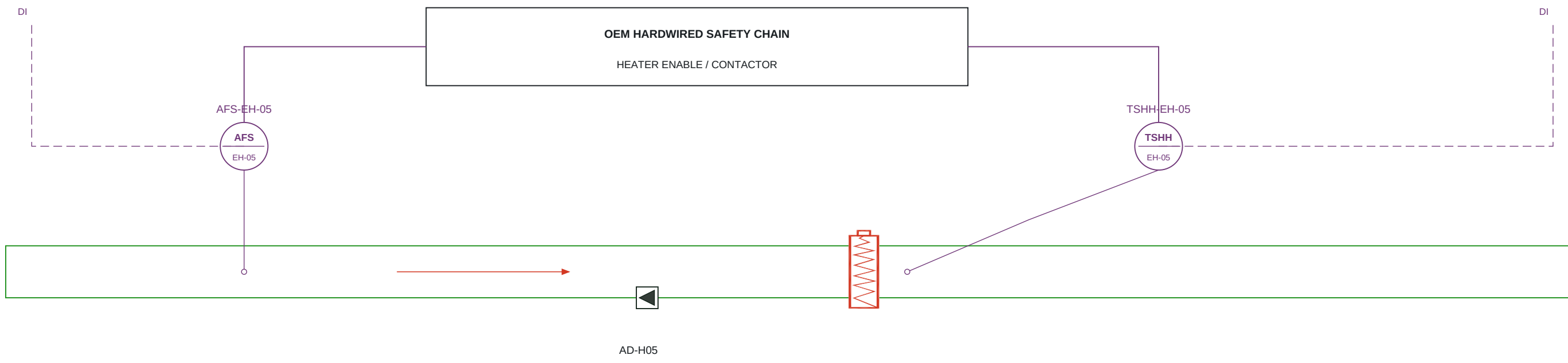
EH-04 | SAFETY / MONITORING INTERFACE



LOSS OF PROOF / HIGH LIMIT: HEATING OFF

PLC heat demand cannot override OEM safety. Sensor technology, setpoints, manual reset and terminal wiring: OEM / I&C approval.

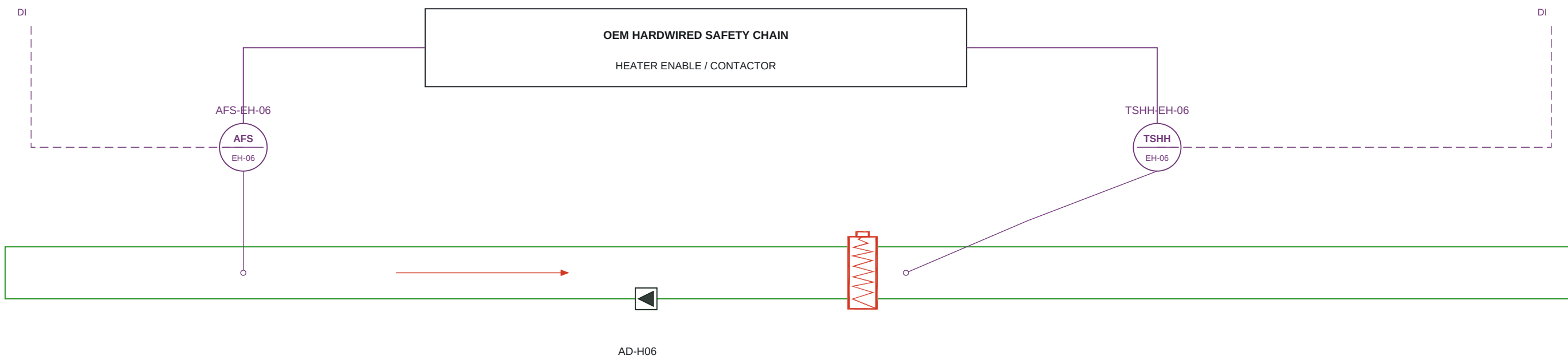
EH-05 | SAFETY / MONITORING INTERFACE



LOSS OF PROOF / HIGH LIMIT: HEATING OFF

PLC heat demand cannot override OEM safety. Sensor technology, setpoints, manual reset and terminal wiring: OEM / I&C approval.

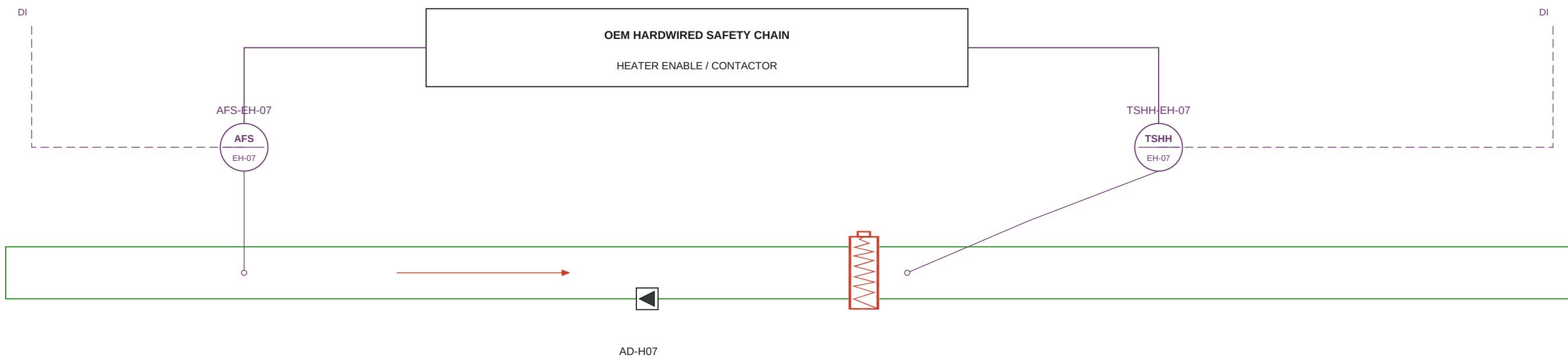
EH-06 | SAFETY / MONITORING INTERFACE



LOSS OF PROOF / HIGH LIMIT: HEATING OFF

PLC heat demand cannot override OEM safety. Sensor technology, setpoints, manual reset and terminal wiring: OEM / I&C approval.

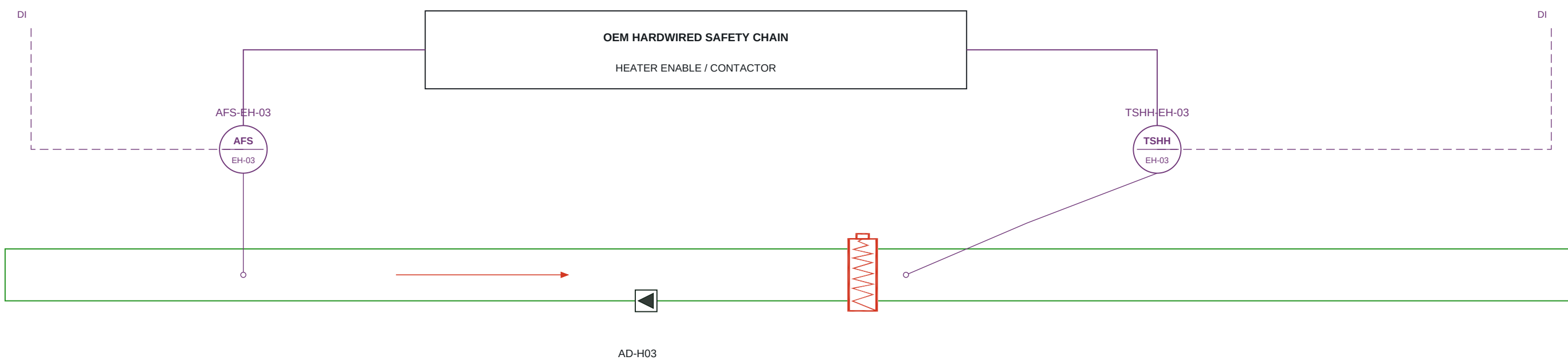
EH-07 | SAFETY / MONITORING INTERFACE



LOSS OF PROOF / HIGH LIMIT: HEATING OFF

PLC heat demand cannot override OEM safety. Sensor technology, setpoints, manual reset and terminal wiring: OEM / I&C approval.

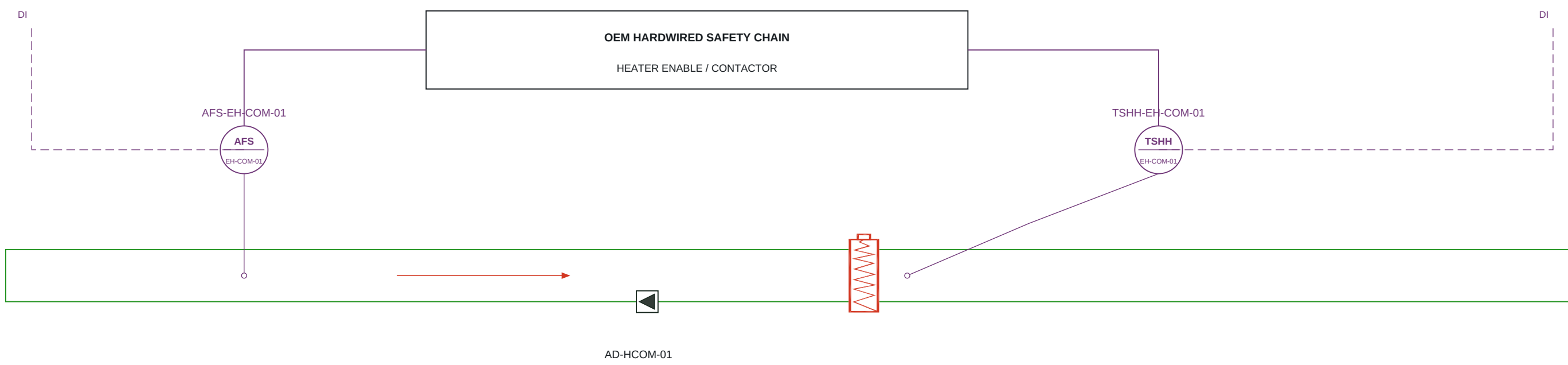
EH-03 | SAFETY / MONITORING INTERFACE



LOSS OF PROOF / HIGH LIMIT: HEATING OFF

PLC heat demand cannot override OEM safety. Sensor technology, setpoints, manual reset and terminal wiring: OEM / I&C approval.

EH-COM-01 | SAFETY / MONITORING INTERFACE



LOSS OF PROOF / HIGH LIMIT: HEATING OFF

PLC heat demand cannot override OEM safety. Sensor technology, setpoints, manual reset and terminal wiring: OEM / I&C approval.

Functional interfaces, not certified wiring diagrams. Six airflow-proof DI and six high-limit DI tags are retained. Protection circuits must be independently implemented and approved by the heater manufacturer.

TAQA TRANSMISSION

N-19082-RO-012 | PUMPING STATION STANDARDIZATION

ADMINISTRATION HVAC

DESIGN BASIS: C-H-SS-001 Rev0 / HAP REFERENCE

DRAWING TITLE

ELECTRIC HEATER PROTECTION AND CONTROL

FOR COORDINATION | CHECKED / APPROVED: PENDING

DRAWING NUMBER

ADM-HVAC-D-014

SCALE: NTS AT A1

REVISION

11

07 OCT 2026

SHEET 14 OF 22

01 INSTRUMENT / SIGNAL CONVENTIONS

- TT

Temperature transmitter
- RHT

Humidity transmitter
- DPT

Differential pressure transmitter
- FIT

Airflow transmitter
- POT

Remote filter DP transmitter
- POI

Local differential pressure indication
- VS

Vibration sensing
- AFS

Airflow proving switch
- TSHH

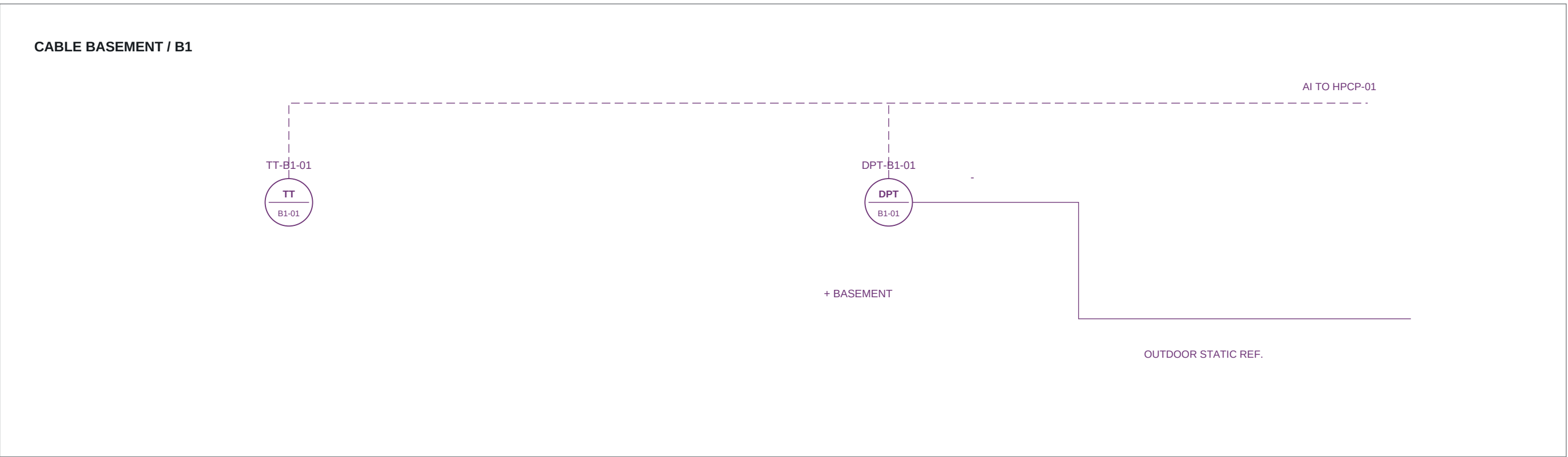
Heater high-temperature limit

Sensing / physical interface (route schematic)

Electrical / communications signal (schematic)

AI / AO / DI / DO are point classes, not selected signal ranges. 4-20 mA, contacts, network protocol, cable types and terminal numbers are not inferred.

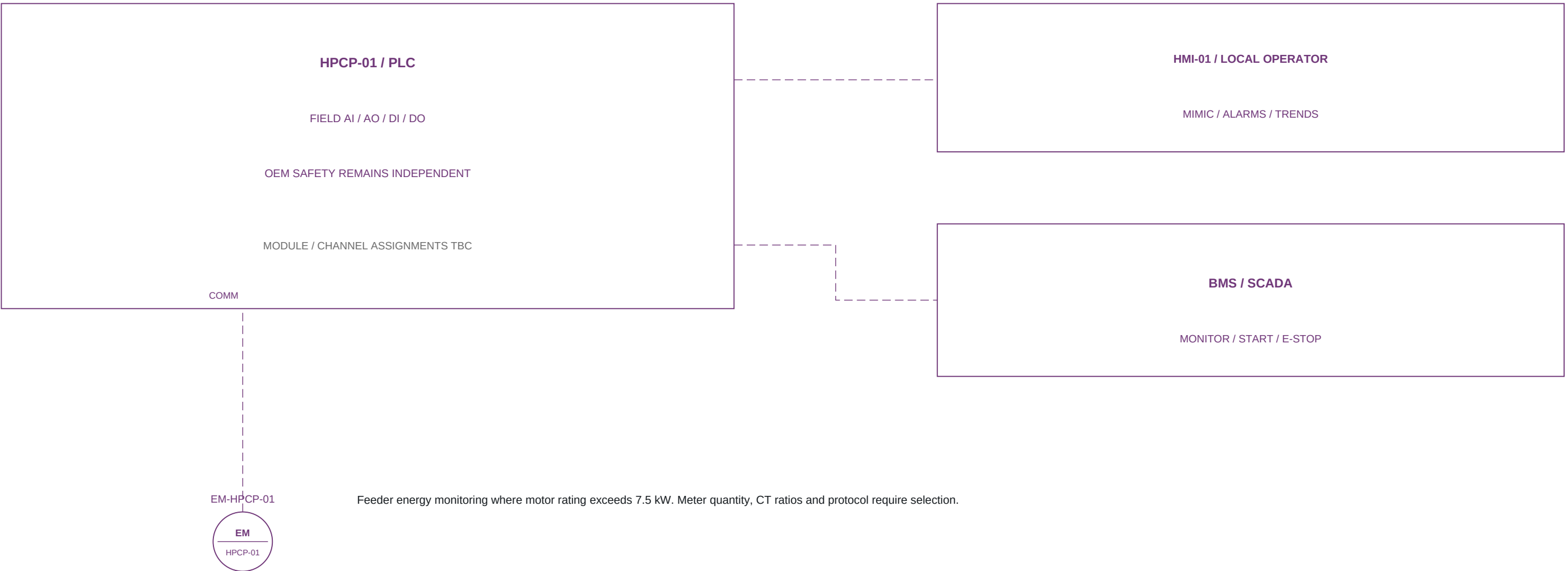
02 BASEMENT FIELD SENSING - PROPOSED



Basement temperature, volume, fan duties and actual pressure target remain open. Fan instrumentation is fully tagged on D-013; no basement architectural layout was supplied.

All controls shown are functional design intent. Approved OEM wiring and project I&C schedules govern implementation. D-016 provides an instrument-to-drawing cross-reference for every one of the 85 scheduled instruments.

03 HPCP / HMI / BMS INTERFACE



04 POINT CLASSES / MINIMUM SPARE

Class	Scheduled	20% spare	Min. channels
AI	65	13	78
AO	22	5	27
DI	141	29	170
DO	44	9	53

COMM interfaces are separate from hardwired channels. Local PDI instruments have no PLC channel. Additional proposals are identified in the instrument index, D-016.

05 FIRE / STARTUP DESIGN INTENT

Confirmed fire: stop every HVAC fan and heater. Restart after fire ESD requires local manual reset. Damper fire positions are governed by the approved cause-and-effect, not this drawing.

Normal operation: two duty 50% PAUs with an available alternate. Prove selected air paths before starting fans. Duty rotation, changeover, timing and final permissives require approved OEM logic.

TAQA TRANSMISSION

N-19082-RO-012 | PUMPING STATION STANDARDIZATION

ADMINISTRATION HVAC

DESIGN BASIS: C-H-SS-001 Rev0 / HAP REFERENCE

DRAWING TITLE

FIELD SIGNALS, BASEMENT AND CONTROL INTERFACES

FOR COORDINATION | CHECKED / APPROVED: PENDING

DRAWING NUMBER

ADM-HVAC-D-015

SCALE: NTS AT A1

REVISION

11

07 OCT 2026

SHEET 15 OF 22

ADMINISTRATION BUILDING - COMPLETE INSTRUMENT INDEX AND DRAWING REFERENCES

Instrument tag	I/O	Service / location	Status	D-sheet
TT-GF-02	AI	GF-02 Elec. panel	Required	007
RHT-GF-02	AI	GF-02 Elec. panel	Required	007
DPT-GF-02	AI	GF-02 Elec. panel	Proposal	007
TT-GF-04	AI	GF-04 Meeting	Required	007
RHT-GF-04	AI	GF-04 Meeting	Required	007
DPT-GF-04	AI	GF-04 Meeting	Proposal	007
TSHH-EH-04	DI	EH-04	OEM safety	014
TT-GF-08	AI	GF-08 Telecom	Required	007
RHT-GF-08	AI	GF-08 Telecom	Required	007
DPT-GF-08	AI	GF-08 Telecom	Proposal	007
TT-GF-05	AI	GF-05 Operator	Required	007
RHT-GF-05	AI	GF-05 Operator	Required	007
DPT-GF-05	AI	GF-05 Operator	Proposal	007
TSHH-EH-05	DI	EH-05	OEM safety	014
TT-GF-06	AI	GF-06 Corridor	Required	007
RHT-GF-06	AI	GF-06 Corridor	Required	007
DPT-GF-06	AI	GF-06 Corridor	Required	007
TSHH-EH-06	DI	EH-06	OEM safety	014
TT-GF-09	AI	GF-09 Toilets	Required	007
RHT-GF-09	AI	GF-09 Toilets	Required	007
DPT-GF-09	AI	GF-09 Toilets	Proposal	007
TT-GF-07	AI	GF-07 Rest room	Required	007
RHT-GF-07	AI	GF-07 Rest room	Required	007
DPT-GF-07	AI	GF-07 Rest room	Proposal	007
TSHH-EH-07	DI	EH-07	OEM safety	014
TT-GF-03	AI	GF-03 Kitchen	Required	007
RHT-GF-03	AI	GF-03 Kitchen	Required	007
DPT-GF-03	AI	GF-03 Kitchen	Proposal	007
TSHH-EH-03	DI	EH-03	OEM safety	014
TT-GF-01	AI	GF-01 Clean agent	Required	007
RHT-GF-01	AI	GF-01 Clean agent	Proposal	007
DPT-GF-01	AI	GF-01 Clean agent	Proposal	007
TT-SA-01	AI	Common supply	Required	008 / 012
TT-RA-01	AI	Common return	Required	008 / 012
RHT-RA-01	AI	Common return	Required	008 / 012
FIT-SA-01	AI	Common SA	Required	008 / 012
FIT-RA-01	AI	Common RA	Required	008 / 012
PDT-OA-01	AI	F-OA-01	Proposal	012
PDI-OA-01	LOCAL	F-OA-01	Proposal	012
TSHH-EH-COM-01	DI	EH-COM-01	OEM safety	014
PDT-F1-A	AI	PAU-A filter 1	Required	012
PDI-F1-A	LOCAL	PAU-A filter 1	Required	012
PDT-F2-A	AI	PAU-A filter 2	Required	012

Instrument tag	I/O	Service / location	Status	D-sheet
PDI-F2-A	LOCAL	PAU-A filter 2	Required	012
DPT-SF-A	AI	PAU-A fan	Required	012
VS-PAU-A	AI	PAU-A frame / fan bearing	Required	012
PDT-F1-B	AI	PAU-B filter 1	Required	012
PDI-F1-B	LOCAL	PAU-B filter 1	Required	012
PDT-F2-B	AI	PAU-B filter 2	Required	012
PDI-F2-B	LOCAL	PAU-B filter 2	Required	012
DPT-SF-B	AI	PAU-B fan	Required	012
VS-PAU-B	AI	PAU-B frame / fan bearing	Required	012
PDT-F1-C	AI	PAU-C filter 1	Required	012
PDI-F1-C	LOCAL	PAU-C filter 1	Required	012
PDT-F2-C	AI	PAU-C filter 2	Required	012
PDI-F2-C	LOCAL	PAU-C filter 2	Required	012
DPT-SF-C	AI	PAU-C fan	Required	012
VS-PAU-C	AI	PAU-C frame / fan bearing	Required	012
DPT-EF-03-01	AI	EF-03-01	Proposal	013
VS-EF-03-01	AI	EF-03-01	Required	013
DPT-EF-03-02	AI	EF-03-02	Proposal	013
VS-EF-03-02	AI	EF-03-02	Required	013
DPT-EF-09-01	AI	EF-09-01	Proposal	013
VS-EF-09-01	AI	EF-09-01	Required	013
DPT-EF-09-02	AI	EF-09-02	Proposal	013
VS-EF-09-02	AI	EF-09-02	Required	013
DPT-EF-01-01	AI	EF-01-01	Proposal	013
VS-EF-01-01	AI	EF-01-01	Required	013
DPT-SF-B1-01	AI	SF-B1-01	Proposal	013
VS-SF-B1-01	AI	SF-B1-01	Required	013
DPT-SF-B1-02	AI	SF-B1-02	Proposal	013
VS-SF-B1-02	AI	SF-B1-02	Required	013
DPT-EF-B1-01	AI	EF-B1-01	Proposal	013
VS-EF-B1-01	AI	EF-B1-01	Required	013
DPT-EF-B1-02	AI	EF-B1-02	Proposal	013
VS-EF-B1-02	AI	EF-B1-02	Required	013
TT-B1-01	AI	Cable basement	Required	015
DPT-B1-01	AI	Cable basement	Proposal	015
EM-HPCP-01	COMM	HPCP feeders >7.5 kW	Required*	015
AFS-EH-04	DI	Electric heater EH-04	Proposal	014
AFS-EH-05	DI	Electric heater EH-05	Proposal	014
AFS-EH-06	DI	Electric heater EH-06	Proposal	014
AFS-EH-07	DI	Electric heater EH-07	Proposal	014
AFS-EH-03	DI	Electric heater EH-03	Proposal	014
AFS-EH-COM-01	DI	Electric heater EH-COM-01	Proposal	014

85 unique instruments; each has a tagged graphic and drawing reference. Proposal / OEM safety / conditional metering statuses are retained. D-sheet references are ADM-HVAC-D-nnn; local PDI instruments are not wired AI points.

TAQA TRANSMISSION

N-19082-RO-012 | PUMPING STATION STANDARDIZATION

ADMINISTRATION HVAC

DESIGN BASIS: C-H-SS-001 Rev0 / HAP REFERENCE

DRAWING TITLE

COMPLETE INSTRUMENT INDEX AND DRAWING REFERENCES

FOR COORDINATION | CHECKED / APPROVED: PENDING

DRAWING NUMBER

ADM-HVAC-D-016

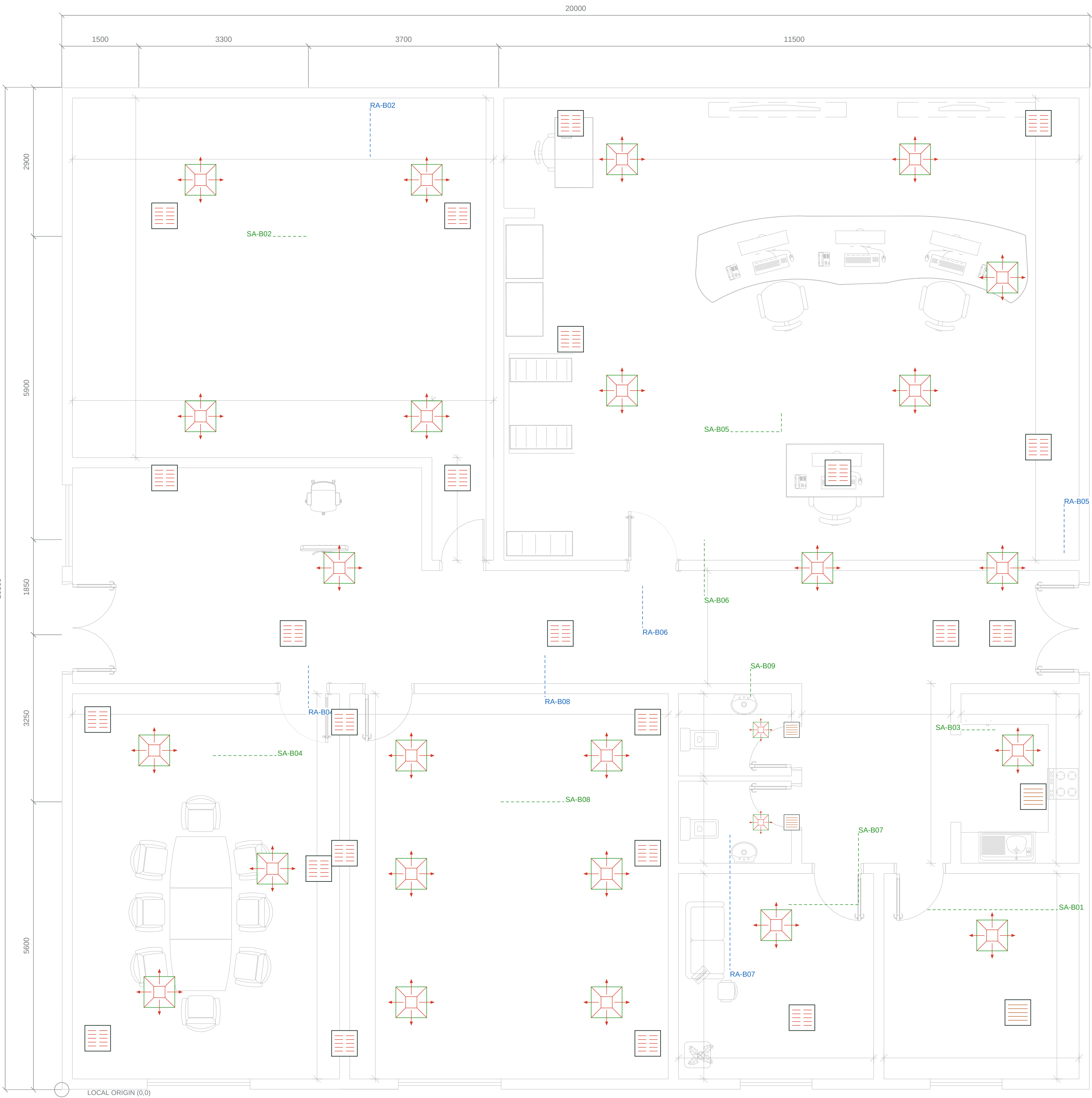
SCALE: NTS AT A1

REVISION

11

07 OCT 2026

SHEET 16 OF 22



SOURCE PLAN COORDINATES / DATUM

Dimensions and coordinates are derived from the supplied architectural plan. Local (0,0) is a drawing origin, not a surveyed site benchmark. Confirm all setting-out against the coordinated ceiling and architecture.

GF	Terminal X range mm	Terminal Y range mm
02	2000 - 7700	11900 - 17700
05	9900 - 19000	12000 - 18800
04	700 - 5000	1000 - 7200
08	5500 - 11400	900 - 7150
06	4500 - 18300	8875 - 10150
07	13900 - 14400	1400 - 3200
09	13600 - 14200	5200 - 7000
03	18600 - 18900	5700 - 6600
01	18100 - 18600	1500 - 3000

Individual terminal coordinates: Terminals register. Room branch centerlines: Duct_sections register. Native CAD dimensions are provided for the known overall extents and room setting-out offsets.

User: structural underside +4.000 m; minimum ceiling +3.300 m. Proposed indoor bare BOD +3.400 m; insulation 50 mm assumed; deepest bare duct 400 mm. D-020 shows the single-layer envelope. Roof routing and penetrations: D-022, pending coordination.

INSPECTION / COORDINATION NOTES

- Verify dimensions on site; do not scale NTS details.
- Match device tags to registers before procurement.
- Reserve access to dampers, heaters and filters.
- Obtain selected-fitting losses and OEM duties.
- Close pressure / outdoor-air design holds.

ADMINISTRATION BUILDING - DRAWING INDEX, DESIGN BASIS AND ISSUE NOTES

DETAILED DRAWING INDEX

Drawing number	Drawing title	Scale
ADM-HVAC-D-001	COORDINATED DUCT PLAN	1:50
ADM-HVAC-D-002	SUPPLY AIR DUCT PLAN	1:50
ADM-HVAC-D-003	RETURN AIR AND EXTRACT DUCT PLAN	1:50
ADM-HVAC-D-004	NORTH ROOMS - DUCT ENLARGEMENT	1:35
ADM-HVAC-D-005	SOUTH-WEST ROOMS - DUCT ENLARGEMENT	1:25
ADM-HVAC-D-006	SOUTH-EAST ROOMS - DUCT ENLARGEMENT	1:25
ADM-HVAC-D-007	ROOM INSTRUMENTATION AND CONTROL LOCATIONS	1:50
ADM-HVAC-D-008	PAU BANK AND PROPOSED ROOF COMMON DISTRIBUTION	NTS
ADM-HVAC-D-009	DUCT SYMBOLS AND INSTALLATION DETAILS	NTS
ADM-HVAC-D-010	DUCT AND TERMINAL SCHEDULES	NTS
ADM-HVAC-D-011	TAPERED FITTINGS AND MAINTENANCE ACCESS	NTS
ADM-HVAC-D-012	PLANT INSTRUMENTATION AND HEATER PROTECTION	NTS
ADM-HVAC-D-013	EXTRACT AND BASEMENT FAN INSTRUMENTATION	NTS
ADM-HVAC-D-014	ELECTRIC HEATER PROTECTION AND CONTROL	NTS
ADM-HVAC-D-015	FIELD SIGNALS, BASEMENT AND CONTROL INTERFACES	NTS
ADM-HVAC-D-016	COMPLETE INSTRUMENT INDEX AND DRAWING REFERENCES	NTS
ADM-HVAC-D-017	PLAN DIMENSIONS AND TERMINAL SETTING-OUT	1:50
ADM-HVAC-D-018	DRAWING INDEX, DESIGN BASIS AND ISSUE NOTES	NTS
ADM-HVAC-D-019	ACTUAL DUCT JUNCTIONS - ENLARGED PLAN DETAILS	AS SHOWN
ADM-HVAC-D-020	CROSSING REGISTER / VERTICAL COORDINATION	NTS
ADM-HVAC-D-021	REV11 TERMINAL / SIZE CHANGES	NTS
ADM-HVAC-D-022	ROOF DROPS / FUNCTIONAL DISTRIBUTION OPTION	NTS

DESIGN BASIS / OPEN ENGINEERING ITEMS

Supply 3697.80 L/s; assumed return 3006.68 L/s. Pressure targets +50 Pa normal rooms, +25 Pa kitchen / toilets / clean-agent; actual balance unverified.

OA reference 420.70 L/s is below 691.12 L/s implied makeup. Toilet net SA-EA is -18.69 L/s; kitchen net is zero. Neither proves the +25 Pa target. Resolve transfer air / extract balance and outdoor-air compliance.

Fan ESP / curves, fitting K, vertical lengths, levels, support and insulation, diffuser throw / noise, access envelopes and I&C settings remain pending.

ISSUE STATUS: FOR ENGINEERING COORDINATION - NOT FOR CONSTRUCTION. Drawn / checked / approved identities and authority are pending. Reference styling does not establish design compliance or professional sign-off.

REVISION 11 - ROOF DISTRIBUTION OPTION

Ceiling constraint: 3.300 m clear below 4.000 m slab. Proposed roof common mains with independent room drops eliminate all indoor duct crossings; local bare duct depth limited to 400 mm. Roof option selection and structural survey pending.

Complete tagged fan / heater protection and basement / energy interfaces are added. Existing 85 instrument requirements and room duties are retained. Proposed section size changes and terminal relocations are scheduled on D-021.

Reference D-113391-ADDC-AES-ME-AC-001_REV.1 is a drafting benchmark; its building duties and instrument requirements are not substituted for this Administration design.

DRAWING / COORDINATION CONVENTIONS

All duct sizes: clear internal width x height in mm; D denotes round diameter. Flows: L/s; velocities: m/s. Add approved insulation outside clear sizes.

RS tags identify proposed room roof drops; each matches D-022 and the riser register. No indoor service crossings. D-020 records proposed bare BOD +3.400 m and the envelope assumptions; actual component envelopes remain pending.

MFD* graphic corresponds to conditional FSD-prefixed register tags. Approval of rated boundaries governs quantity, rating and fail positions.

TT / RHT / DPT room locations are proposed. Pressure references are outdoors. Dashed I&C lines are functional signals, not installed cable routes.

TAQA TRANSMISSION

N-19082-RO-012 | PUMPING STATION STANDARDIZATION

ADMINISTRATION HVAC

DESIGN BASIS: C-H-SS-001 Rev0 / HAP REFERENCE

DRAWING TITLE

DRAWING INDEX, DESIGN BASIS AND ISSUE NOTES

FOR COORDINATION | CHECKED / APPROVED: PENDING

DRAWING NUMBER

ADM-HVAC-D-018

SCALE: NTS AT A1

REVISION

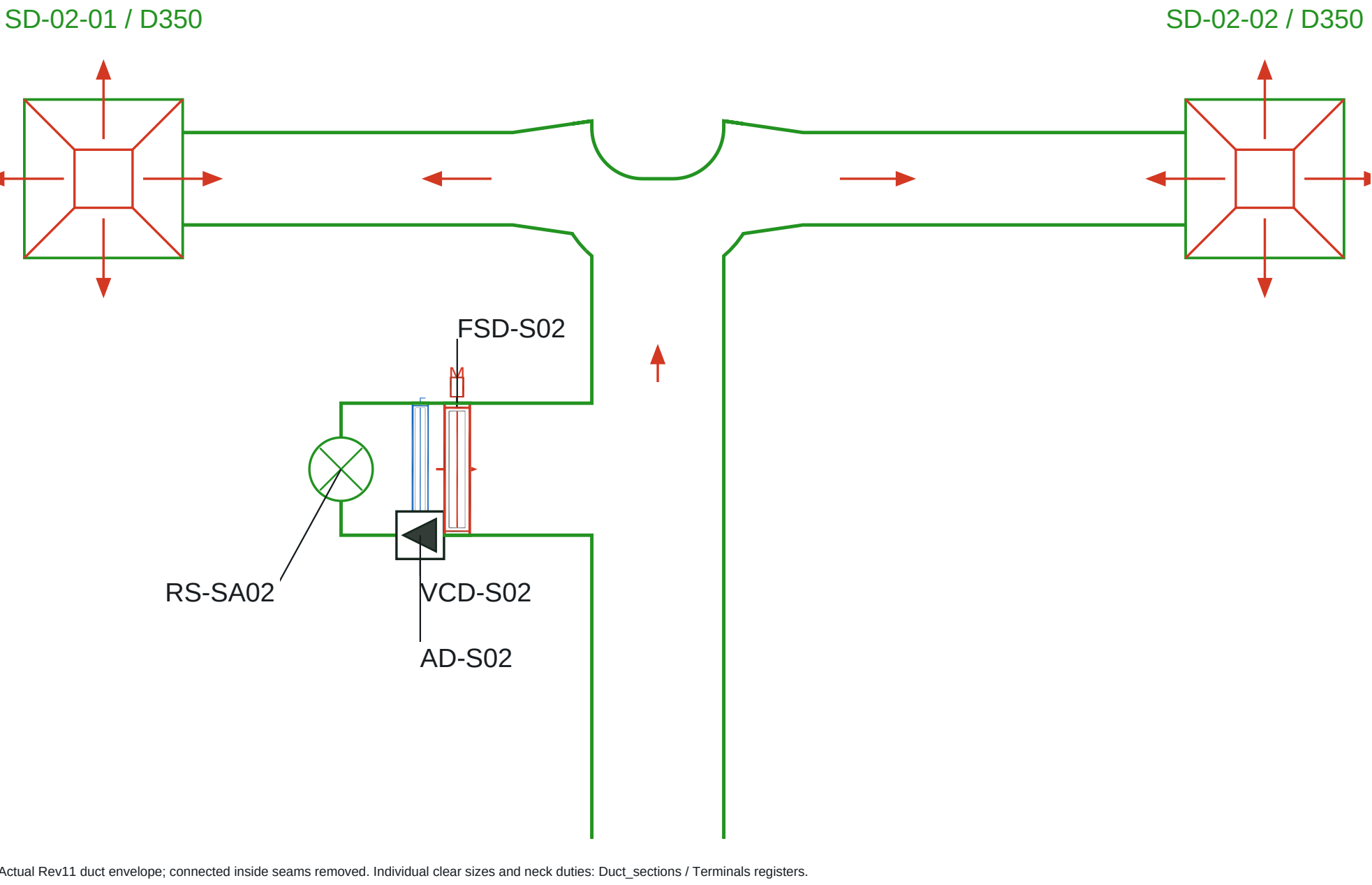
11

07 OCT 2026

SHEET 18 OF 22

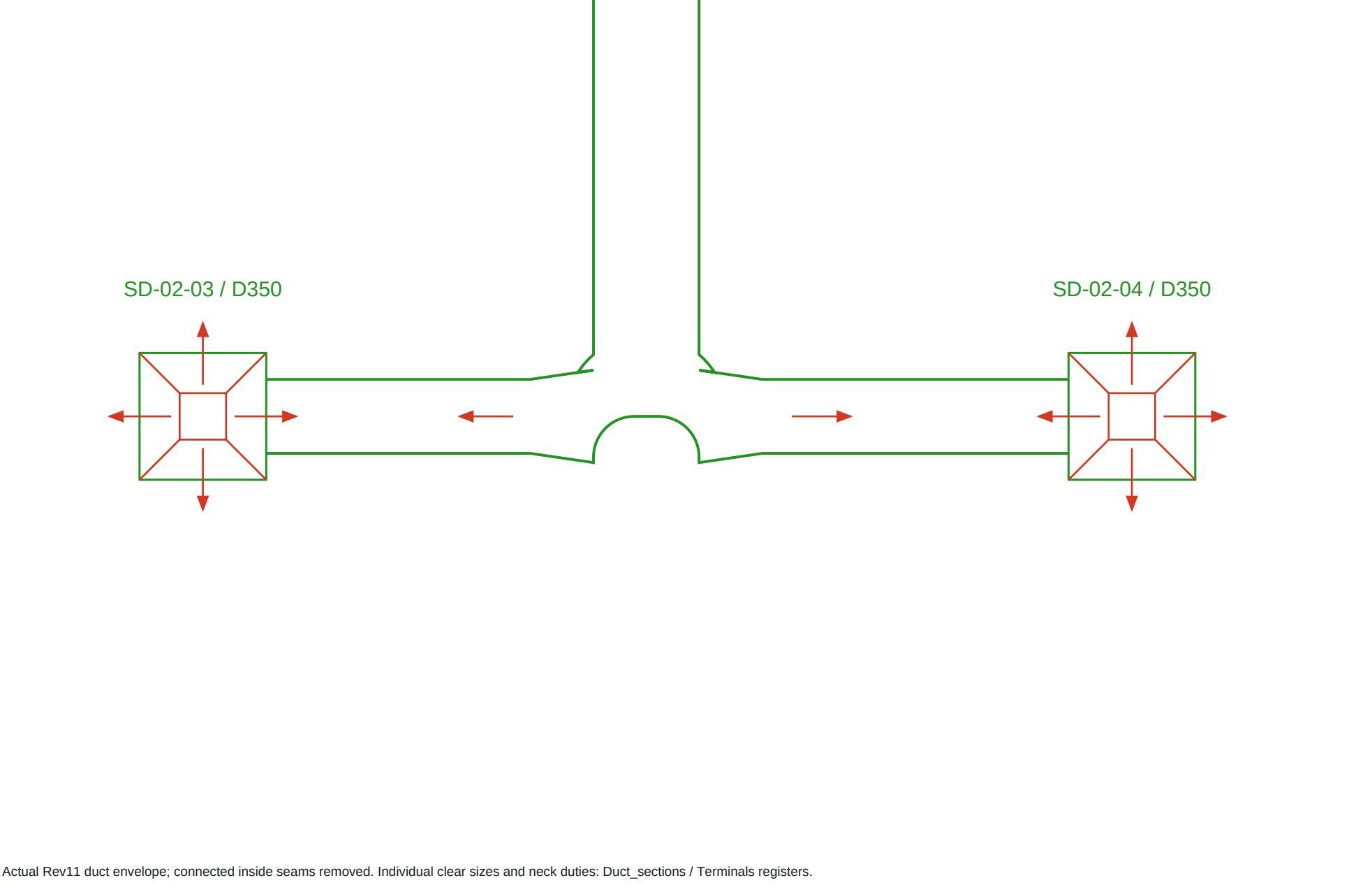
ADMINISTRATION BUILDING - ACTUAL DUCT JUNCTIONS - ENLARGED PLAN DETAILS

A GF-02 ROOF DROP / BRANCH / HEADER LEGS / 1:20



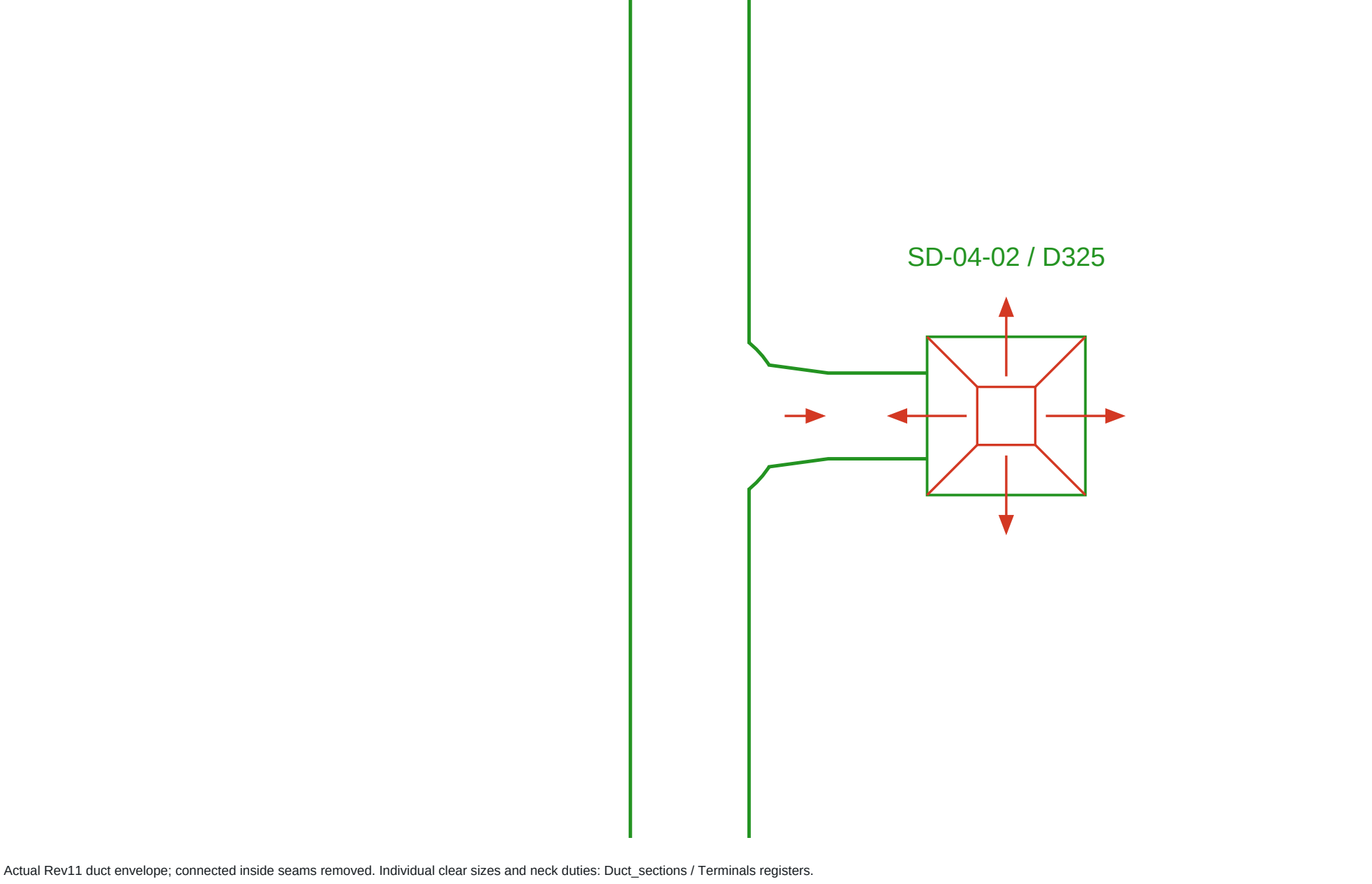
Formed fitting / access / face-to-face dimensions require selected manufacturer confirmation. Other services omitted in this enlargement; single-layer envelope on D-020, roof drops on D-022.

B GF-02 END TEE / TWO INDIVIDUAL NECKS / 1:25



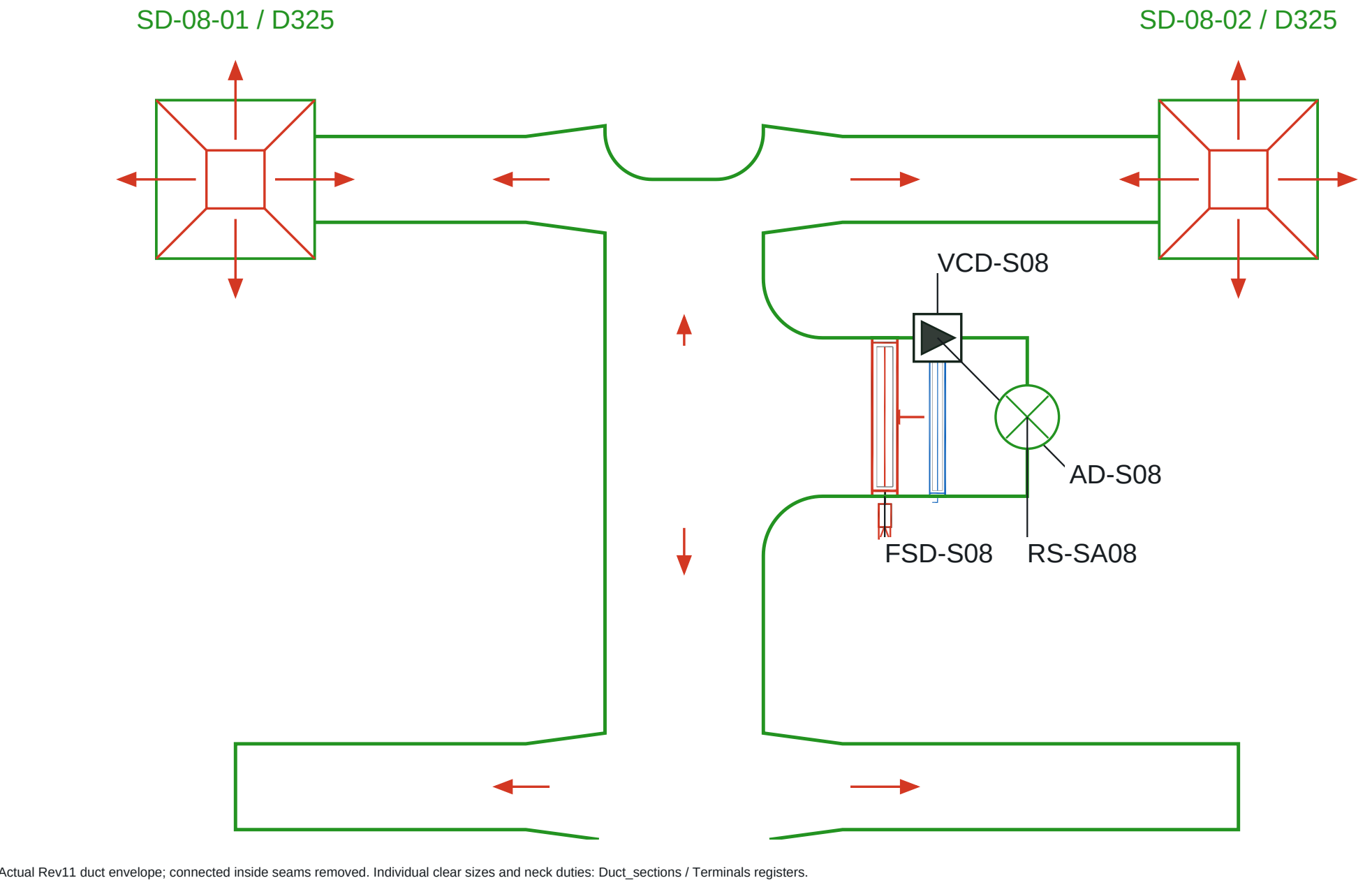
Formed fitting / access / face-to-face dimensions require selected manufacturer confirmation. Other services omitted in this enlargement; single-layer envelope on D-020, roof drops on D-022.

C MEETING CONTINUING HEADER / SIDE NECK / 1:20



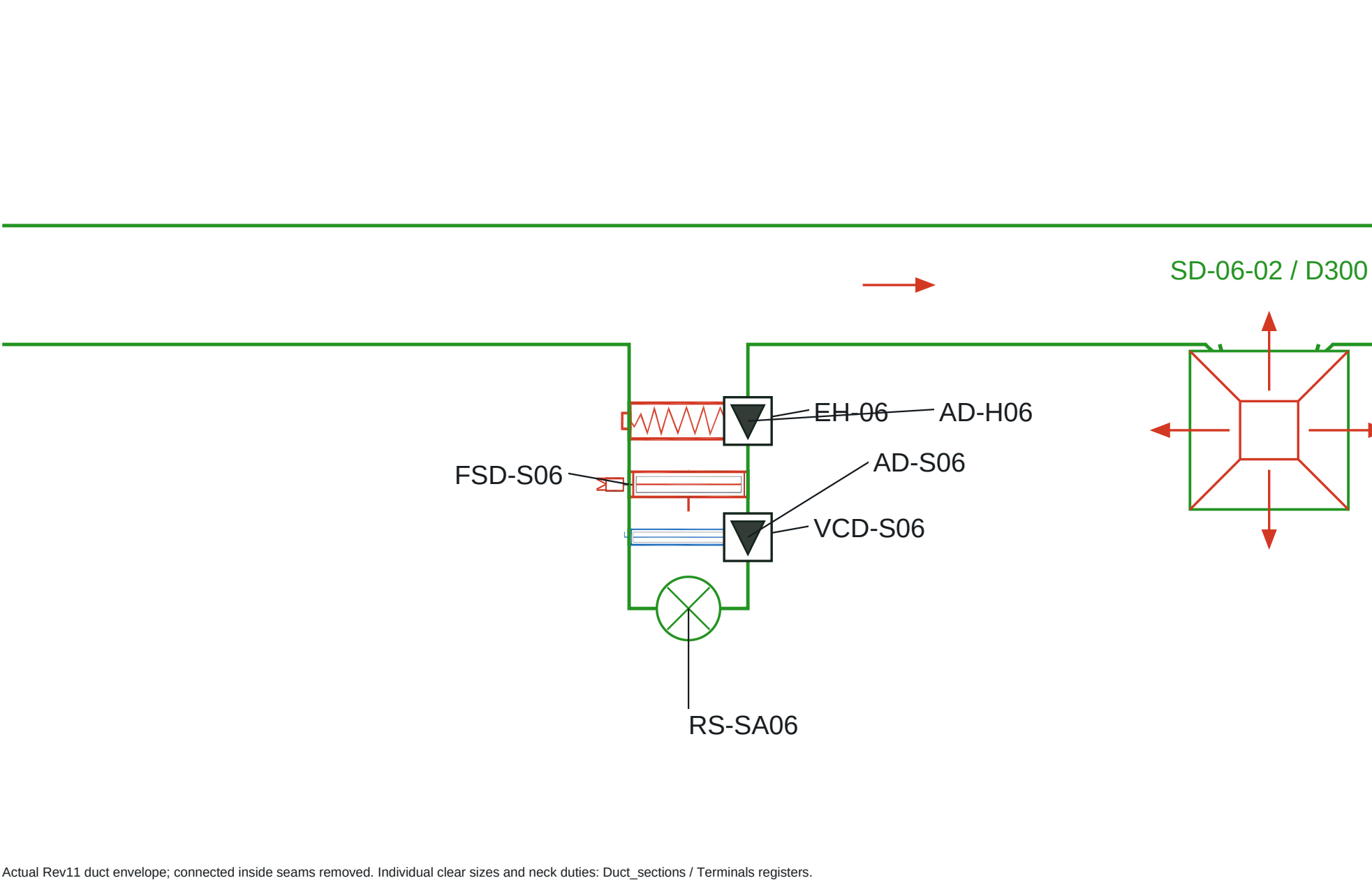
Formed fitting / access / face-to-face dimensions require selected manufacturer confirmation. Other services omitted in this enlargement; single-layer envelope on D-020, roof drops on D-022.

D TELECOM FEEDER / UP AND DOWN HEADERS / 1:20



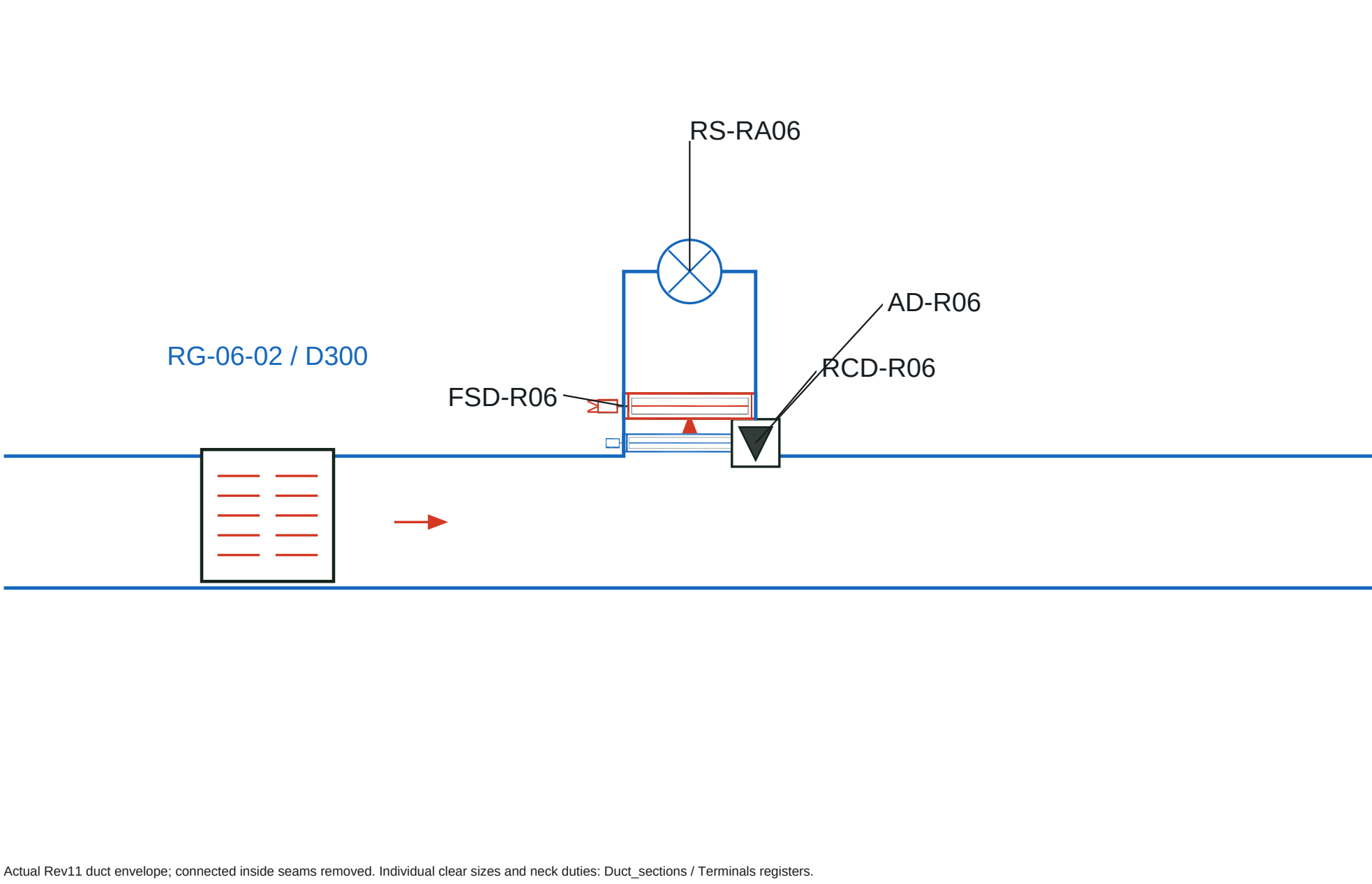
Formed fitting / access / face-to-face dimensions require selected manufacturer confirmation. Other services omitted in this enlargement; single-layer envelope on D-020, roof drops on D-022.

E CORRIDOR ROOM DROP BRANCH / HEATER / HEADER / 1:20



Formed fitting / access / face-to-face dimensions require selected manufacturer confirmation. Other services omitted in this enlargement; single-layer envelope on D-020, roof drops on D-022.

F CORRIDOR COLLECTION / RETURN DEVICE STATION / 1:20



Formed fitting / access / face-to-face dimensions require selected manufacturer confirmation. Other services omitted in this enlargement; single-layer envelope on D-020, roof drops on D-022.

These details are generated from the delivered routing geometry, not independent typical symbols. Continuous exteriors, flared necks and concave fitting throats are checked. Short necks use an integral plenum / vertical drop; their selected construction is pending.

TAQA TRANSMISSION

N-19082-RO-012 | PUMPING STATION STANDARDIZATION

ADMINISTRATION HVAC

DESIGN BASIS: C-H-SS-001 Rev0 / HAP REFERENCE

DRAWING TITLE

ACTUAL DUCT JUNCTIONS - ENLARGED PLAN DETAILS

FOR COORDINATION | CHECKED / APPROVED: PENDING

DRAWING NUMBER

ADM-HVAC-D-019

SCALE: AS SHOWN AT A1

REVISION

11

07 OCT 2026

SHEET 19 OF 22

ADMINISTRATION BUILDING - 3.30 m CEILING - SINGLE-LAYER INDOOR DUCT ENVELOPE

USER DATUMS AND PROPOSED INDOOR LEVELS

Item	Value	Basis / status
Structural underside	+4.000 m AFFL	User confirmed slab / roof
Minimum ceiling / clear height	+3.300 m AFFL	User requirement
Available total space	700 mm	Between confirmed datums
Proposed bare duct bottom	+3.400 m AFFL	All indoor trees; not installed
Indoor insulation	50 mm each face	Assumed; project specification TBC
Largest bare duct depth	400 mm	Round necks max D350 retained
Lowest insulated bottom	+3.350 m AFFL	50 mm above required ceiling
Highest insulated top	+3.850 m AFFL	150 mm below slab for deepest duct

WHAT THE REVISED ROUTES RESOLVE

Deep common mains move to a proposed roof distribution system. Fifteen independent room supply / return drops replace the indoor common-main feeders.

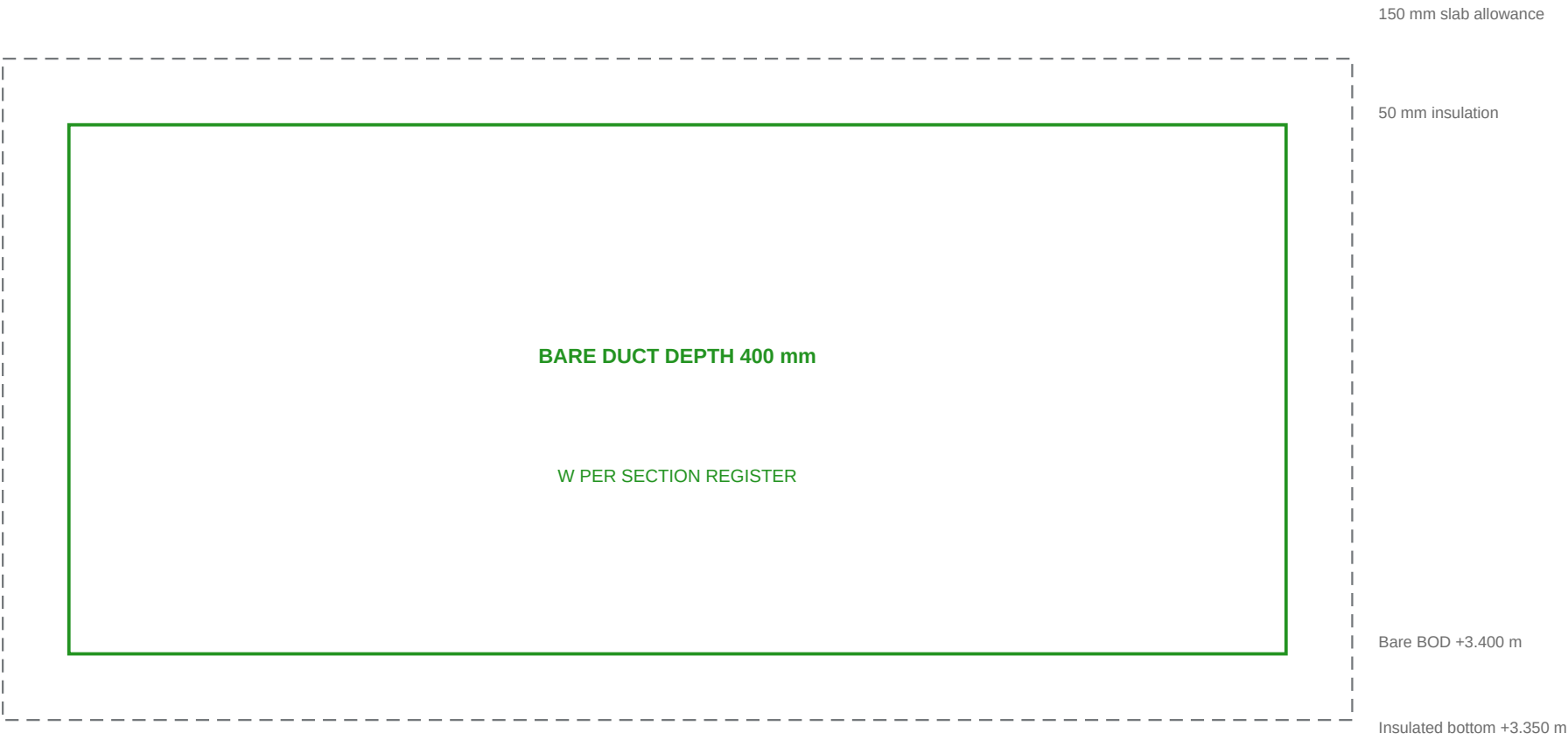
Electrical and telecom returns collect at room perimeters. Every room tree is connected, every collection node conserves airflow, and distinct room trees stay separate.

No indoor bare duct crossings or insulated separate-service overlaps. Continuous duct walls, formed junctions and individual terminal necks are generated from the real revised plan geometry.

Rev10 CX-08 had its lowest insulated bottom at +2.050 m, 1250 mm below this requirement. The Rev10 stacked arrangement is superseded by this roof option; it must not be installed at a 3.30 m ceiling.

DEEPEST DUCT ENVELOPE - NTS

SLAB / ROOF UNDERSIDE +4.000 m



CEILING / CLEAR HEIGHT +3.300 m

Stack: 150 mm slab / support allowance + 50 mm insulation + 400 mm bare duct + 50 mm insulation + 50 mm ceiling gap = 700 mm.

This checks scheduled duct envelopes only. Beam downstands, other services, flanges / stiffeners, damper actuators, heater casing, supports, access hatches and terminal plenums are not manufacturer-selected or surveyed.

The 50 mm gap below insulation is not a general maintenance clearance or a selected plenum dimension. Terminal drops / plenums must use the available connection space within the duct envelope; selected assemblies may require a local redesign.

Roof option selection pending. Do not cut the slab from these drawings. New penetrations, curbs, weatherproofing, roof loads and fire boundaries require structural / architectural / fire coordination.

PROPOSED LEVELS, NOT INSTALLATION APPROVAL. AFFL datum requires site confirmation. 50 mm insulation is an indoor assumption, not a verified UAE standard. D-022 identifies all proposed roof drops; actual roof routing and equipment envelopes remain open.

TAQA TRANSMISSION

N-19082-RO-012 | PUMPING STATION STANDARDIZATION

ADMINISTRATION HVAC

DESIGN BASIS: C-H-SS-001 Rev0 / HAP REFERENCE

DRAWING TITLE

3.30 m CEILING - SINGLE-LAYER INDOOR DUCT ENVELOPE

FOR COORDINATION | CHECKED / APPROVED: PENDING

DRAWING NUMBER

ADM-HVAC-D-020

SCALE: NTS AT A1

REVISION

11

07 OCT 2026

SHEET 20 OF 22

ADMINISTRATION BUILDING - REV11 TERMINAL MOVES AND SHALLOW DUCT SIZE CHANGES

PROPOSED TERMINAL MOVES FROM REV10

Terminal	Rev10 X, Y mm	Rev11 X, Y mm	L/s
SD-02-01	2700, 18000	2700, 17700	166.2
SD-02-02	7100, 18000	7100, 17700	166.2
RG-02-01	5400, 17000	2000, 17000	151.36584740090174
RG-02-03	5400, 11900	2000, 11900	151.36584740090174
SD-08-01	6500, 6500	6800, 6500	147.43333333333334
SD-08-03	6500, 4200	6800, 4200	147.43333333333334
SD-08-05	6500, 1700	6800, 1700	147.43333333333334
RG-08-01	6400, 7150	5500, 7150	137.54389826726785
RG-08-02	11500, 7150	11400, 7150	137.54389826726785
RG-08-03	5900, 4600	5500, 4600	137.54389826726785
RG-08-04	11500, 4600	11400, 4600	137.54389826726785
RG-08-05	5900, 900	5500, 900	137.54389826726785
RG-08-06	11500, 900	11400, 900	137.54389826726785

Individual duties, neck diameters and face sizes are retained. Moves make room for perimeter return trees and insulation; coordinate reflected ceiling, throw / noise, lighting and maintenance access before adoption.

TRACEABLE ROUTING / AIRFLOW

116 indoor section paths; 15 proposed roof drops. Local branches, headers and necks remain connected to their allocated room riser. Roof-main sizes / flows are separately listed as a functional basis; actual roof lengths and levels are TBC.

All indoor section velocities, plan centerline lengths and straight-duct friction are recomputed. Wider shallow sections replace the deep returns. Selected elbow / riser / transition losses and revised total fan ESP remain pending; straight friction is not a system pressure calculation.

Rev11 is a roof-distribution OPTION developed for the new 3.30 m clear-height requirement. Rev10 remains preserved as superseded for this ceiling constraint. These are engineering coordination proposals; final ceiling / OEM / pressure / structural approvals are pending.

CLEAR SIZE REVISIONS FROM REV10

Section	Rev10 mm	Rev11 mm
RA-B02	600x400	650x400
RA-H02-U01	600x400	450x350
RA-H02-D01	600x400	450x350
RA-B08	500x650	850x400
RA-H08-U01	500x650	500x350
RA-H08-D01	500x650	500x350
RA-H08-D02	500x650	500x350
RA-B05	500x525	700x400
RA-H05-U01	500x525	700x400
RA-H05-U02	500x525	700x400
RA-H05-U03	500x525	700x400
RA-H05-D01	500x525	700x400
RA-B06	350x550	500x400
RA-H06-D01	350x550	500x400
RA-H05-L01	500x525	700x400
RA-H05-N01	500x525	700x400
RA-H05-E01	500x525	700x400
RA-H06-L01	350x550	500x400
RA-H06-L02	350x550	500x400
RA-H06-E01	350x550	500x400

Largest indoor bare depth: 400 mm. Outdoor common mains retain their flow / clear-size basis; weather insulation, actual roof routes and fitting / level selection are pending.

85 instrument requirements and all existing I/O duties are retained. Common heater / probes / conditional humidifier move to the proposed roof plant, with weather-protected equipment and mounting still to be selected.

TAQA TRANSMISSION

N-19082-RO-012 | PUMPING STATION STANDARDIZATION

ADMINISTRATION HVAC

DESIGN BASIS: C-H-SS-001 Rev0 / HAP REFERENCE

DRAWING TITLE

REV11 TERMINAL MOVES AND SHALLOW DUCT SIZE CHANGES

FOR COORDINATION | CHECKED / APPROVED: PENDING

DRAWING NUMBER

ADM-HVAC-D-021

SCALE: NTS AT A1

REVISION

11

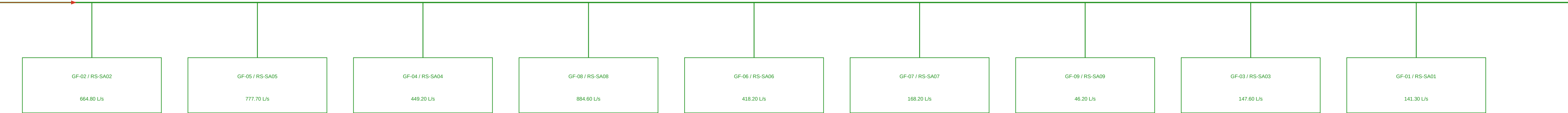
07 OCT 2026

SHEET 21 OF 22

ADMINISTRATION BUILDING - ROOF DISTRIBUTION OPTION AND ROOM RISER REGISTER

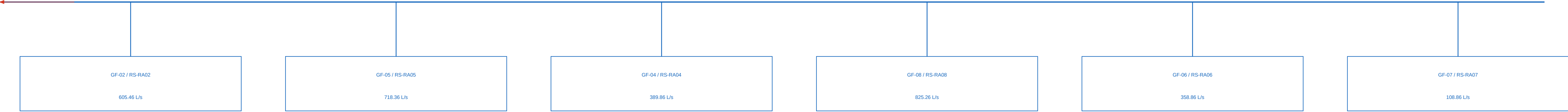
FUNCTIONAL ROOF CONNECTIONS - NTS / ACTUAL ROOF ROUTING TBC

SUPPLY FROM ROOF COMMON PLANT D-008 / 3697.80 L/s



Separate supply drops connect to each room VCD / conditional MFD / heater station; no ceiling common main.

RETURN TO ROOF COMMON PLANT D-008 / 3006.68 L/s ASSUMED



PROPOSED ROOF DROP CENTERS - SOURCE PLAN X / Y, mm

Riser	Branch / GF	X, Y mm	Clear W x H	L/s	Indoor bare BOD	Roof level / vertical length
RS-SA02	SA-B02 / 02	3600, 16600	500x350	664.80	+3.400 m	TBC / TBC
RS-RA02	RA-B02 / 02	6000, 18150	650x400	605.46	+3.400 m	TBC / TBC
RS-SA04	SA-B04 / 04	4200, 6500	350x400	449.20	+3.400 m	TBC / TBC
RS-RA04	RA-B04 / 04	4800, 8250	500x350	389.86	+3.400 m	TBC / TBC
RS-SA08	SA-B08 / 08	9800, 5600	600x400	884.60	+3.400 m	TBC / TBC
RS-RA08	RA-B08 / 08	9400, 8450	850x400	825.26	+3.400 m	TBC / TBC
RS-SA05	SA-B05 / 05	12500, 12800	600x350	777.70	+3.400 m	TBC / TBC
RS-RA05	RA-B05 / 05	19500, 10400	700x400	718.36	+3.400 m	TBC / TBC
RS-SA06	SA-B06 / 06	12500, 9475	450x300	418.20	+3.400 m	TBC / TBC
RS-RA06	RA-B06 / 06	11300, 9800	500x400	358.86	+3.400 m	TBC / TBC
RS-SA09	SA-B09 / 09	13400, 8200	200x150	46.20	+3.400 m	TBC / TBC
RS-SA07	SA-B07 / 07	15500, 5000	300x250	168.20	+3.400 m	TBC / TBC
RS-RA07	RA-B07 / 07	13000, 5000	350x250	108.86	+3.400 m	TBC / TBC
RS-SA03	SA-B03 / 03	17000, 7000	300x200	147.60	+3.400 m	TBC / TBC
RS-SA01	SA-B01 / 01	19400, 3500	300x200	141.30	+3.400 m	TBC / TBC

Roof flow/size basis is functional, not a fabricated plan: no roof survey, beam map, selected curb / sleeve / fire assembly or common heater weather enclosure was supplied. All 15 penetrations are proposals; coordinate spacing, framing, access and roof drainage before setting them out.

DESIGN OPTION SELECTION PENDING. Do not treat this functional schematic as a physical roof routing / clash clearance drawing. The indoor plan is geometrically checked; roof branches, levels, insulation, lengths, losses, support loads and OEM clearances remain to be designed from site information.

TAQA TRANSMISSION N-19082-RO-012 PUMPING STATION STANDARDIZATION ADMINISTRATION HVAC DESIGN BASIS: C-H-SS-001 Rev0 / HAP REFERENCE	DRAWING TITLE ROOF DISTRIBUTION OPTION AND ROOM RISER REGISTER FOR COORDINATION CHECKED / APPROVED: PENDING	DRAWING NUMBER ADM-HVAC-D-022 SCALE: NTS AT A1	REVISION 11 07 OCT 2026 SHEET 22 OF 22
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